

Part IA — Vectors and Matrices

Based on lectures by N. Peake

Notes taken by Dexter Chua

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These notes are not endorsed by the lecturers, and I have modified them (often significantly) after lectures. They are nowhere near accurate representations of what was actually lectured, and in particular, all errors are almost surely mine.

Complex numbers

Review of complex numbers, including complex conjugate, inverse, modulus, argument and Argand diagram. Informal treatment of complex logarithm, n -th roots and complex powers. de Moivre's theorem. [2]

Vectors

Review of elementary algebra of vectors in $\mathbb{R}^3$, including scalar product. Brief discussion of vectors in $\mathbb{R}^n$ and $\mathbb{C}^n$; scalar product and the Cauchy-Schwarz inequality. Concepts of linear span, linear independence, subspaces, basis and dimension.

Suffix notation: including summation convention, δ_{ij} and ε_{ijk} . Vector product and triple product: definition and geometrical interpretation. Solution of linear vector equations. Applications of vectors to geometry, including equations of lines, planes and spheres. [5]

Matrices

Elementary algebra of 3×3 matrices, including determinants. Extension to $n \times n$ complex matrices. Trace, determinant, non-singular matrices and inverses. Matrices as linear transformations; examples of geometrical actions including rotations, reflections, dilations, shears; kernel and image. [4]

Simultaneous linear equations: matrix formulation; existence and uniqueness of solutions, geometric interpretation; Gaussian elimination. [3]

Symmetric, anti-symmetric, orthogonal, hermitian and unitary matrices. Decomposition of a general matrix into isotropic, symmetric trace-free and antisymmetric parts. [1]

Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors; geometric significance. [2]

Proof that eigenvalues of hermitian matrix are real, and that distinct eigenvalues give an orthogonal basis of eigenvectors. The effect of a general change of basis (similarity transformations). Diagonalization of general matrices: sufficient conditions; examples of matrices that cannot be diagonalized. Canonical forms for 2×2 matrices. [5]

Discussion of quadratic forms, including change of basis. Classification of conics, cartesian and polar forms. [1]

Rotation matrices and Lorentz transformations as transformation groups. [1]

Contents

0	Introduction	4
1	Complex numbers	5
1.1	Basic properties	5
1.2	Complex exponential function	7
1.3	Roots of unity	9
1.4	Complex logarithm and power	9
1.5	De Moivre's theorem	9
1.6	Lines and circles in $\mathbb{C}$	10
2	Vectors	12
2.1	Definition and basic properties	12
2.2	Scalar product	13
2.2.1	Geometric picture ($\mathbb{R}^2$ and $\mathbb{R}^3$ only)	13
2.2.2	General algebraic definition	14
2.3	Cauchy-Schwarz inequality	14
2.4	Vector product	15
2.5	Scalar triple product	17
2.6	Spanning sets and bases	18
2.6.1	2D space	18
2.6.2	3D space	18
2.6.3	$\mathbb{R}^n$ space	19
2.6.4	$\mathbb{C}^n$ space	20
2.7	Vector subspaces	21
2.8	Suffix notation	22
2.9	Geometry	25
2.9.1	Lines	25
2.9.2	Plane	26
2.10	Vector equations	27
3	Linear maps	28
3.1	Examples	28
3.1.1	Rotation in $\mathbb{R}^3$	28
3.1.2	Reflection in $\mathbb{R}^3$	29
3.2	Linear Maps	29
3.3	Rank and nullity	31
3.4	Matrices	32
3.4.1	Examples	33
3.4.2	Matrix Algebra	33
3.4.3	Decomposition of an $n \times n$ matrix	35
3.4.4	Matrix inverse	36
3.5	Determinants	37
3.5.1	Permutations	38
3.5.2	Properties of determinants	39
3.5.3	Minors and Cofactors	42

4	Matrices and linear equations	45
4.1	Simple example, 2×2	45
4.2	Inverse of an $n \times n$ matrix	46
4.3	Homogeneous and inhomogeneous equations	47
4.3.1	Gaussian elimination	48
4.4	Matrix rank	50
4.5	Homogeneous problem $A\mathbf{x} = \mathbf{0}$	52
4.5.1	Geometrical interpretation	52
4.5.2	Linear mapping view of $A\mathbf{x} = \mathbf{0}$	53
4.6	General solution of $A\mathbf{x} = \mathbf{d}$	53
5	Eigenvalues and eigenvectors	56
5.1	Preliminaries and definitions	56
5.2	Linearly independent eigenvectors	58
5.3	Transformation matrices	61
5.3.1	Transformation law for vectors	62
5.3.2	Transformation law for matrices	63
5.4	Similar matrices	65
5.5	Diagonalizable matrices	66
5.6	Canonical (Jordan normal) form	68
5.7	Cayley-Hamilton Theorem	70
5.8	Eigenvalues and eigenvectors of a Hermitian matrix	72
5.8.1	Eigenvalues and eigenvectors	72
5.8.2	Gram-Schmidt orthogonalization (non-examinable)	73
5.8.3	Unitary transformations	73
5.8.4	Diagonalization of $n \times n$ Hermitian matrices	74
5.8.5	Normal matrices	75
6	Quadratic forms and conics	77
6.1	Quadrics and conics	78
6.1.1	Quadrics	78
6.1.2	Conic sections ($n = 2$)	79
6.2	Focus-directrix property	80
7	Transformation groups	83
7.1	The orthogonal group	83
7.2	Length-preserving transformations	83
7.3	Lorentz transformations	84

0 Introduction

Vectors and matrices are the language in which a lot of mathematics is written in. In physics, many variables such as position and momentum are expressed as vectors. Heisenberg also formulated quantum mechanics in terms of vectors and matrices. In statistics, one might pack all the results of all experiments into a single vector, and work with a large vector instead of many small quantities. In group theory, matrices are used to represent the symmetries of space (as well as many other groups).

So what is a vector? Vectors are very general objects, and can in theory represent very complex objects. However, in this course, our focus is on vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$. We can think of each of these as an array of n real or complex numbers. For example, $(1, 6, 4)$ is a vector in $\mathbb{R}^3$. These vectors are added in the obvious way. For example, $(1, 6, 4) + (3, 5, 2) = (4, 11, 6)$. We can also multiply vectors by numbers, say $2(1, 6, 4) = (2, 12, 8)$. Often, these vectors represent points in an n -dimensional space.

Matrices, on the other hand, represent *functions* between vectors, i.e. a function that takes in a vector and outputs another vector. These, however, are not arbitrary functions. Instead matrices represent *linear functions*. These are functions that satisfy the equality $f(\lambda \mathbf{x} + \mu \mathbf{y}) = \lambda f(\mathbf{x}) + \mu f(\mathbf{y})$ for arbitrary numbers λ, μ and vectors $\mathbf{x}, \mathbf{y}$. It is important to note that the function $\mathbf{x} \mapsto \mathbf{x} + \mathbf{c}$ for some constant vector $\mathbf{c}$ is *not* linear according to this definition, even though it might look linear.

It turns out that for each linear function from $\mathbb{R}^n$ to $\mathbb{R}^m$, we can represent the function uniquely by an $m \times n$ array of numbers, which is what we call the *matrix*. Expressing a linear function as a matrix allows us to conveniently study many of its properties, which is why we usually talk about matrices instead of the function itself.

1 Complex numbers

In $\mathbb{R}$, not every polynomial equation has a solution. For example, there does not exist any x such that $x^2 + 1 = 0$, since for any x , x^2 is non-negative, and $x^2 + 1$ can never be 0. To solve this problem, we introduce the “number” i that satisfies $i^2 = -1$. Then i is a solution to the equation $x^2 + 1 = 0$. Similarly, $-i$ is also a solution to the equation.

We can add and multiply numbers with i . For example, we can obtain numbers $3 + i$ or $1 + 3i$. These numbers are known as *complex numbers*. It turns out that by adding this single number i , *every* polynomial equation will have a root. In fact, for an n th order polynomial equation, we will later see that there will always be n roots, if we account for multiplicity. We will go into details in Chapter 5.

Apart from solving equations, complex numbers have a lot of rather important applications. For example, they are used in electronics to represent alternating currents, and form an integral part in the formulation of quantum mechanics.

1.1 Basic properties

Definition (Complex number). A *complex number* is a number $z \in \mathbb{C}$ of the form $z = a + ib$ with $a, b \in \mathbb{R}$, where $i^2 = -1$. We write $a = \text{Re}(z)$ and $b = \text{Im}(z)$.

To perform operations on complex numbers, we use the following algebraic rules. Let $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ be two complex numbers. Then:

$$\begin{aligned} z_1 \pm z_2 &= (a_1 + ib_1) \pm (a_2 + ib_2) \\ &= (a_1 \pm a_2) + i(b_1 \pm b_2) \\ z_1 z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= (a_1 a_2 - b_1 b_2) + i(b_1 a_2 + a_1 b_2) \\ z^{-1} &= \frac{1}{a + ib} \\ &= \frac{a - ib}{a^2 + b^2} \end{aligned}$$

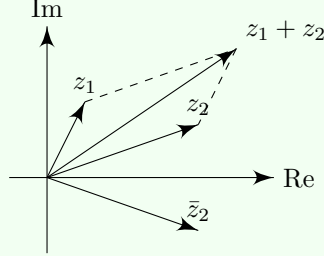
Addition and subtraction are performed component-wise. For multiplication, we expand the product using $i^2 = -1$. For the inverse, we multiply both numerator and denominator by the complex conjugate.

Definition (Complex conjugate). The *complex conjugate* of $z = a + ib$ is $a - ib$. It is written as $\bar{z}$ or z^* .

It is often helpful to visualize complex numbers in a diagram:

Definition (Argand diagram). An *Argand diagram* is a diagram in which a complex number $z = x + iy$ is represented by a vector $\mathbf{p} = \begin{pmatrix} x \\ y \end{pmatrix}$.

Addition of vectors corresponds to vector addition and $\bar{z}$ is the reflection of z in the x -axis.



Definition (Modulus and argument of complex number). The *modulus* of $z = x + iy$ is $r = |z| = \sqrt{x^2 + y^2}$. The *argument* $\theta = \arg z$ is the angle between z and the positive real axis, measured counter-clockwise. It satisfies $\tan \theta = y/x$, but the quadrant must be determined from the signs of x and y (since $\tan^{-1}(y/x)$ alone cannot distinguish, for example, $-1 + i$ from $1 - i$). The modulus is the length of the vector in the Argand diagram. We have

$$z = r(\cos \theta + i \sin \theta)$$

Clearly the pair (r, θ) uniquely describes a complex number z , but each complex number $z \in \mathbb{C}$ can be described by many different θ since $\sin(2\pi + \theta) = \sin \theta$ and $\cos(2\pi + \theta) = \cos \theta$. Often we take the *principal value* $\theta \in (-\pi, \pi]$.

Proposition (Multiplication in polar form). When multiplying two complex numbers, the moduli multiply and the arguments add. That is, if $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)].$$

Proof. We compute

$$\begin{aligned} z_1 z_2 &= r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \sin \theta_2 \cos \theta_1)] \\ &= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)] \end{aligned}$$

where we have used the addition formulas for cosine and sine. $\square$

Proposition (Modulus squared formula). $z \bar{z} = a^2 + b^2 = |z|^2$.

Proposition (Complex number inverse formula). $z^{-1} = \bar{z}/|z|^2$.

Theorem (Triangle inequality). For all $z_1, z_2 \in \mathbb{C}$, we have

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

Alternatively, we have $|z_1 - z_2| \geq ||z_1| - |z_2||$.

1.2 Complex exponential function

Exponentiation was originally defined for integer powers as repeated multiplication. This is then extended to rational powers using roots. We can also extend this to any real number since real numbers can be approximated arbitrarily accurately by rational numbers. However, what does it mean to take an exponent of a complex number?

To do so, we use the Taylor series definition of the exponential function:

Definition (Exponential function). The *exponential function* is defined as

$$\exp(z) = e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

This automatically allows taking exponents of arbitrary complex numbers. Having defined exponentiation this way, we want to check that it satisfies the usual properties, such as $\exp(z + w) = \exp(z)\exp(w)$. To prove this, we will first need a helpful lemma.

Lemma (Double sum rearrangement). For a family (a_{mn}) of complex numbers, we have

$$\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} a_{mn} = \sum_{r=0}^{\infty} \sum_{m=0}^r a_{r-m,m}.$$

Proof. The idea is to rearrange the order of summation. The sum on the left traverses the terms a_{mn} row by row (for each fixed n , we sum over all m). We can visualize this as the sum of all terms in an infinite array. Instead of summing row by row, we will sum along the diagonals. The diagonal r contains the terms $(a_{n,m})$ such that $n + m = r$, that is, the terms $a_{r,0}, a_{r-1,1}, \dots, a_{0,r}$. Formally:

$$\begin{aligned} \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} a_{mn} &= a_{00} + a_{01} + a_{02} + \cdots \\ &\quad + a_{10} + a_{11} + a_{12} + \cdots \\ &\quad + a_{20} + a_{21} + a_{22} + \cdots \\ &= (a_{00}) + (a_{10} + a_{01}) + (a_{20} + a_{11} + a_{02}) + \cdots \\ &= \sum_{r=0}^{\infty} \sum_{m=0}^r a_{r-m,m}. \quad \square \end{aligned}$$

This is not exactly a rigorous proof, since we should not hand-wave about infinite sums so casually. But in fact, we did not even show that the definition of $\exp(z)$ is well defined for all numbers z , since the sum might diverge. All these will be done in the IA Analysis I course.

Theorem (Exponential addition formula). $\exp(z_1) \exp(z_2) = \exp(z_1 + z_2)$

Proof.

$$\begin{aligned}
 \exp(z_1) \exp(z_2) &= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{z_1^m}{m!} \frac{z_2^n}{n!} \\
 &= \sum_{r=0}^{\infty} \sum_{m=0}^r \frac{z_1^{r-m}}{(r-m)!} \frac{z_2^m}{m!} \\
 &= \sum_{r=0}^{\infty} \frac{1}{r!} \sum_{m=0}^r \frac{r!}{(r-m)!m!} z_1^{r-m} z_2^m \\
 &= \sum_{r=0}^{\infty} \frac{(z_1 + z_2)^r}{r!} \quad \square
 \end{aligned}$$

Again, to define the sine and cosine functions, instead of referring to “angles” (since it doesn’t make much sense to refer to complex “angles”), we again use a series definition.

Definition (Sine and cosine functions). Define, for all $z \in \mathbb{C}$,

$$\begin{aligned}
 \sin z &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1} = z - \frac{1}{3!} z^3 + \frac{1}{5!} z^5 + \dots \\
 \cos z &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} = 1 - \frac{1}{2!} z^2 + \frac{1}{4!} z^4 + \dots
 \end{aligned}$$

One very important result is the relationship between $\exp$, $\sin$ and $\cos$.

Theorem (Euler’s formula). $e^{iz} = \cos z + i \sin z$.

Alternatively, since $\sin(-z) = -\sin z$ and $\cos(-z) = \cos z$, we have

$$\begin{aligned}
 \cos z &= \frac{e^{iz} + e^{-iz}}{2}, \\
 \sin z &= \frac{e^{iz} - e^{-iz}}{2i}.
 \end{aligned}$$

Proof.

$$\begin{aligned}
 e^{iz} &= \sum_{n=0}^{\infty} \frac{i^n}{n!} z^n \\
 &= \sum_{n=0}^{\infty} \frac{i^{2n}}{(2n)!} z^{2n} + \sum_{n=0}^{\infty} \frac{i^{2n+1}}{(2n+1)!} z^{2n+1} \\
 &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} + i \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1} \\
 &= \cos z + i \sin z \quad \square
 \end{aligned}$$

Thus we can write $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$.

1.3 Roots of unity

Definition (Roots of unity). The *nth roots of unity* are the roots to the equation $z^n = 1$ for $n \in \mathbb{N}$. Since this is a polynomial of order n , there are n roots of unity. In fact, the *nth roots of unity* are $\exp\left(2\pi i \frac{k}{n}\right)$ for $k = 0, 1, 2, 3 \dots n-1$.

Proposition (Sum of roots of unity). If $\omega = \exp\left(\frac{2\pi i}{n}\right)$, then $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$

Proof. Two proofs are provided:

- (i) Consider the equation $z^n = 1$, or equivalently $z^n - 1 = 0$. By Vieta's formulas, the sum of all roots equals the negative of the coefficient of z^{n-1} . Since the coefficient of z^{n-1} is 0, the sum of all roots $= 1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$.
- (ii) Since $\omega^n - 1 = (\omega - 1)(1 + \omega + \dots + \omega^{n-1})$ and $\omega \neq 1$, dividing by $(\omega - 1)$, we have $1 + \omega + \dots + \omega^{n-1} = (\omega^n - 1)/(\omega - 1) = 0$. $\square$

1.4 Complex logarithm and power

Definition (Complex logarithm). The *complex logarithm* $w = \log z$ is a solution to $e^w = z$, i.e. $w = \log z$. Writing $z = re^{i\theta}$, we have $\log z = \log(re^{i\theta}) = \log r + i\theta$. This can be multi-valued for different values of θ and, as above, we should select the θ that satisfies $-\pi < \theta \leq \pi$.

Example. $\log 2i = \log 2 + i\frac{\pi}{2}$

Definition (Complex power). The *complex power* z^α for $z, \alpha \in \mathbb{C}$ is defined as $z^\alpha = e^{\alpha \log z}$. This, again, can be multi-valued, as $z^\alpha = e^{\alpha \log |z|} e^{i\alpha\theta} e^{2in\pi\alpha}$ (there are finitely many values if $\alpha \in \mathbb{Q}$, infinitely many otherwise). Nevertheless, we make z^α single-valued by insisting $-\pi < \theta \leq \pi$.

1.5 De Moivre's theorem

Theorem (De Moivre's theorem).

$$\cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n.$$

Proof. First prove for the $n \geq 0$ case by induction. The $n = 0$ case is true since it merely reads $1 = 1$. We then have

$$\begin{aligned} (\cos \theta + i \sin \theta)^{n+1} &= (\cos \theta + i \sin \theta)^n (\cos \theta + i \sin \theta) \\ &= (\cos n\theta + i \sin n\theta)(\cos \theta + i \sin \theta) \\ &= \cos(n+1)\theta + i \sin(n+1)\theta \end{aligned}$$

If $n < 0$, let $m = -n$. Then $m > 0$ and

$$\begin{aligned} (\cos \theta + i \sin \theta)^{-m} &= (\cos m\theta + i \sin m\theta)^{-1} \\ &= \frac{\cos m\theta - i \sin m\theta}{(\cos m\theta + i \sin m\theta)(\cos m\theta - i \sin m\theta)} \\ &= \frac{\cos(-m\theta) + i \sin(-m\theta)}{\cos^2 m\theta + \sin^2 m\theta} \\ &= \cos(-m\theta) + i \sin(-m\theta) \\ &= \cos n\theta + i \sin n\theta \end{aligned} \quad \square$$

Note that “ $\cos n\theta + i \sin n\theta = e^{in\theta} = (e^{i\theta})^n = (\cos \theta + i \sin \theta)^n$ ” is *not* a valid proof of De Moivre’s theorem, since we do not know yet that $e^{in\theta} = (e^{i\theta})^n$. In fact, De Moivre’s theorem tells us that this is a valid rule to apply.

Example. We have $\cos 5\theta + i \sin 5\theta = (\cos \theta + i \sin \theta)^5$. By binomial expansion of the RHS and taking real and imaginary parts, we have

$$\begin{aligned} \cos 5\theta &= 5 \cos \theta - 20 \cos^3 \theta + 16 \cos^5 \theta \\ \sin 5\theta &= 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta \end{aligned}$$

1.6 Lines and circles in $\mathbb{C}$

Since complex numbers can be regarded as points on the 2D plane, we can often use complex numbers to represent two dimensional objects.

Suppose that we want to represent a straight line through $z_0 \in \mathbb{C}$ parallel to $w \in \mathbb{C}$. The obvious way to do so is to let $z = z_0 + \lambda w$ where λ can take any real value. However, this is not an optimal way of doing so, since we are not using the power of complex numbers fully. This is just the same as the vector equation for straight lines, which you may or may not know from your A levels.

Instead, we arrange the equation to give $\lambda = \frac{z - z_0}{w}$. We take the complex conjugate of this expression to obtain $\bar{\lambda} = \frac{\bar{z} - \bar{z}_0}{\bar{w}}$. The trick here is to realize that λ is a real number. So we must have $\lambda = \bar{\lambda}$. This means that we must have

$$\begin{aligned} \frac{z - z_0}{w} &= \frac{\bar{z} - \bar{z}_0}{\bar{w}} \\ z\bar{w} - \bar{z}w &= z_0\bar{w} - \bar{z}_0w. \end{aligned}$$

Theorem (Equation of straight line). The equation of a straight line through z_0 and parallel to w is given by

$$z\bar{w} - \bar{z}w = z_0\bar{w} - \bar{z}_0w.$$

The equation of a circle, on the other hand, is rather straightforward. Suppose that we want a circle with center $c \in \mathbb{C}$ and radius $\rho \in \mathbb{R}^+$. By definition of a circle, a point z is on the circle iff its distance to c is ρ , i.e. $|z - c| = \rho$. Recalling that $|z|^2 = z\bar{z}$, we obtain,

$$\begin{aligned} |z - c| &= \rho \\ |z - c|^2 &= \rho^2 \\ (z - c)(\bar{z} - \bar{c}) &= \rho^2 \\ z\bar{z} - \bar{c}z - c\bar{z} &= \rho^2 - c\bar{c} \end{aligned}$$

Theorem (Equation of a circle). The general equation of a circle with center $c \in \mathbb{C}$ and radius $\rho \in \mathbb{R}^+$ can be given by

$$z\bar{z} - \bar{c}z - c\bar{z} = \rho^2 - c\bar{c}.$$

2 Vectors

Having studied complex numbers, which generalize the real numbers, we now turn to another fundamental algebraic structure: vector spaces. Although the complex numbers $\mathbb{C}$ themselves form a vector space of dimension 2 over $\mathbb{R}$, we will develop the general theory of vectors in spaces of arbitrary dimension.

One often first learns about vectors as arrays of numbers, and then defines addition and multiplication in terms of the individual components. However, this is not the approach we will take here. The array of numbers is merely a *representation* of the vector, not the vector itself.

In this chapter, we will define vectors axiomatically, that is, in terms of their fundamental properties. The various operations will then be defined by the axioms they must satisfy.

2.1 Definition and basic properties

Definition (Vector). A *vector space* over $\mathbb{R}$ or $\mathbb{C}$ is a collection of vectors $\mathbf{v} \in V$, together with two operations: addition of two vectors and multiplication of a vector with a scalar (i.e. a number from $\mathbb{R}$ or $\mathbb{C}$, respectively).

Vector addition has to satisfy the following axioms:

- (i) $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ (commutativity)
- (ii) $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ (associativity)
- (iii) There is a vector $\mathbf{0}$ such that $\mathbf{a} + \mathbf{0} = \mathbf{a}$. (identity)
- (iv) For all vectors $\mathbf{a}$, there is a vector $(-\mathbf{a})$ such that $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$ (inverse)

Scalar multiplication has to satisfy the following axioms:

- (i) $\lambda(\mathbf{a} + \mathbf{b}) = \lambda\mathbf{a} + \lambda\mathbf{b}$.
- (ii) $(\lambda + \mu)\mathbf{a} = \lambda\mathbf{a} + \mu\mathbf{a}$.
- (iii) $\lambda(\mu\mathbf{a}) = (\lambda\mu)\mathbf{a}$.
- (iv) $1\mathbf{a} = \mathbf{a}$.

Often, vectors can be associated with a notion of length (or *norm*) and direction. In this case, we can visualize a vector as an “arrow” in space. Note that $\lambda\mathbf{a}$ is either parallel ($\lambda \geq 0$) or anti-parallel ($\lambda \leq 0$) to $\mathbf{a}$. The precise definition of length requires the introduction of the scalar product, which we will study in the following section.

Example. $\mathbb{R}^n$ is a vector space with component-wise addition and scalar multiplication. Note that the vector space $\mathbb{R}$ is a line, but not all lines are vector spaces. For example, $x + y = 1$ is not a vector space since it does not contain $\mathbf{0}$.

2.2 Scalar product

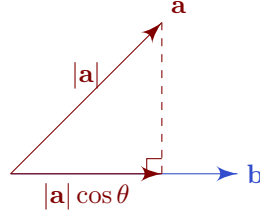
In a vector space, we can define the *scalar product* of two vectors, which returns a scalar (i.e. a real or complex number). We will first look at the usual scalar product defined for $\mathbb{R}^n$, and then define the scalar product axiomatically.

2.2.1 Geometric picture ($\mathbb{R}^2$ and $\mathbb{R}^3$ only)

Definition (Scalar/dot product). $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. It satisfies the following properties:

- (i) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- (ii) $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2 \geq 0$
- (iii) $\mathbf{a} \cdot \mathbf{a} = 0$ iff $\mathbf{a} = \mathbf{0}$
- (iv) If $\mathbf{a} \cdot \mathbf{b} = 0$ and $\mathbf{a}, \mathbf{b} \neq \mathbf{0}$, then $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

Intuitively, this is the product of the parts of $\mathbf{a}$ and $\mathbf{b}$ that are parallel.



Using the dot product, we can write the projection of $\mathbf{b}$ onto $\mathbf{a}$ as $(|\mathbf{b}| \cos \theta) \hat{\mathbf{a}} = (\hat{\mathbf{a}} \cdot \mathbf{b}) \hat{\mathbf{a}}$.

Proposition (Cosine law from scalar product). For a triangle ABC , if θ is the angle at vertex A , then

$$|\overrightarrow{BC}|^2 = |\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2 - 2|\overrightarrow{AB}||\overrightarrow{AC}| \cos \theta.$$

Proof. We calculate $|\overrightarrow{BC}|^2$ using vectors:

$$\begin{aligned} |\overrightarrow{BC}|^2 &= |\overrightarrow{AC} - \overrightarrow{AB}|^2 \\ &= (\overrightarrow{AC} - \overrightarrow{AB}) \cdot (\overrightarrow{AC} - \overrightarrow{AB}) \\ &= |\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2 - 2(\overrightarrow{AB} \cdot \overrightarrow{AC}) \\ &= |\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2 - 2|\overrightarrow{AB}||\overrightarrow{AC}| \cos \theta \end{aligned}$$

where θ is the angle between the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. □

We will see later a convenient algebraic method for evaluating this type of scalar product.

2.2.2 General algebraic definition

Definition (Inner/scalar product). In a real vector space V , an *inner product* or *scalar product* is a map $V \times V \rightarrow \mathbb{R}$ that satisfies the following axioms. It is written as $\mathbf{x} \cdot \mathbf{y}$ or $\langle \mathbf{x} | \mathbf{y} \rangle$ (the latter notation, called *Dirac notation* or *bra-ket notation*, is common in quantum mechanics).

- (i) $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$ (symmetry)
- (ii) $\mathbf{x} \cdot (\lambda \mathbf{y} + \mu \mathbf{z}) = \lambda \mathbf{x} \cdot \mathbf{y} + \mu \mathbf{x} \cdot \mathbf{z}$ (linearity in 2nd argument)
- (iii) $\mathbf{x} \cdot \mathbf{x} \geq 0$ with equality iff $\mathbf{x} = \mathbf{0}$ (positive definite)

Note that this is a definition only for *real* vector spaces, where the scalars are real. We will have a different set of definitions for complex vector spaces.

In particular, here we can use (i) and (ii) together to show linearity in 1st argument. However, this is generally not true for complex vector spaces.

Definition (Norm of a vector). The *norm* of a vector, written as $|\mathbf{a}|$ or $\|\mathbf{a}\|$, is defined as

$$|\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}}.$$

The norm generalizes the notion of length to abstract vector spaces.

Definition (Unit vector). A *unit vector* is a vector with norm equal to 1, i.e. $|\mathbf{v}| = 1$. We write a unit vector as $\hat{\mathbf{v}}$. For any nonzero vector $\mathbf{v}$, we can construct a unit vector in the same direction by $\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}$ (a process called *normalization*).

Example. Instead of the usual $\mathbb{R}^n$ vector space, we can consider the set of all real (integrable) functions as a vector space. We can define the following inner product:

$$\langle f | g \rangle = \int_0^1 f(x)g(x) \, dx.$$

2.3 Cauchy-Schwarz inequality

Theorem (Cauchy-Schwarz inequality). For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| |\mathbf{y}|.$$

Proof. The strategy of this proof is to exploit the fact that the square of a norm is always non-negative. Consider the expression $|\mathbf{x} - \lambda \mathbf{y}|^2$, where λ is an arbitrary real parameter. This quantity, being the square of a norm, must be

non-negative:

$$\begin{aligned} |\mathbf{x} - \lambda \mathbf{y}|^2 &\geq 0 \\ (\mathbf{x} - \lambda \mathbf{y}) \cdot (\mathbf{x} - \lambda \mathbf{y}) &\geq 0 \\ \lambda^2 |\mathbf{y}|^2 - \lambda(2\mathbf{x} \cdot \mathbf{y}) + |\mathbf{x}|^2 &\geq 0. \end{aligned}$$

This last expression is a quadratic polynomial in λ . Since it is non-negative for all $\lambda \in \mathbb{R}$, this polynomial cannot have two distinct real roots. Consequently, its discriminant must satisfy $\Delta \leq 0$. We thus obtain:

$$\begin{aligned} 4(\mathbf{x} \cdot \mathbf{y})^2 &\leq 4|\mathbf{y}|^2|\mathbf{x}|^2 \\ (\mathbf{x} \cdot \mathbf{y})^2 &\leq |\mathbf{x}|^2|\mathbf{y}|^2 \\ |\mathbf{x} \cdot \mathbf{y}| &\leq |\mathbf{x}||\mathbf{y}|. \end{aligned} \quad \square$$

Note that we proved this using the axioms of the scalar product. So this result holds for *all* possible scalar products on *any* (real) vector space.

Example. Let $\mathbf{x} = (\alpha, \beta, \gamma)$ and $\mathbf{y} = (1, 1, 1)$. Then by the Cauchy-Schwarz inequality, we have

$$\begin{aligned} \alpha + \beta + \gamma &\leq \sqrt{3}\sqrt{\alpha^2 + \beta^2 + \gamma^2} \\ \alpha^2 + \beta^2 + \gamma^2 &\geq \alpha\beta + \beta\gamma + \gamma\alpha, \end{aligned}$$

with equality if $\alpha = \beta = \gamma$.

Corollary (Triangle inequality).

$$|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|.$$

Proof.

$$\begin{aligned} |\mathbf{x} + \mathbf{y}|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) \\ &= |\mathbf{x}|^2 + 2\mathbf{x} \cdot \mathbf{y} + |\mathbf{y}|^2 \\ &\leq |\mathbf{x}|^2 + 2|\mathbf{x}||\mathbf{y}| + |\mathbf{y}|^2 \\ &= (|\mathbf{x}| + |\mathbf{y}|)^2. \end{aligned}$$

So

$$|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|. \quad \square$$

2.4 Vector product

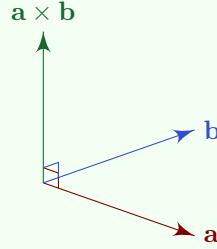
Apart from the scalar product, we can also define the *vector product*. However, this is defined only for $\mathbb{R}^3$ space, but not spaces in general.

Definition (Vector/cross product). Consider $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$. Define the

vector product

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is a unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. Since there are two (opposite) unit vectors that are perpendicular to both of them, we pick $\hat{\mathbf{n}}$ to be the one that is perpendicular to $\mathbf{a}, \mathbf{b}$ in a *right-handed* sense.



The vector product satisfies the following properties:

- (i) $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.
- (ii) $\mathbf{a} \times \mathbf{a} = \mathbf{0}$.
- (iii) $\mathbf{a} \times \mathbf{b} = \mathbf{0} \Rightarrow \mathbf{a} = \lambda \mathbf{b}$ for some $\lambda \in \mathbb{R}$ (or $\mathbf{b} = \mathbf{0}$).
- (iv) $\mathbf{a} \times (\lambda \mathbf{b}) = \lambda(\mathbf{a} \times \mathbf{b})$.
- (v) $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.

The vector product has an important geometric interpretation. If we consider a triangle OAB , its area is given by $\frac{1}{2}|\overrightarrow{OA}||\overrightarrow{OB}| \sin \theta = \frac{1}{2}|\overrightarrow{OA} \times \overrightarrow{OB}|$. We define the *vector area* as $\frac{1}{2}\overrightarrow{OA} \times \overrightarrow{OB}$, a notion particularly useful in vector calculus for the study of surfaces.

To explicitly calculate a vector product in Cartesian coordinates, we use the determinant method. Let $\mathbf{a} = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}}$ and $\mathbf{b} = b_1\hat{\mathbf{i}} + b_2\hat{\mathbf{j}} + b_3\hat{\mathbf{k}}$ be two vectors in $\mathbb{R}^3$ expressed in the canonical basis $\{\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}\}$. Then:

Proposition (Cross product determinant formula).

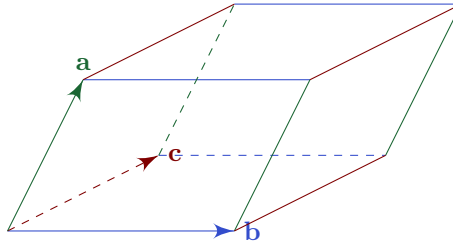
$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= (a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}}) \times (b_1\hat{\mathbf{i}} + b_2\hat{\mathbf{j}} + b_3\hat{\mathbf{k}}) \\ &= (a_2b_3 - a_3b_2)\hat{\mathbf{i}} + \cdots \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \end{aligned}$$

2.5 Scalar triple product

Definition (Scalar triple product). The *scalar triple product* is defined as

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

Proposition (Scalar triple product as volume). If a parallelepiped has sides represented by vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ that form a right-handed system, then the volume of the parallelepiped is given by $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$.



Proof. The area of the base of the parallelepiped is given by $|\mathbf{b}||\mathbf{c}| \sin \theta = |\mathbf{b} \times \mathbf{c}|$. Thus the volume = $|\mathbf{b} \times \mathbf{c}| |\mathbf{a}| \cos \phi = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$, where ϕ is the angle between $\mathbf{a}$ and the normal to $\mathbf{b}$ and $\mathbf{c}$. However, since $\mathbf{a}, \mathbf{b}, \mathbf{c}$ form a right-handed system, we have $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \geq 0$. Therefore the volume is $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. $\square$

Proposition (Cyclic permutation of scalar triple product). The scalar triple product is invariant under cyclic permutations and changes sign under transpositions:

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}] = [\mathbf{c}, \mathbf{a}, \mathbf{b}] = -[\mathbf{b}, \mathbf{a}, \mathbf{c}] = -[\mathbf{a}, \mathbf{c}, \mathbf{b}] = -[\mathbf{c}, \mathbf{b}, \mathbf{a}].$$

Proof. Since the order of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ doesn't affect the volume of the parallelepiped, the value remains the same under cyclic permutations. Under a transposition, the handedness of the system reverses, changing the sign. $\square$

Theorem (Cross product distributivity). $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.

Proof. Let $\mathbf{d} = \mathbf{a} \times (\mathbf{b} + \mathbf{c}) - \mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{c}$. We have

$$\begin{aligned} \mathbf{d} \cdot \mathbf{d} &= \mathbf{d} \cdot [\mathbf{a} \times (\mathbf{b} + \mathbf{c})] - \mathbf{d} \cdot (\mathbf{a} \times \mathbf{b}) - \mathbf{d} \cdot (\mathbf{a} \times \mathbf{c}) \\ &= (\mathbf{b} + \mathbf{c}) \cdot (\mathbf{d} \times \mathbf{a}) - \mathbf{b} \cdot (\mathbf{d} \times \mathbf{a}) - \mathbf{c} \cdot (\mathbf{d} \times \mathbf{a}) \\ &= 0 \end{aligned}$$

Thus $\mathbf{d} = \mathbf{0}$. $\square$

2.6 Spanning sets and bases

2.6.1 2D space

Definition (Spanning set). A set of vectors $\{\mathbf{a}, \mathbf{b}\}$ *spans* $\mathbb{R}^2$ if for all vectors $\mathbf{r} \in \mathbb{R}^2$, there exist some $\lambda, \mu \in \mathbb{R}$ such that $\mathbf{r} = \lambda\mathbf{a} + \mu\mathbf{b}$.

In $\mathbb{R}^2$, two vectors span the space if they are linearly independent, i.e. if they are not parallel.

Theorem (Uniqueness of spanning coefficients in $\mathbb{R}^2$). The coefficients λ, μ are unique.

Proof. Suppose that $\mathbf{r} = \lambda\mathbf{a} + \mu\mathbf{b} = \lambda'\mathbf{a} + \mu'\mathbf{b}$. Take the vector product with $\mathbf{a}$ on both sides to get $(\mu - \mu')\mathbf{a} \times \mathbf{b} = \mathbf{0}$. Since $\mathbf{a} \times \mathbf{b} \neq \mathbf{0}$, then $\mu = \mu'$. Similarly, $\lambda = \lambda'$. $\square$

Definition (Linearly independent vectors in $\mathbb{R}^2$). Two vectors $\mathbf{a}$ and $\mathbf{b}$ are *linearly independent* if for $\alpha, \beta \in \mathbb{R}$, $\alpha\mathbf{a} + \beta\mathbf{b} = \mathbf{0}$ iff $\alpha = \beta = 0$. In $\mathbb{R}^2$, $\mathbf{a}$ and $\mathbf{b}$ are linearly independent if $\mathbf{a} \times \mathbf{b} \neq \mathbf{0}$.

Definition (Basis of $\mathbb{R}^2$). A set of vectors is a *basis* of $\mathbb{R}^2$ if it spans $\mathbb{R}^2$ and are linearly independent.

Example. $\{\hat{\mathbf{i}}, \hat{\mathbf{j}}\} = \{(1, 0), (0, 1)\}$ is a basis of $\mathbb{R}^2$. They are the standard basis of $\mathbb{R}^2$.

2.6.2 3D space

We can extend the above definitions of spanning set and linear independent set to $\mathbb{R}^3$. Here we have

Theorem (Non-coplanar vectors form basis of $\mathbb{R}^3$). If $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ are non-coplanar, i.e. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \neq 0$, then they form a basis of $\mathbb{R}^3$.

Proof. For any $\mathbf{r}$, write $\mathbf{r} = \lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c}$. Performing the scalar product with $\mathbf{b} \times \mathbf{c}$ on both sides, one obtains $\mathbf{r} \cdot (\mathbf{b} \times \mathbf{c}) = \lambda\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mu\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) + \nu\mathbf{c} \cdot (\mathbf{b} \times \mathbf{c}) = \lambda[\mathbf{a}, \mathbf{b}, \mathbf{c}]$. Thus $\lambda = [\mathbf{r}, \mathbf{b}, \mathbf{c}]/[\mathbf{a}, \mathbf{b}, \mathbf{c}]$. The values of μ and ν can be found similarly. Thus each $\mathbf{r}$ can be written as a linear combination of $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$.

By the formula derived above, it follows that if $\alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c} = \mathbf{0}$, then $\alpha = \beta = \gamma = 0$. Thus they are linearly independent. $\square$

Note that while we came up with formulas for λ, μ and ν , we did not actually prove that these coefficients indeed work. This is rather unsatisfactory. We could, of course, expand everything out and show that this indeed works, but

in IB Linear Algebra, we will prove a much more general result, saying that if we have an n -dimensional space and a set of n linear independent vectors, then they form a basis.

In $\mathbb{R}^3$, the standard basis is $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$, or $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$.

2.6.3 $\mathbb{R}^n$ space

In general, we can define

Definition (Linearly independent vectors). A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \cdots \mathbf{v}_m\}$ is *linearly independent* if

$$\sum_{i=1}^m \lambda_i \mathbf{v}_i = \mathbf{0} \Rightarrow (\forall i) \lambda_i = 0.$$

Definition (Spanning set). A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \cdots \mathbf{u}_m\} \subseteq \mathbb{R}^n$ is a *spanning set* of $\mathbb{R}^n$ if

$$(\forall \mathbf{x} \in \mathbb{R}^n)(\exists \lambda_i) \sum_{i=1}^m \lambda_i \mathbf{u}_i = \mathbf{x}$$

Definition (Basis vectors). A *basis* of $\mathbb{R}^n$ is a linearly independent spanning set. The standard basis of $\mathbb{R}^n$ is $\mathbf{e}_1 = (1, 0, 0, \cdots, 0)$, $\mathbf{e}_2 = (0, 1, 0, \cdots, 0)$, $\cdots$, $\mathbf{e}_n = (0, 0, 0, \cdots, 1)$.

Definition (Orthonormal basis). A basis $\{\mathbf{e}_i\}$ is *orthonormal* if $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ if $i \neq j$ and $\mathbf{e}_i \cdot \mathbf{e}_i = 1$ for all i, j .

Using the Kronecker Delta symbol, which we will define later, we can write this condition as $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$.

Definition (Dimension of vector space). The *dimension* of a vector space is the number of vectors in its basis. (Exercise: show that this is well-defined)

We usually denote the components of a vector $\mathbf{x}$ by x_i . So we have $\mathbf{x} = (x_1, x_2, \cdots, x_n)$.

Definition (Scalar product). The *scalar product* of $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as $\mathbf{x} \cdot \mathbf{y} = \sum x_i y_i$.

The reader should check that this definition coincides with the $|\mathbf{x}||\mathbf{y}| \cos \theta$ definition in the case of $\mathbb{R}^2$ and $\mathbb{R}^3$.

2.6.4 $\mathbb{C}^n$ space

$\mathbb{C}^n$ is very similar to $\mathbb{R}^n$, except that we have complex numbers. As a result, we need a different definition of the scalar product. If we still defined $\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i$, then if we let $\mathbf{u} = (0, i)$, then $\mathbf{u} \cdot \mathbf{u} = -1 < 0$. This would be bad if we want to use the scalar product to define a norm.

Definition ($\mathbb{C}^n$). $\mathbb{C}^n = \{(z_1, z_2, \dots, z_n) : z_i \in \mathbb{C}\}$. It has the same standard basis as $\mathbb{R}^n$ but the scalar product is defined differently. For $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$, $\mathbf{u} \cdot \mathbf{v} = \sum u_i^* v_i$. The scalar product has the following properties:

- (i) $\mathbf{u} \cdot \mathbf{v} = (\mathbf{v} \cdot \mathbf{u})^*$
- (ii) $\mathbf{u} \cdot (\lambda \mathbf{v} + \mu \mathbf{w}) = \lambda(\mathbf{u} \cdot \mathbf{v}) + \mu(\mathbf{u} \cdot \mathbf{w})$
- (iii) $\mathbf{u} \cdot \mathbf{u} \geq 0$ and $\mathbf{u} \cdot \mathbf{u} = 0$ iff $\mathbf{u} = \mathbf{0}$

Instead of linearity in the first argument, here we have $(\lambda \mathbf{u} + \mu \mathbf{v}) \cdot \mathbf{w} = \lambda^* \mathbf{u} \cdot \mathbf{w} + \mu^* \mathbf{v} \cdot \mathbf{w}$.

Example.

$$\begin{aligned}
 & \sum_{k=1}^4 (-i)^k |\mathbf{x} + i^k \mathbf{y}|^2 \\
 &= \sum_{k=1}^4 (-i)^k \langle \mathbf{x} + i^k \mathbf{y} | \mathbf{x} + i^k \mathbf{y} \rangle \\
 &= \sum_{k=1}^4 (-i)^k (\langle \mathbf{x} + i^k \mathbf{y} | \mathbf{x} \rangle + i^k \langle \mathbf{x} + i^k \mathbf{y} | \mathbf{y} \rangle) \\
 &= \sum_{k=1}^4 (-i)^k (\langle \mathbf{x} | \mathbf{x} \rangle + (-i)^k \langle \mathbf{y} | \mathbf{x} \rangle + i^k \langle \mathbf{x} | \mathbf{y} \rangle + i^k (-i)^k \langle \mathbf{y} | \mathbf{y} \rangle) \\
 &= \sum_{k=1}^4 (-i)^k [(|\mathbf{x}|^2 + |\mathbf{y}|^2) + (-1)^k \langle \mathbf{y} | \mathbf{x} \rangle + \langle \mathbf{x} | \mathbf{y} \rangle] \\
 &= (|\mathbf{x}|^2 + |\mathbf{y}|^2) \sum_{k=1}^4 (-i)^k + \langle \mathbf{y} | \mathbf{x} \rangle \sum_{k=1}^4 (-1)^k + \langle \mathbf{x} | \mathbf{y} \rangle \sum_{k=1}^4 1 \\
 &= 4 \langle \mathbf{x} | \mathbf{y} \rangle.
 \end{aligned}$$

Theorem (Cauchy-Schwarz inequality for $\mathbb{C}^n$). For all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| |\mathbf{y}|.$$

Proof. The proof follows the same argument as the real case. First multiply $\mathbf{y}$ by some $e^{i\theta}$ so that $\mathbf{x} \cdot (e^{i\theta} \mathbf{y})$ is a real number. Then apply the same technique as in the real proof. The factor of $e^{i\theta}$ will drop off at the end when we take the modulus signs. $\square$

2.7 Vector subspaces

Definition (Vector subspace). A *vector subspace* of a vector space V is a subset of V that is also a vector space under the same operations. Both V and $\{\mathbf{0}\}$ are subspaces of V . All others are proper subspaces. A useful criterion is that a subset $U \subseteq V$ is a subspace iff

- (i) $\mathbf{x}, \mathbf{y} \in U \Rightarrow (\mathbf{x} + \mathbf{y}) \in U$.
- (ii) $\mathbf{x} \in U \Rightarrow \lambda \mathbf{x} \in U$ for all scalars λ .
- (iii) $\mathbf{0} \in U$.

This can be more concisely written as “ U is non-empty and for all $\mathbf{x}, \mathbf{y} \in U$, $(\lambda \mathbf{x} + \mu \mathbf{y}) \in U$ ”.

Example.

- (i) If $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ is a basis of $\mathbb{R}^3$, then $\{\mathbf{a} + \mathbf{c}, \mathbf{b} + \mathbf{c}\}$ is a basis of a 2D subspace.

Suppose $\mathbf{x}, \mathbf{y} \in \text{span}\{\mathbf{a} + \mathbf{c}, \mathbf{b} + \mathbf{c}\}$. Let

$$\begin{aligned}\mathbf{x} &= \alpha_1(\mathbf{a} + \mathbf{c}) + \beta_1(\mathbf{b} + \mathbf{c}); \\ \mathbf{y} &= \alpha_2(\mathbf{a} + \mathbf{c}) + \beta_2(\mathbf{b} + \mathbf{c}).\end{aligned}$$

Then

$$\lambda \mathbf{x} + \mu \mathbf{y} = (\lambda \alpha_1 + \mu \alpha_2)(\mathbf{a} + \mathbf{c}) + (\lambda \beta_1 + \mu \beta_2)(\mathbf{b} + \mathbf{c}) \in \text{span}\{\mathbf{a} + \mathbf{c}, \mathbf{b} + \mathbf{c}\}.$$

Thus this is a subspace of $\mathbb{R}^3$.

Now check that $\mathbf{a} + \mathbf{c}, \mathbf{b} + \mathbf{c}$ is a basis. We only need to check linear independence. If $\alpha(\mathbf{a} + \mathbf{c}) + \beta(\mathbf{b} + \mathbf{c}) = \mathbf{0}$, then $\alpha \mathbf{a} + \beta \mathbf{b} + (\alpha + \beta) \mathbf{c} = \mathbf{0}$. Since $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ is a basis of $\mathbb{R}^3$, therefore $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are linearly independent and $\alpha = \beta = 0$. Therefore $\mathbf{a} + \mathbf{c}, \mathbf{b} + \mathbf{c}$ is a basis and the subspace has dimension 2.

- (ii) Given a set of numbers α_i , let $U = \{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n \alpha_i x_i = 0\}$. We show that this is a vector subspace of $\mathbb{R}^n$: Take $\mathbf{x}, \mathbf{y} \in U$, then consider $\lambda \mathbf{x} + \mu \mathbf{y}$. We have $\sum \alpha_i (\lambda x_i + \mu y_i) = \lambda \sum \alpha_i x_i + \mu \sum \alpha_i y_i = 0$. Thus $\lambda \mathbf{x} + \mu \mathbf{y} \in U$.

The dimension of the subspace is $n - 1$ as we can freely choose x_i for $i = 1, \dots, n - 1$ and then x_n is uniquely determined by the previous x_i 's.

- (iii) Let $W = \{\mathbf{x} \in \mathbb{R}^n : \sum \alpha_i x_i = 1\}$. Then $\sum \alpha_i (\lambda x_i + \mu y_i) = \lambda + \mu \neq 1$. Therefore W is not a vector subspace.

2.8 Suffix notation

In vector calculations, it is often tedious to manipulate explicit sums. *Suffix notation* (also called *index notation*), developed notably by Einstein, allows us to greatly simplify the writing and manipulation of vector and tensor expressions. This notation is particularly useful in fluid mechanics, general relativity, and theoretical physics.

Let us begin by establishing the basic conventions. Let $\mathbf{v} \in \mathbb{R}^3$. We can write $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3 = (v_1, v_2, v_3)$, where $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ denotes the canonical basis of $\mathbb{R}^3$. In general, the i -th component of $\mathbf{v}$ is written v_i . We can thus express vector equations in index form. For example, $\mathbf{a} = \mathbf{b}$ is written $a_i = b_i$ for all i , and $\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$ is written $c_i = \alpha a_i + \beta b_i$. We say that a vector has one *free* index, denoted i , while a scalar has none.

Notation (Einstein's summation convention). Consider a sum $\mathbf{x} \cdot \mathbf{y} = \sum x_i y_i$. The *summation convention* says that we can drop the $\sum$ symbol and simply write $\mathbf{x} \cdot \mathbf{y} = x_i y_i$. If suffixes are repeated once, summation is understood.

Note that i is a dummy suffix and doesn't matter what it's called, i.e. $x_i y_i = x_j y_j = x_k y_k$ etc.

The rules of this convention are:

- (i) Suffix appears once in a term: free suffix
- (ii) Suffix appears twice in a term: dummy suffix and is summed over
- (iii) Suffix appears three times or more: WRONG!

Example. $[(\mathbf{a} \cdot \mathbf{b})\mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b}]_i = a_j b_j c_i - a_j c_j b_i$ summing over j understood.

It is possible for an item to have more than one index. These objects are known as *tensors*, which will be studied in depth in the IA Vector Calculus course.

Here we will define two important tensors:

Definition (Kronecker delta).

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

We have

$$\begin{pmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}.$$

So the Kronecker delta represents an identity matrix.

Example.

- (i) $a_i \delta_{i1} = a_1$. In general, $a_i \delta_{ij} = a_j$ (i is dummy, j is free).
- (ii) $\delta_{ij} \delta_{jk} = \delta_{ik}$
- (iii) $\delta_{ii} = n$ if we are in $\mathbb{R}^n$.
- (iv) $a_p \delta_{pq} b_q = a_p b_p$ with p, q both dummy suffices and summed over.

Definition (Alternating symbol ε_{ijk}). Consider rearrangements of 1, 2, 3. We can divide them into even and odd permutations. Even permutations include (1, 2, 3), (2, 3, 1) and (3, 1, 2). These are permutations obtained by performing two (or no) swaps of the elements of (1, 2, 3). (Alternatively, it is any “rotation” of (1, 2, 3)) The odd permutations are (2, 1, 3), (1, 3, 2) and (3, 2, 1). They are the permutations obtained by one swap only.

Define

$$\varepsilon_{ijk} = \begin{cases} +1 & ijk \text{ is even permutation} \\ -1 & ijk \text{ is odd permutation} \\ 0 & \text{otherwise (i.e. repeated suffices)} \end{cases}$$

ε_{ijk} has 3 free suffices.

We have $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = +1$ and $\varepsilon_{213} = \varepsilon_{132} = \varepsilon_{321} = -1$.
 $\varepsilon_{112} = \varepsilon_{111} = \dots = 0$.

We have

- (i) $\varepsilon_{ijk} \delta_{jk} = \varepsilon_{ijj} = 0$
- (ii) If $a_{jk} = a_{kj}$ (i.e. a_{ij} is symmetric), then $\varepsilon_{ijk} a_{jk} = \varepsilon_{ijk} a_{kj} = -\varepsilon_{ikj} a_{kj}$. Since $\varepsilon_{ijk} a_{jk} = \varepsilon_{ikj} a_{kj}$ (we simply renamed dummy suffices), we have $\varepsilon_{ijk} a_{jk} = 0$.

Proposition (Cross product in terms of Levi-Civita symbol). $(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k$

Proof. By expansion of formula □

Theorem (Levi-Civita contraction identity). $\varepsilon_{ijk} \varepsilon_{ipq} = \delta_{jp} \delta_{kq} - \delta_{jq} \delta_{kp}$

Proof. Proof by exhaustion:

$$\text{RHS} = \begin{cases} +1 & \text{if } j = p \text{ and } k = q \\ -1 & \text{if } j = q \text{ and } k = p \\ 0 & \text{otherwise} \end{cases}$$

LHS: Summing over i , the only non-zero terms are when $j, k \neq i$ and $p, q \neq i$. If $j = p$ and $k = q$, LHS is $(-1)^2$ or $(+1)^2 = 1$. If $j = q$ and $k = p$, LHS is $(+1)(-1)$ or $(-1)(+1) = -1$. All other possibilities result in 0. □

Corollary (Alternative Levi-Civita contractions). By permuting the contracted indices, we also have

$$\varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{jp}\delta_{iq}$$

and

$$\varepsilon_{ijk}\varepsilon_{pqj} = \delta_{ip}\delta_{kq} - \delta_{iq}\delta_{kp}.$$

Proposition (Cyclic permutation of scalar triple product).

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})$$

Proof. In suffix notation, we have

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = a_i(\mathbf{b} \times \mathbf{c})_i = \varepsilon_{ijk}b_jc_ka_i = \varepsilon_{jki}b_jc_ka_i = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}). \quad \square$$

Theorem (Vector triple product).

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

Proof.

$$\begin{aligned} [\mathbf{a} \times (\mathbf{b} \times \mathbf{c})]_i &= \varepsilon_{ijk}a_j(b \times c)_k \\ &= \varepsilon_{ijk}\varepsilon_{kpq}a_jb_pc_q \\ &= \varepsilon_{ijk}\varepsilon_{pqk}a_jb_pc_q \\ &= (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp})a_jb_pc_q \\ &= a_jb_ic_j - a_jc_ib_j \\ &= (\mathbf{a} \cdot \mathbf{c})b_i - (\mathbf{a} \cdot \mathbf{b})c_i \end{aligned} \quad \square$$

Similarly, $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$.

Spherical trigonometry

Proposition (Dot product of cross products).
 $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{c}) - (\mathbf{a} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{c}).$

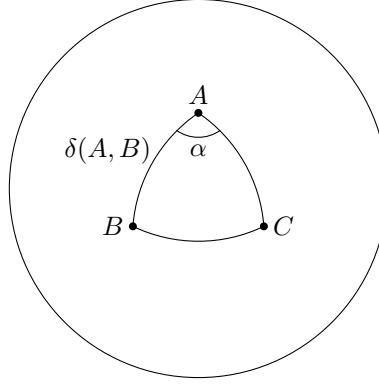
Proof.

$$\begin{aligned} \text{LHS} &= (\mathbf{a} \times \mathbf{b})_i(\mathbf{a} \times \mathbf{c})_i \\ &= \varepsilon_{ijk}a_jb_k\varepsilon_{ipq}a_pa_q \\ &= (\delta_{jp}\delta_{kq} - \delta_{jq}\delta_{kp})a_jb_ka_pc_q \\ &= a_jb_ka_jc_k - a_jb_ka_kc_j \\ &= (\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{c}) - (\mathbf{a} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{c}) \end{aligned} \quad \square$$

Theorem (Spherical cosine rule). Consider the unit sphere, center O , with points A, B, C on the surface corresponding to unit vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Let

$\delta(A, B)$ denote the arc length distance between A and B (i.e., $\angle AOB$), and let α be the angle at A in the spherical triangle ABC . Then

$$\cos \alpha \sin \delta(A, B) \sin \delta(A, C) = \cos \delta(B, C) - \cos \delta(A, B) \cos \delta(A, C).$$



Proof. Since $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors on the surface of the sphere, we have $\mathbf{a} \cdot \mathbf{b} = \cos \angle AOB = \cos \delta(A, B)$. Similarly for other dot products. Also, $|\mathbf{a} \times \mathbf{b}| = \sin \delta(A, B)$.

The angle α satisfies

$$\begin{aligned} \cos \alpha &= \frac{(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{a} \times \mathbf{c})}{|\mathbf{a} \times \mathbf{b}| |\mathbf{a} \times \mathbf{c}|} \\ &= \frac{\mathbf{b} \cdot \mathbf{c} - (\mathbf{a} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{c})}{|\mathbf{a} \times \mathbf{b}| |\mathbf{a} \times \mathbf{c}|} \end{aligned}$$

by the proposition on dot products of cross products. Substituting our expressions for the dot and cross products yields the result. $\square$

What does this spherical geometry look like?

Consider a spherical equilateral triangle. Using the spherical cosine rule,

$$\cos \alpha = \frac{\cos \delta - \cos^2 \delta}{\sin^2 \delta} = 1 - \frac{1}{1 + \cos \delta}.$$

Since $\cos \delta \leq 1$, we have $\cos \alpha \leq \frac{1}{2}$ and $\alpha \geq 60^\circ$. Equality holds iff $\delta = 0$, i.e. the triangle is simply a point. So on a sphere, each angle of an equilateral triangle is greater than 60° , and the angle sum of a triangle is greater than 180° .

2.9 Geometry

2.9.1 Lines

Any line through $\mathbf{a}$ and parallel to $\mathbf{t}$ can be written as

$$\mathbf{x} = \mathbf{a} + \lambda \mathbf{t}.$$

By crossing both sides of the equation with $\mathbf{t}$, we have

Theorem (Straight line equation using cross product). The equation of a straight line through $\mathbf{a}$ and parallel to $\mathbf{t}$ is

$$(\mathbf{x} - \mathbf{a}) \times \mathbf{t} = \mathbf{0} \text{ or } \mathbf{x} \times \mathbf{t} = \mathbf{a} \times \mathbf{t}.$$

2.9.2 Plane

To define a plane Π , we need a normal $\mathbf{n}$ to the plane and a fixed point $\mathbf{b}$. For any $\mathbf{x} \in \Pi$, the vector $\mathbf{x} - \mathbf{b}$ is contained in the plane and is thus normal to $\mathbf{n}$, i.e. $(\mathbf{x} - \mathbf{b}) \cdot \mathbf{n} = 0$.

Theorem (Plane equation). The equation of a plane through $\mathbf{b}$ with normal $\mathbf{n}$ is given by

$$\mathbf{x} \cdot \mathbf{n} = \mathbf{b} \cdot \mathbf{n}.$$

If $\mathbf{n} = \hat{\mathbf{n}}$ is a unit normal, then $d = \mathbf{x} \cdot \hat{\mathbf{n}} = \mathbf{b} \cdot \hat{\mathbf{n}}$ is the perpendicular distance from the origin to Π .

Alternatively, if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ lie in the plane, then the equation of the plane is

$$(\mathbf{x} - \mathbf{a}) \cdot [(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})] = 0.$$

Example.

- (i) Consider the intersection between a line $\mathbf{x} \times \mathbf{t} = \mathbf{a} \times \mathbf{t}$ with the plane $\mathbf{x} \cdot \mathbf{n} = \mathbf{b} \cdot \mathbf{n}$. Cross $\mathbf{n}$ on the right with the line equation to obtain

$$(\mathbf{x} \cdot \mathbf{n})\mathbf{t} - (\mathbf{t} \cdot \mathbf{n})\mathbf{x} = (\mathbf{a} \times \mathbf{t}) \times \mathbf{n}$$

Eliminate $\mathbf{x} \cdot \mathbf{n}$ using $\mathbf{x} \cdot \mathbf{n} = \mathbf{b} \cdot \mathbf{n}$

$$(\mathbf{t} \cdot \mathbf{n})\mathbf{x} = (\mathbf{b} \cdot \mathbf{n})\mathbf{t} - (\mathbf{a} \times \mathbf{t}) \times \mathbf{n}$$

Provided $\mathbf{t} \cdot \mathbf{n}$ is non-zero, the point of intersection is

$$\mathbf{x} = \frac{(\mathbf{b} \cdot \mathbf{n})\mathbf{t} - (\mathbf{a} \times \mathbf{t}) \times \mathbf{n}}{\mathbf{t} \cdot \mathbf{n}}.$$

Exercise: what if $\mathbf{t} \cdot \mathbf{n} = 0$?

- (ii) Shortest distance between two lines. Let L_1 be $(\mathbf{x} - \mathbf{a}_1) \times \mathbf{t}_1 = \mathbf{0}$ and L_2 be $(\mathbf{x} - \mathbf{a}_2) \times \mathbf{t}_2 = \mathbf{0}$.

The distance of closest approach s is along a line perpendicular to both L_1 and L_2 , i.e. the line of closest approach is perpendicular to both lines and thus parallel to $\mathbf{t}_1 \times \mathbf{t}_2$. The distance s can then be found by projecting $\mathbf{a}_1 - \mathbf{a}_2$ onto $\mathbf{t}_1 \times \mathbf{t}_2$. Thus $s = \left| (\mathbf{a}_1 - \mathbf{a}_2) \cdot \frac{\mathbf{t}_1 \times \mathbf{t}_2}{|\mathbf{t}_1 \times \mathbf{t}_2|} \right|$.

2.10 Vector equations

Example. $\mathbf{x} - (\mathbf{x} \times \mathbf{a}) \times \mathbf{b} = \mathbf{c}$. Strategy: take the dot or cross of the equation with suitable vectors. The equation can be expanded to form

$$\mathbf{x} - (\mathbf{x} \cdot \mathbf{b})\mathbf{a} + (\mathbf{a} \cdot \mathbf{b})\mathbf{x} = \mathbf{c}.$$

Dot this with $\mathbf{b}$ to obtain

$$\begin{aligned}\mathbf{x} \cdot \mathbf{b} - (\mathbf{x} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{b}) + (\mathbf{a} \cdot \mathbf{b})(\mathbf{x} \cdot \mathbf{b}) &= \mathbf{c} \cdot \mathbf{b} \\ \mathbf{x} \cdot \mathbf{b} &= \mathbf{c} \cdot \mathbf{b}.\end{aligned}$$

Substituting this into the original equation, we have

$$\mathbf{x}(1 + \mathbf{a} \cdot \mathbf{b}) = \mathbf{c} + (\mathbf{c} \cdot \mathbf{b})\mathbf{a}$$

If $(1 + \mathbf{a} \cdot \mathbf{b})$ is non-zero, then

$$\mathbf{x} = \frac{\mathbf{c} + (\mathbf{c} \cdot \mathbf{b})\mathbf{a}}{1 + \mathbf{a} \cdot \mathbf{b}}$$

Otherwise, when $(1 + \mathbf{a} \cdot \mathbf{b}) = 0$, if $\mathbf{c} + (\mathbf{c} \cdot \mathbf{b})\mathbf{a} \neq \mathbf{0}$, then a contradiction is reached. Otherwise, $\mathbf{x} \cdot \mathbf{b} = \mathbf{c} \cdot \mathbf{b}$ is the most general solution, which is a plane of solutions.

3 Linear maps

Having studied vectors and their algebraic and geometric properties, we now turn our attention to functions that preserve the vector space structure. A *linear map* is a function $f : V \rightarrow W$ between two vector spaces that satisfies the following property: for all vectors $\mathbf{a}, \mathbf{b} \in V$ and all scalars λ, μ , we have

$$f(\lambda \mathbf{a} + \mu \mathbf{b}) = \lambda f(\mathbf{a}) + \mu f(\mathbf{b}).$$

This property expresses that f respects the operations of the vector space: it commutes with vector addition and scalar multiplication.

Linear maps play a fundamental role in mathematics and physics. Indeed, in many contexts, these are the only transformations that preserve the essential properties of the systems being studied. For example, rotations and reflections in Euclidean space are linear maps.

We will begin by studying two important examples of linear maps — rotations and reflections — before studying their properties in a formal manner.

3.1 Examples

3.1.1 Rotation in $\mathbb{R}^3$

In $\mathbb{R}^3$, first consider the simple cases where we rotate about the z axis by θ . We call this rotation R and write $\mathbf{x}' = R(\mathbf{x})$.

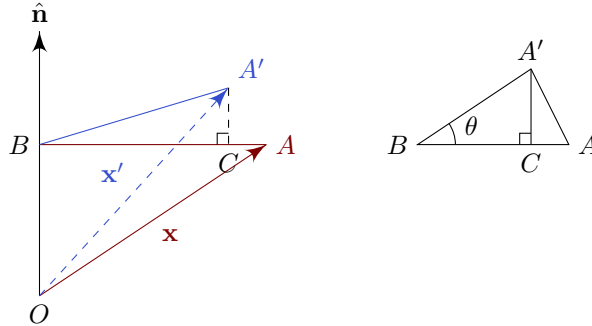
Suppose that initially, $\mathbf{x} = (x, y, z) = (r \cos \phi, r \sin \phi, z)$. Then after a rotation by θ , we get

$$\begin{aligned} \mathbf{x}' &= (r \cos(\phi + \theta), r \sin(\phi + \theta), z) \\ &= (r \cos \phi \cos \theta - r \sin \phi \sin \theta, r \sin \phi \cos \theta + r \cos \phi \sin \theta, z) \\ &= (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta, z). \end{aligned}$$

We can represent this by a matrix R such that $x'_i = R_{ij}x_j$. Using our formula above, we obtain

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Now consider the general case where we rotate by θ about $\hat{\mathbf{n}}$.



We have $\mathbf{x}' = \overrightarrow{OB} + \overrightarrow{BC} + \overrightarrow{CA'}$. We know that

$$\begin{aligned}\overrightarrow{OB} &= (\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}} \\ \overrightarrow{BC} &= \overrightarrow{BA} \cos \theta \\ &= (\overrightarrow{BO} + \overrightarrow{OA}) \cos \theta \\ &= -(\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}} + \mathbf{x} \cos \theta\end{aligned}$$

Finally, to get $\overrightarrow{CA'}$, we know that $|\overrightarrow{CA'}| = |\overrightarrow{BA'}| \sin \theta = |\overrightarrow{BA}| \sin \theta = |\hat{\mathbf{n}} \times \mathbf{x}| \sin \theta$. Also, $\overrightarrow{CA'}$ is parallel to $\hat{\mathbf{n}} \times \mathbf{x}$. So we must have $\overrightarrow{CA'} = (\hat{\mathbf{n}} \times \mathbf{x}) \sin \theta$.

Thus $\mathbf{x}' = \mathbf{x} \cos \theta + (1 - \cos \theta)(\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}} + \hat{\mathbf{n}} \times \mathbf{x} \sin \theta$. In components,

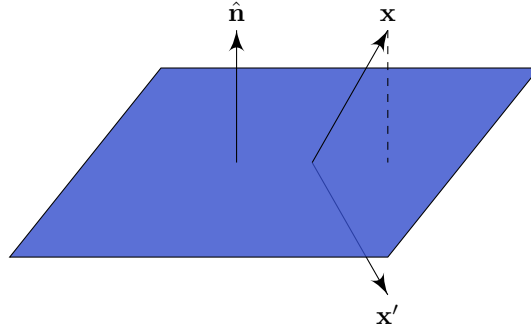
$$x'_i = x_i \cos \theta + (1 - \cos \theta)n_j x_j n_i - \varepsilon_{ijk} x_j n_k \sin \theta.$$

We want to find an R such that $x'_i = R_{ij}x_j$. So

$$R_{ij} = \delta_{ij} \cos \theta + (1 - \cos \theta)n_i n_j - \varepsilon_{ijk} n_k \sin \theta.$$

3.1.2 Reflection in $\mathbb{R}^3$

Suppose we want to reflect through a plane through O with normal $\hat{\mathbf{n}}$. First of all the projection of $\mathbf{x}$ onto $\hat{\mathbf{n}}$ is given by $(\mathbf{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$. So we get $\mathbf{x}' = \mathbf{x} - 2(\mathbf{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$. In suffix notation, we have $x'_i = x_i - 2x_j n_j n_i$. So our reflection matrix is $R_{ij} = \delta_{ij} - 2n_i n_j$.



3.2 Linear Maps

Definition (Domain, codomain and image of map). Consider sets A and B and mapping $T : A \rightarrow B$ such that each $x \in A$ is mapped into a unique $x' = T(x) \in B$. A is the *domain* of T and B is the *co-domain* of T . Typically, we have $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ or $T : \mathbb{C}^n \rightarrow \mathbb{C}^m$.

Definition (Linear map). Let V, W be real (or complex) vector spaces, and $T : V \rightarrow W$. Then T is a *linear map* if

- (i) $T(\mathbf{a} + \mathbf{b}) = T(\mathbf{a}) + T(\mathbf{b})$ for all $\mathbf{a}, \mathbf{b} \in V$.

(ii) $T(\lambda \mathbf{a}) = \lambda T(\mathbf{a})$ for all $\lambda \in \mathbb{R}$ (or $\mathbb{C}$).

Equivalently, we have $T(\lambda \mathbf{a} + \mu \mathbf{b}) = \lambda T(\mathbf{a}) + \mu T(\mathbf{b})$.

Example.

- (i) Consider a translation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $T(\mathbf{x}) = \mathbf{x} + \mathbf{a}$ for some fixed, given $\mathbf{a}$. This is *not* a linear map since $T(\lambda \mathbf{x} + \mu \mathbf{y}) \neq \lambda \mathbf{x} + \mu \mathbf{y} + (\lambda + \mu)\mathbf{a}$.
- (ii) Rotation, reflection and projection are linear transformations.

Definition (Image and kernel of map). The *image* of a map $f : U \rightarrow V$ is the subset of V $\{f(\mathbf{u}) : \mathbf{u} \in U\}$. The *kernel* is the subset of U $\{\mathbf{u} \in U : f(\mathbf{u}) = \mathbf{0}\}$.

Example.

- (i) Consider $S : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with $S(x, y, z) = (x + y, 2x - z)$. Simple yet tedious algebra shows that this is linear. Now consider the effect of S on the standard basis. $S(1, 0, 0) = (1, 2)$, $S(0, 1, 0) = (1, 0)$ and $S(0, 0, 1) = (0, -1)$. Clearly these are linearly dependent, but they do span the whole of $\mathbb{R}^2$. We can say $S(\mathbb{R}^3) = \mathbb{R}^2$. So the image is $\mathbb{R}^2$.
Now solve $S(x, y, z) = \mathbf{0}$. We need $x + y = 0$ and $2x - z = 0$. Thus $\mathbf{x} = (x, -x, 2x)$, i.e. it is parallel to $(1, -1, 2)$. So the set $\{\lambda(1, -1, 2) : \lambda \in \mathbb{R}\}$ is the kernel of S .
- (ii) Consider a rotation in $\mathbb{R}^3$. The kernel is the zero vector and the image is $\mathbb{R}^3$.
- (iii) Consider a projection of $\mathbf{x}$ onto a plane with normal $\hat{\mathbf{n}}$. The image is the plane itself, and the kernel is any vector parallel to $\hat{\mathbf{n}}$.

Theorem (Image and kernel are subspaces). Consider a linear map $f : U \rightarrow V$, where U, V are vector spaces. Then $\text{im}(f)$ is a subspace of V , and $\ker(f)$ is a subspace of U .

Proof. Both are non-empty since $f(\mathbf{0}) = \mathbf{0}$.

If $\mathbf{x}, \mathbf{y} \in \text{im}(f)$, then $\exists \mathbf{a}, \mathbf{b} \in U$ such that $\mathbf{x} = f(\mathbf{a}), \mathbf{y} = f(\mathbf{b})$. Then $\lambda \mathbf{x} + \mu \mathbf{y} = \lambda f(\mathbf{a}) + \mu f(\mathbf{b}) = f(\lambda \mathbf{a} + \mu \mathbf{b})$. Now $\lambda \mathbf{a} + \mu \mathbf{b} \in U$ since U is a vector space, so there is an element in U that maps to $\lambda \mathbf{x} + \mu \mathbf{y}$. So $\lambda \mathbf{x} + \mu \mathbf{y} \in \text{im}(f)$ and $\text{im}(f)$ is a subspace of V .

Suppose $\mathbf{x}, \mathbf{y} \in \ker(f)$, i.e. $f(\mathbf{x}) = f(\mathbf{y}) = \mathbf{0}$. Then $f(\lambda \mathbf{x} + \mu \mathbf{y}) = \lambda f(\mathbf{x}) + \mu f(\mathbf{y}) = \lambda \mathbf{0} + \mu \mathbf{0} = \mathbf{0}$. Therefore $\lambda \mathbf{x} + \mu \mathbf{y} \in \ker(f)$. $\square$

3.3 Rank and nullity

Definition (Rank of linear map). The *rank* of a linear map $f : U \rightarrow V$, denoted by $r(f)$, is the dimension of the image of f .

Definition (Nullity of linear map). The *nullity* of f , denoted $n(f)$ is the dimension of the kernel of f .

Example. For the projection onto a plane in $\mathbb{R}^3$, the image is the whole plane and the rank is 2. The kernel is a line so the nullity is 1.

Theorem (Rank-nullity theorem). For a linear map $f : U \rightarrow V$,

$$r(f) + n(f) = \dim(U).$$

Proof. (Non-examinable) Write $\dim(U) = n$ and $n(f) = m$. If $m = n$, then f is the zero map, and the proof is trivial, since $r(f) = 0$. Otherwise, assume $m < n$.

Suppose $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\}$ is a basis of $\ker f$. Extend this to a basis of the whole of U to get $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m, \mathbf{e}_{m+1}, \dots, \mathbf{e}_n\}$. To prove the theorem, we need to prove that $\{f(\mathbf{e}_{m+1}), f(\mathbf{e}_{m+2}), \dots, f(\mathbf{e}_n)\}$ is a basis of $\text{im}(f)$.

- (i) First show that it spans $\text{im}(f)$. Take $\mathbf{y} \in \text{im}(f)$. Thus $\exists \mathbf{x} \in U$ such that $\mathbf{y} = f(\mathbf{x})$. Then

$$\mathbf{y} = f(\alpha_1 \mathbf{e}_1 + \alpha_2 \mathbf{e}_2 + \dots + \alpha_n \mathbf{e}_n),$$

since $\mathbf{e}_1, \dots, \mathbf{e}_n$ is a basis of U . Thus

$$\mathbf{y} = \alpha_1 f(\mathbf{e}_1) + \alpha_2 f(\mathbf{e}_2) + \dots + \alpha_m f(\mathbf{e}_m) + \alpha_{m+1} f(\mathbf{e}_{m+1}) + \dots + \alpha_n f(\mathbf{e}_n).$$

The first m terms map to $\mathbf{0}$, since $\mathbf{e}_1, \dots, \mathbf{e}_m$ is the basis of the kernel of f . Thus

$$\mathbf{y} = \alpha_{m+1} f(\mathbf{e}_{m+1}) + \dots + \alpha_n f(\mathbf{e}_n).$$

- (ii) To show that they are linearly independent, suppose

$$\alpha_{m+1} f(\mathbf{e}_{m+1}) + \dots + \alpha_n f(\mathbf{e}_n) = \mathbf{0}.$$

Then

$$f(\alpha_{m+1} \mathbf{e}_{m+1} + \dots + \alpha_n \mathbf{e}_n) = \mathbf{0}.$$

Thus $\alpha_{m+1} \mathbf{e}_{m+1} + \dots + \alpha_n \mathbf{e}_n \in \ker(f)$. Since $\{\mathbf{e}_1, \dots, \mathbf{e}_m\}$ span $\ker(f)$, there exist some $\alpha_1, \alpha_2, \dots, \alpha_m$ such that

$$\alpha_{m+1} \mathbf{e}_{m+1} + \dots + \alpha_n \mathbf{e}_n = \alpha_1 \mathbf{e}_1 + \dots + \alpha_m \mathbf{e}_m.$$

But $\mathbf{e}_1 \dots \mathbf{e}_n$ is a basis of U and are linearly independent. So $\alpha_i = 0$ for all i . Then the only solution to the equation $\alpha_{m+1} f(\mathbf{e}_{m+1}) + \dots + \alpha_n f(\mathbf{e}_n) = \mathbf{0}$ is $\alpha_i = 0$, and they are linearly independent by definition. $\square$

Example. Calculate the kernel and image of $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, defined by $f(x, y, z) = (x + y + z, 2x - y + 5z, x + 2z)$.

First find the kernel: we've got the system of equations:

$$\begin{aligned}x + y + z &= 0 \\2x - y + 5z &= 0 \\x + 2z &= 0\end{aligned}$$

Note that the first and second equation add to give $3x + 6z = 0$, which is identical to the third. Then using the first and third equation, we have $y = -x - z = z$. So the kernel is any vector in the form $(-2z, z, z)$ and is the span of $(-2, 1, 1)$.

To find the image, extend the basis of $\ker(f)$ to a basis of the whole of $\mathbb{R}^3$: $\{(-2, 1, 1), (0, 1, 0), (0, 0, 1)\}$. Apply f to this basis to obtain $(0, 0, 0)$, $(1, -1, 0)$ and $(1, 5, 2)$. From the proof of the rank-nullity theorem, we know that $f(0, 1, 0)$ and $f(0, 0, 1)$ is a basis of the image.

To get the standard form of the image, we know that the normal to the plane is parallel to $(1, -1, 0) \times (1, 5, 2) \parallel (1, 1, -3)$. Since $\mathbf{0} \in \text{im}(f)$, the equation of the plane is $x + y - 3z = 0$.

3.4 Matrices

In the previous examples, we have represented our linear maps by an object R such that $x'_i = R_{ij}x_j$. We call R the *matrix* of the linear map. We will now formalize this construction and show that every linear map can be represented uniquely by a matrix.

Let $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map, and denote $\mathbf{x}' = \alpha(\mathbf{x})$. Choose a basis $\{\mathbf{e}_i\}$ of $\mathbb{R}^n$. Every vector $\mathbf{x} \in \mathbb{R}^n$ can be written $\mathbf{x} = x_j \mathbf{e}_j$ for some scalars x_j (where summation over j is implicit by Einstein's convention). By linearity of α , we obtain:

$$\mathbf{x}' = \alpha(x_j \mathbf{e}_j) = x_j \alpha(\mathbf{e}_j).$$

Thus, the image $\mathbf{x}'$ is entirely determined by the images of the basis vectors $\alpha(\mathbf{e}_j)$. Denoting the i -th component of $\alpha(\mathbf{e}_j)$ by $[\alpha(\mathbf{e}_j)]_i$, we have:

$$x'_i = [\alpha(\mathbf{e}_j)]_i x_j.$$

We then define the matrix coefficients $A_{ij} = [\alpha(\mathbf{e}_j)]_i$, which gives $x'_i = A_{ij}x_j$. We write

$$A = \{A_{ij}\} = \begin{pmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & A_{ij} & \vdots \\ A_{m1} & \cdots & A_{mn} \end{pmatrix}$$

Here A_{ij} is the entry in the i th row of the j th column. We say that A is an $m \times n$ matrix, and write $\mathbf{x}' = A\mathbf{x}$.

We see that the columns of the matrix are the images of the standard basis vectors under the mapping α .

Example.

3.4.1 Examples

- (i) In $\mathbb{R}^2$, consider a reflection in a line with an angle θ to the x axis. We know that $\hat{\mathbf{i}} \mapsto \cos 2\theta \hat{\mathbf{i}} + \sin 2\theta \hat{\mathbf{j}}$, with $\hat{\mathbf{j}} \mapsto \sin 2\theta \hat{\mathbf{i}} - \cos 2\theta \hat{\mathbf{j}}$. Then the matrix is
$$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

- (ii) In $\mathbb{R}^3$, as we've previously seen, a rotation by θ about the z axis is given by

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

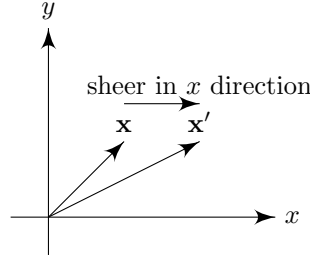
- (iii) In $\mathbb{R}^3$, a reflection in plane with normal $\hat{\mathbf{n}}$ is given by $R_{ij} = \delta_{ij} - 2\hat{n}_i\hat{n}_j$. Written as a matrix, we have

$$\begin{pmatrix} 1 - 2\hat{n}_1^2 & -2\hat{n}_1\hat{n}_2 & -2\hat{n}_1\hat{n}_3 \\ -2\hat{n}_2\hat{n}_1 & 1 - 2\hat{n}_2^2 & -2\hat{n}_2\hat{n}_3 \\ -2\hat{n}_3\hat{n}_1 & -2\hat{n}_3\hat{n}_2 & 1 - 2\hat{n}_3^2 \end{pmatrix}$$

- (iv) Dilation ("stretching") $\alpha : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is given by a map $(x, y, z) \mapsto (\lambda x, \mu y, \nu z)$ for some λ, μ, ν . The matrix is

$$\begin{pmatrix} \lambda & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & \nu \end{pmatrix}$$

- (v) Shear: Consider $S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that shears in the x direction:



We have $(x, y, z) \mapsto (x + \lambda y, y, z)$. Then

$$S = \begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

3.4.2 Matrix Algebra

This part is mostly on a whole lot of definitions, saying what we can do with matrices and classifying them into different types.

Definition (Addition of matrices). Consider two linear maps $\alpha, \beta : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The sum of α and β is defined by

$$(\alpha + \beta)(\mathbf{x}) = \alpha(\mathbf{x}) + \beta(\mathbf{x})$$

In terms of the matrix, we have

$$(A + B)_{ij}x_j = A_{ij}x_j + B_{ij}x_j,$$

or

$$(A + B)_{ij} = A_{ij} + B_{ij}.$$

Definition (Scalar multiplication of matrices). Define $(\lambda\alpha)\mathbf{x} = \lambda[\alpha(\mathbf{x})]$. So $(\lambda A)_{ij} = \lambda A_{ij}$.

Definition (Matrix multiplication). Consider maps $\alpha : \mathbb{R}^\ell \rightarrow \mathbb{R}^n$ and $\beta : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The composition is $\beta\alpha : \mathbb{R}^\ell \rightarrow \mathbb{R}^m$. Take $\mathbf{x} \in \mathbb{R}^\ell \mapsto \mathbf{x}'' \in \mathbb{R}^m$. Then $\mathbf{x}'' = (BA)\mathbf{x} = B\mathbf{x}'$, where $\mathbf{x}' = A\mathbf{x}$. Using suffix notation, we have $x''_i = (B\mathbf{x}')_i = b_{ik}x'_k = B_{ik}A_{kj}x_j$. But $x''_i = (BA)_{ij}x_j$. So

$$(BA)_{ij} = B_{ik}A_{kj}.$$

Generally, an $m \times n$ matrix multiplied by an $n \times \ell$ matrix gives an $m \times \ell$ matrix. $(BA)_{ij}$ is given by the i th row of B dotted with the j th column of A .

Note that the number of columns of B has to be equal to the number of rows of A for multiplication to be defined. If $\ell = m$ as well, then both BA and AB make sense, but $AB \neq BA$ in general. In fact, they don't even have to have the same dimensions.

Also, since function composition is associative, we get $A(BC) = (AB)C$.

Definition (Transpose of matrix). If A is an $m \times n$ matrix, the *transpose* A^T is an $n \times m$ matrix defined by $(A^T)_{ij} = A_{ji}$.

Proposition (Matrix transpose properties).

(i) $(A^T)^T = A$.

(ii) If $\mathbf{x}$ is a column vector $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $\mathbf{x}^T$ is a row vector $(x_1 \ x_2 \ \cdots \ x_n)$.

(iii) $(AB)^T = B^T A^T$ since $(AB)^T_{ij} = (AB)_{ji} = A_{jk}B_{ki} = B_{ki}A_{jk}$

$$= (B^T)_{ik}(A^T)_{kj} = (B^T A^T)_{ij}.$$

Definition (Hermitian conjugate). Define $A^\dagger = (A^T)^*$. Similarly, $(AB)^\dagger = B^\dagger A^\dagger$.

Definition (Symmetric matrix). A matrix is *symmetric* if $A^T = A$.

Definition (Hermitian matrix). A matrix is *Hermitian* if $A^\dagger = A$. (The diagonal of a Hermitian matrix must be real).

Definition (Anti/skew symmetric matrix). A matrix is *anti-symmetric* or *skew symmetric* if $A^T = -A$. The diagonals are all zero.

Definition (Skew-Hermitian matrix). A matrix is *skew-Hermitian* if $A^\dagger = -A$. The diagonals are pure imaginary.

Definition (Trace of matrix). The *trace* of an $n \times n$ matrix A is the sum of the diagonal. $\text{tr}(A) = A_{ii}$.

Example. Consider the reflection matrix $R_{ij} = \delta_{ij} - 2\hat{n}_i\hat{n}_j$. We have $\text{tr}(A) = R_{ii} = 3 - 2\hat{n} \cdot \hat{n} = 3 - 2 = 1$.

Proposition (Trace commutativity). $\text{tr}(BC) = \text{tr}(CB)$

Proof. $\text{tr}(BC) = B_{ik}C_{ki} = C_{ki}B_{ik} = (CB)_{kk} = \text{tr}(CB)$ $\square$

Definition (Identity matrix). $I = \delta_{ij}$.

3.4.3 Decomposition of an $n \times n$ matrix

Any $n \times n$ matrix B can be split as a sum of symmetric and antisymmetric parts. Write

$$B_{ij} = \underbrace{\frac{1}{2}(B_{ij} + B_{ji})}_{S_{ij}} + \underbrace{\frac{1}{2}(B_{ij} - B_{ji})}_{A_{ij}}.$$

We have $S_{ij} = S_{ji}$, so S is symmetric, while $A_{ji} = -A_{ij}$, and A is antisymmetric. So $B = S + A$.

Furthermore, we can decompose S into an isotropic part (a scalar multiple of the identity) plus a trace-less part (i.e. sum of diagonal = 0). Write

$$S_{ij} = \underbrace{\frac{1}{n} \operatorname{tr}(S) \delta_{ij}}_{\text{isotropic part}} + \underbrace{\left(S_{ij} - \frac{1}{n} \operatorname{tr}(S) \delta_{ij}\right)}_{T_{ij}}.$$

We have $\operatorname{tr}(T) = T_{ii} = S_{ii} - \frac{1}{n} \operatorname{tr}(S) \delta_{ii} = \operatorname{tr}(S) - \frac{1}{n} \operatorname{tr}(S)(n) = 0$.

Putting all these together,

$$B = \frac{1}{n} \operatorname{tr}(B) I + \left\{ \frac{1}{2} (B + B^T) - \frac{1}{n} \operatorname{tr}(B) I \right\} + \frac{1}{2} (B - B^T).$$

In three dimensions, we can write the antisymmetric part A in terms of a single vector: we have

$$A = \begin{pmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{pmatrix}$$

and we can consider

$$\varepsilon_{ijk} \omega_k = \begin{pmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{pmatrix}$$

So if we have $\omega = (c, b, a)$, then $A_{ij} = \varepsilon_{ijk} \omega_k$.

This decomposition can be useful in certain physical applications. For example, if the matrix represents the stress of a system, different parts of the decomposition will correspond to different types of stresses.

3.4.4 Matrix inverse

Definition (Inverse of matrix). Consider an $m \times n$ matrix A and $n \times m$ matrices B and C . If $BA = I$, then we say B is the *left inverse* of A . If $AC = I$, then we say C is the *right inverse* of A . If A is square ($n \times n$), then $B = B(AC) = (BA)C = C$, i.e. the left and right inverses coincide. Both are denoted by A^{-1} , the *inverse* of A . Therefore we have

$$AA^{-1} = A^{-1}A = I.$$

Note that not all square matrices have inverses. For example, the zero matrix clearly has no inverse.

Definition (Invertible matrix). If A has an inverse, then A is *invertible*.

Proposition (Inverse of product formula). $(AB)^{-1} = B^{-1}A^{-1}$

Proof. $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I.$ □

Definition (Orthogonal and unitary matrices). A real $n \times n$ matrix is *orthogonal* if $A^T A = A A^T = I$, i.e. $A^T = A^{-1}$. A complex $n \times n$ matrix is *unitary* if $U^\dagger U = U U^\dagger = I$, i.e. $U^\dagger = U^{-1}$.

Note that an orthogonal matrix A satisfies $A_{ik}(A_{kj}^T) = \delta_{ij}$, i.e. $A_{ik}A_{jk} = \delta_{ij}$. We can see this as saying “the scalar product of two distinct rows is 0, and the scalar product of a row with itself is 1”. Alternatively, the rows (and columns — by considering A^T) of an orthogonal matrix form an orthonormal set.

Similarly, for a unitary matrix, $U_{ik}U_{jk}^\dagger = \delta_{ij}$, i.e. $u_{ik}u_{jk}^* = \delta_{ij}$. i.e. the rows are orthonormal, using the definition of complex scalar product.

Example.

- (i) The reflection in a plane is an orthogonal matrix. Since $R_{ij} = \delta_{ij} - 2n_i n_j$, We have

$$\begin{aligned} R_{ik}R_{jk} &= (\delta_{ik} - 2n_i n_k)(\delta_{jk} - 2n_j n_k) \\ &= \delta_{ik}\delta_{jk} - 2\delta_{jk}n_i n_k - 2\delta_{ik}n_j n_k + 2n_i n_k n_j n_k \\ &= \delta_{ij} - 2n_i n_j - 2n_j n_i + 4n_i n_j (n_k n_k) \\ &= \delta_{ij} \end{aligned}$$

- (ii) The rotation is an orthogonal matrix. We could multiply out using suffix notation, but it would be cumbersome to do so. Alternatively, denote rotation matrix by θ about $\hat{\mathbf{n}}$ as $R(\theta, \hat{\mathbf{n}})$. Clearly, $R(\theta, \hat{\mathbf{n}})^{-1} = R(-\theta, \hat{\mathbf{n}})$. We have

$$\begin{aligned} R_{ij}(-\theta, \hat{\mathbf{n}}) &= (\cos \theta)\delta_{ij} + n_i n_j (1 - \cos \theta) + \varepsilon_{ijk} n_k \sin \theta \\ &= (\cos \theta)\delta_{ji} + n_j n_i (1 - \cos \theta) - \varepsilon_{jik} n_k \sin \theta \\ &= R_{ji}(\theta, \hat{\mathbf{n}}) \end{aligned}$$

In other words, $R(-\theta, \hat{\mathbf{n}}) = R(\theta, \hat{\mathbf{n}})^T$. So $R(\theta, \hat{\mathbf{n}})^{-1} = R(\theta, \hat{\mathbf{n}})^T$.

3.5 Determinants

Consider a linear map $\alpha : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. The standard basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is mapped to $\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3$ with $\mathbf{e}'_i = A\mathbf{e}_i$. Thus the unit cube formed by $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is mapped to the parallelepiped with volume

$$\begin{aligned} [\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3] &= \varepsilon_{ijk}(e'_1)_i(e'_2)_j(e'_3)_k \\ &= \varepsilon_{ijk}A_{i\ell}\underbrace{(e_1)_\ell}_{\delta_{1\ell}}A_{jm}\underbrace{(e_2)_m}_{\delta_{2m}}A_{kn}\underbrace{(e_3)_n}_{\delta_{3n}} \\ &= \varepsilon_{ijk}A_{i1}A_{j2}A_{k3} \end{aligned}$$

We call this the determinant and write as

$$\det(A) = \begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix}$$

3.5.1 Permutations

To define the determinant for square matrices of arbitrary size, we first have to consider *permutations*.

Definition (Permutation). A *permutation* of a set S is a bijection $\varepsilon : S \rightarrow S$.

Notation. Consider the set S_n of all permutations of $1, 2, 3, \dots, n$. S_n contains $n!$ elements. Consider $\rho \in S_n$ with $i \mapsto \rho(i)$. We write

$$\rho = \begin{pmatrix} 1 & 2 & \cdots & n \\ \rho(1) & \rho(2) & \cdots & \rho(n) \end{pmatrix}.$$

Definition (Fixed point). A *fixed point* of ρ is a k such that $\rho(k) = k$. e.g. in $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix}$, 3 is the fixed point. By convention, we can omit the fixed point and write as $\begin{pmatrix} 1 & 2 & 4 \\ 4 & 1 & 2 \end{pmatrix}$.

Definition (Disjoint permutation). Two permutations are *disjoint* if numbers moved by one are fixed by the other, and vice versa. e.g. $\begin{pmatrix} 1 & 2 & 4 & 5 & 6 \\ 5 & 6 & 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 6 & 2 \end{pmatrix} \begin{pmatrix} 1 & 4 & 5 \\ 5 & 1 & 4 \end{pmatrix}$, and the two cycles on the right hand side are disjoint. Disjoint permutations commute, but in general non-disjoint permutations do not.

Definition (Transposition and k -cycle). $\begin{pmatrix} 2 & 6 \\ 6 & 2 \end{pmatrix}$ is a *2-cycle* or a *transposition*, and we can simply write $(2\ 6)$. $\begin{pmatrix} 1 & 4 & 5 \\ 5 & 1 & 4 \end{pmatrix}$ is a *3-cycle*, and we can simply write $(1\ 5\ 4)$. (1 is mapped to 5; 5 is mapped to 4; 4 is mapped to 1)

Proposition (q -cycle as product of 2-cycles). Any q -cycle can be written as a product of 2-cycles.

Proof. $(1\ 2\ 3\ \cdots\ n) = (1\ 2)(2\ 3)(3\ 4)\cdots(n-1\ n)$. □

Definition (Sign of permutation). The *sign* of a permutation $\varepsilon(\rho)$ is $(-1)^r$, where r is the number of 2-cycles when ρ is written as a product of 2-cycles. If $\varepsilon(\rho) = +1$, it is an even permutation. Otherwise, it is an odd permutation. Note that $\varepsilon(\rho\sigma) = \varepsilon(\rho)\varepsilon(\sigma)$ and $\varepsilon(\rho^{-1}) = \varepsilon(\rho)$.

The proof that this is well-defined can be found in IA Groups.

Definition (Levi-Civita symbol). The *Levi-Civita* symbol is defined by

$$\varepsilon_{j_1 j_2 \dots j_n} = \begin{cases} +1 & \text{if } j_1 j_2 j_3 \dots j_n \text{ is an even permutation of } 1, 2, \dots, n \\ -1 & \text{if it is an odd permutation} \\ 0 & \text{if any 2 of them are equal} \end{cases}$$

Clearly, $\varepsilon_{\rho(1)\rho(2)\dots\rho(n)} = \varepsilon(\rho)$.

Definition (Determinant). The *determinant* of an $n \times n$ matrix A is defined as:

$$\det(A) = \sum_{\sigma \in S_n} \varepsilon(\sigma) A_{\sigma(1)1} A_{\sigma(2)2} \dots A_{\sigma(n)n},$$

or equivalently,

$$\det(A) = \varepsilon_{j_1 j_2 \dots j_n} A_{j_1 1} A_{j_2 2} \dots A_{j_n n}.$$

Proposition (Determinant of 2×2 matrix).

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

3.5.2 Properties of determinants

Proposition (Determinant of transpose). $\det(A) = \det(A^T)$.

Proof. Take a single term $A_{\sigma(1)1} A_{\sigma(2)2} \dots A_{\sigma(n)n}$ and let ρ be another permutation in S_n . We have

$$A_{\sigma(1)1} A_{\sigma(2)2} \dots A_{\sigma(n)n} = A_{\sigma(\rho(1))\rho(1)} A_{\sigma(\rho(2))\rho(2)} \dots A_{\sigma(\rho(n))\rho(n)}$$

since the right hand side is just re-ordering the order of multiplication. Choose $\rho = \sigma^{-1}$ and note that $\varepsilon(\sigma) = \varepsilon(\rho)$. Then

$$\det(A) = \sum_{\rho \in S_n} \varepsilon(\rho) A_{1\rho(1)} A_{2\rho(2)} \dots A_{n\rho(n)} = \det(A^T). \quad \square$$

Proposition (Row scaling effect on determinant). If matrix B is formed by multiplying every element in a single row of A by a scalar λ , then $\det(B) = \lambda \det(A)$. Consequently, $\det(\lambda A) = \lambda^n \det(A)$.

Proof. When one row is multiplied by λ , each term in the determinant contains exactly one element from that row, so each term is multiplied by λ once, giving $\det(B) = \lambda \det(A)$. For $\det(\lambda A)$, every entry is multiplied by λ , which is equivalent to multiplying each of the n rows by λ , giving $\lambda^n \det(A)$. $\square$

Proposition (Identical rows gives zero determinant). If 2 rows (or 2 columns) of A are identical, the determinant is 0.

Proof. wlog, suppose columns 1 and 2 are the same. Then

$$\det(A) = \sum_{\sigma \in S_n} \varepsilon(\sigma) A_{\sigma(1)1} A_{\sigma(2)2} \cdots A_{\sigma(n)n}.$$

Now write an arbitrary σ in the form $\sigma = \rho(1\ 2)$. Then $\varepsilon(\sigma) = \varepsilon(\rho)\varepsilon((1\ 2)) = -\varepsilon(\rho)$. So

$$\det(A) = \sum_{\rho \in S_n} -\varepsilon(\rho) A_{\rho(2)1} A_{\rho(1)2} A_{\rho(3)3} \cdots A_{\rho(n)n}.$$

But columns 1 and 2 are identical, so $A_{\rho(2)1} = A_{\rho(2)2}$ and $A_{\rho(1)2} = A_{\rho(1)1}$. So $\det(A) = -\det(A)$ and $\det(A) = 0$. $\square$

Proposition (Linearly dependent rows gives zero determinant). If 2 rows or 2 columns of a matrix are linearly dependent, then the determinant is zero.

Proof. Suppose in A , $(\text{column } r) + \lambda(\text{column } s) = 0$. Define

$$B_{ij} = \begin{cases} A_{ij} & j \neq r \\ A_{ij} + \lambda A_{is} & j = r \end{cases}.$$

Then $\det(B) = \det(A) + \lambda \det(\text{matrix with column } r = \text{column } s) = \det(A)$. Then we can see that the r th column of B is all zeroes. So each term in the sum contains one zero and $\det(A) = \det(B) = 0$. $\square$

Even if we don't have linearly dependent rows or columns, we can still run the exact same proof as above, and still get that $\det(B) = \det(A)$. Linear dependence is only required to show that $\det(B) = 0$. So in general, we can add a linear multiple of a column (or row) onto another column (or row) without changing the determinant.

Proposition (Adding row multiple preserves determinant). Given a matrix A , if B is a matrix obtained by adding a multiple of a column (or row) of A to another column (or row) of A , then $\det A = \det B$.

Corollary (Row swap negates determinant). Swapping two rows or columns of a matrix negates the determinant.

Proof. We do the column case only. Let $A = (\mathbf{a}_1 \cdots \mathbf{a}_i \cdots \mathbf{a}_j \cdots \mathbf{a}_n)$. Then

$$\begin{aligned} \det(\mathbf{a}_1 \cdots \mathbf{a}_i \cdots \mathbf{a}_j \cdots \mathbf{a}_n) &= \det(\mathbf{a}_1 \cdots \mathbf{a}_i + \mathbf{a}_j \cdots \mathbf{a}_j \cdots \mathbf{a}_n) \\ &= \det(\mathbf{a}_1 \cdots \mathbf{a}_i + \mathbf{a}_j \cdots \mathbf{a}_j - (\mathbf{a}_i + \mathbf{a}_j) \cdots \mathbf{a}_n) \\ &= \det(\mathbf{a}_1 \cdots \mathbf{a}_i + \mathbf{a}_j \cdots - \mathbf{a}_i \cdots \mathbf{a}_n) \\ &= \det(\mathbf{a}_1 \cdots \mathbf{a}_j \cdots - \mathbf{a}_i \cdots \mathbf{a}_n) \\ &= -\det(\mathbf{a}_1 \cdots \mathbf{a}_j \cdots \mathbf{a}_i \cdots \mathbf{a}_n) \end{aligned}$$

Alternatively, we can prove this from the definition directly, using the fact that the sign of a transposition is -1 (and that the sign is multiplicative). $\square$

Proposition (Determinant of product). $\det(AB) = \det(A) \det(B)$.

Proof. First note that $\sum_{\sigma} \varepsilon(\sigma) A_{\sigma(1)\rho(1)} A_{\sigma(2)\rho(2)} = \varepsilon(\rho) \det(A)$, i.e. swapping columns (or rows) an even/odd number of times gives a factor ± 1 respectively. We can prove this by writing $\sigma = \mu\rho$.

Now

$$\begin{aligned} \det AB &= \sum_{\sigma} \varepsilon(\sigma) (AB)_{\sigma(1)1} (AB)_{\sigma(2)2} \cdots (AB)_{\sigma(n)n} \\ &= \sum_{\sigma} \varepsilon(\sigma) \sum_{k_1, k_2, \dots, k_n}^n A_{\sigma(1)k_1} B_{k_1 1} \cdots A_{\sigma(n)k_n} B_{k_n n} \\ &= \sum_{k_1, \dots, k_n} B_{k_1 1} \cdots B_{k_n n} \underbrace{\sum_{\sigma} \varepsilon(\sigma) A_{\sigma(1)k_1} A_{\sigma(2)k_2} \cdots A_{\sigma(n)k_n}}_S \end{aligned}$$

Now consider the many different S 's. If in S , two of k_1 and k_n are equal, then S is a determinant of a matrix with two columns the same, i.e. $S = 0$. So we only have to consider the sum over distinct k_i 's. Thus the k_i 's are a permutation of $1, \dots, n$, say $k_i = \rho(i)$. Then we can write

$$\begin{aligned} \det AB &= \sum_{\rho} B_{\rho(1)1} \cdots B_{\rho(n)n} \sum_{\sigma} \varepsilon(\sigma) A_{\sigma(1)\rho(1)} \cdots A_{\sigma(n)\rho(n)} \\ &= \sum_{\rho} B_{\rho(1)1} \cdots B_{\rho(n)n} (\varepsilon(\rho) \det A) \\ &= \det A \sum_{\rho} \varepsilon(\rho) B_{\rho(1)1} \cdots B_{\rho(n)n} \\ &= \det A \det B \end{aligned} \quad \square$$

Corollary (Determinant of orthogonal matrix). If A is orthogonal, $\det A = \pm 1$.

Proof.

$$\begin{aligned} AA^T &= I \\ \det AA^T &= \det I \\ \det A \det A^T &= 1 \\ (\det A)^2 &= 1 \\ \det A &= \pm 1 \end{aligned} \quad \square$$

Corollary (Determinant of unitary matrix). If U is unitary, $|\det U| = 1$.

Proof. We have $\det U^\dagger = (\det U^T)^* = \det(U)^*$. Since $UU^\dagger = I$, we have $\det(U) \det(U)^* = 1$. $\square$

Proposition (Orthogonal matrices in $\mathbb{R}^3$ are rotations or reflections). In $\mathbb{R}^3$, orthogonal matrices represent either a rotation ($\det = 1$) or a reflection ($\det = -1$).

3.5.3 Minors and Cofactors

Definition (Minor and cofactor). For an $n \times n$ matrix A , define A^{ij} to be the $(n-1) \times (n-1)$ matrix in which row i and column j of A have been removed.

The *minor* of the ij th element of A is $M_{ij} = \det A^{ij}$.

The *cofactor* of the ij th element of A is $\Delta_{ij} = (-1)^{i+j} M_{ij}$.

Notation. We use $\bar{}$ to denote a symbol which has been missed out of a natural sequence.

Example. $1, 2, 3, 5 = 1, 2, 3, \bar{4}, 5$.

The significance of these definitions is that we can use them to provide a systematic way of evaluating determinants. We will also use them to find inverses of matrices.

Theorem (Laplace expansion formula). For any particular fixed i ,

$$\det A = \sum_{j=1}^n A_{ji} \Delta_{ji}.$$

Proof.

$$\det A = \sum_{j_i=1}^n A_{j_i i} \sum_{j_1, \dots, \bar{j}_i, \dots, j_n}^n \varepsilon_{j_1 j_2 \dots j_n} A_{j_1 1} A_{j_2 2} \dots \overline{A_{j_i i}} \dots A_{j_n n}$$

Let $\sigma \in S_n$ be the permutation which moves j_i to the i th position, and leave everything else in its natural order, i.e.

$$\sigma = \begin{pmatrix} 1 & \dots & i & i+1 & i+2 & \dots & j_i-1 & j_i & j_i+1 & \dots & n \\ 1 & \dots & j_i & i & i+1 & \dots & j_i-2 & j_i-1 & j_i+1 & \dots & n \end{pmatrix}$$

if $j_i > i$, and similarly for other cases. To perform this permutation, $|i - j_i|$ transpositions are made. So $\varepsilon(\sigma) = (-1)^{i-j_i}$.

Now consider the permutation $\rho \in S_n$

$$\rho = \begin{pmatrix} 1 & \dots & \dots & \bar{j}_i & \dots & n \\ j_1 & \dots & \bar{j}_i & \dots & \dots & j_n \end{pmatrix}$$

The composition $\rho\sigma$ reorders $(1, \dots, n)$ to $(j_1, j_2, \dots, j_n)$. So $\varepsilon(\rho\sigma) = \varepsilon_{j_1 \dots j_n} =$

$\varepsilon(\rho)\varepsilon(\sigma) = (-1)^{i-j_i}\varepsilon_{j_1\cdots\bar{j}_i\cdots j_n}$. Hence the original equation becomes

$$\begin{aligned}\det A &= \sum_{j_i=1}^n A_{j_i i} \sum_{j_1\cdots\bar{j}_i\cdots j_n} (-1)^{i-j_i}\varepsilon_{j_1\cdots\bar{j}_i\cdots j_n} A_{j_1 1} \cdots \overline{A_{j_i i}} \cdots A_{j_n n} \\ &= \sum_{j_i=1}^n A_{j_i i} (-1)^{i-j_i} M_{j_i i} \\ &= \sum_{j_i=1}^n A_{j_i i} \Delta_{j_i i} \\ &= \sum_{j=1}^n A_{j i} \Delta_{j i}\end{aligned}\quad \square$$

Example. $\det A = \begin{vmatrix} 2 & 4 & 2 \\ 3 & 2 & 1 \\ 2 & 0 & 1 \end{vmatrix}$. We can pick the first row and have

$$\begin{aligned}\det A &= 2 \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} - 4 \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} + 2 \begin{vmatrix} 3 & 2 \\ 2 & 0 \end{vmatrix} \\ &= 2(2 - 0) - 4(3 - 2) + 2(0 - 4) \\ &= -8.\end{aligned}$$

Alternatively, we can pick the second column and have

$$\begin{aligned}\det A &= -4 \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 2 \\ 2 & 1 \end{vmatrix} - 0 \begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} \\ &= -4(3 - 2) + 2(2 - 4) - 0 \\ &= -8.\end{aligned}$$

In practical terms, we use a combination of properties of determinants with a sensible choice of i to evaluate $\det(A)$.

Example. Consider $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$. Row 1 - row 2 gives

$$\begin{vmatrix} 0 & a-b & a^2-b^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b) \begin{vmatrix} 0 & 1 & a+b \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$$

Do row 2 - row 3. We obtain

$$(a-b)(b-c) \begin{vmatrix} 0 & 1 & a+b \\ 0 & 1 & b+c \\ 1 & c & c^2 \end{vmatrix}.$$

Row 1 - row 2 gives

$$(a-b)(b-c)(a-c) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & b+c \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(a-c).$$

4 Matrices and linear equations

4.1 Simple example, 2×2

Remark. We begin by studying the inverse of a 2×2 matrix through the lens of solving linear systems. This concrete case motivates the general theory developed in the following subsections.

Example (Solving a 2×2 system by elimination). Let $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ be a 2×2 matrix, and consider the system of linear equations

$$A_{11}x_1 + A_{12}x_2 = d_1 \quad (\text{a})$$

$$A_{21}x_1 + A_{22}x_2 = d_2, \quad (\text{b})$$

where $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\mathbf{d} = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$ are column vectors. In matrix form, this system is $A\mathbf{x} = \mathbf{d}$.

To solve for x_1 and x_2 , we eliminate variables. Computing $(\text{a}) \times A_{22} - (\text{b}) \times A_{12}$ and similarly the other way round, we obtain

$$\begin{aligned} (A_{11}A_{22} - A_{12}A_{21})x_1 &= A_{22}d_1 - A_{12}d_2 \\ \underbrace{(A_{11}A_{22} - A_{12}A_{21})}_{\det A} x_2 &= A_{11}d_2 - A_{21}d_1. \end{aligned}$$

Assuming $\det A \neq 0$, we may divide by $\det A$ and write the solution in matrix form:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1}{\det A} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}.$$

Proposition (Inverse of a 2×2 matrix). Let $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ be a 2×2 matrix. Then A^{-1} exists if and only if $\det A \neq 0$, in which case

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix}.$$

Proof. Suppose $\det A \neq 0$. The preceding example shows that the unique solution to $A\mathbf{x} = \mathbf{d}$ is

$$\mathbf{x} = \frac{1}{\det A} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix} \mathbf{d}.$$

On the other hand, if A^{-1} exists, then multiplying $A\mathbf{x} = \mathbf{d}$ on the left by A^{-1} gives $\mathbf{x} = A^{-1}\mathbf{d}$. Comparing these expressions, we conclude that

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix}.$$

The converse, that $\det A = 0$ implies A^{-1} does not exist, follows from the general theory in the next subsection. $\square$

4.2 Inverse of an $n \times n$ matrix

Remark. For larger matrices, the formula for the inverse is similar to the 2×2 case, but slightly more complicated (and costly to evaluate). The key ingredient is the following lemma.

Lemma (Cofactor sum formula). Let A be an $n \times n$ matrix with cofactors Δ_{jk} . Then

$$\sum_{k=1}^n A_{ik} \Delta_{jk} = \delta_{ij} \det A.$$

Proof. If $i \neq j$, then consider an $n \times n$ matrix B , which is identical to A except the j th row is replaced by the i th row of A . So Δ_{jk} of $B = \Delta_{jk}$ of A , since Δ_{jk} does not depend on the elements in row j . Since B has a duplicate row, we know that

$$0 = \det B = \sum_{k=1}^n B_{jk} \Delta_{jk} = \sum_{k=1}^n A_{ik} \Delta_{jk}.$$

If $i = j$, then the expression is $\det A$ by the Laplace expansion formula. $\square$

Theorem (Matrix inverse via cofactors). Let A be an $n \times n$ matrix. If $\det A \neq 0$, then A^{-1} exists and is given by

$$(A^{-1})_{ij} = \frac{\Delta_{ji}}{\det A},$$

where Δ_{ji} denotes the (j, i) -cofactor of A .

Proof. Using the Einstein summation convention, we compute

$$(A^{-1})_{ik} A_{kj} = \frac{\Delta_{ki}}{\det A} A_{kj} = \frac{\delta_{ij} \det A}{\det A} = \delta_{ij},$$

where the second equality follows from the cofactor sum formula. Hence $A^{-1}A = I$. $\square$

Corollary. Let A be an $n \times n$ matrix. If $\det A = 0$, then A^{-1} does not exist.

Proof. Suppose for contradiction that B is an inverse of A . Then $AB = I$, so $\det(AB) = \det I = 1$. But by the multiplicative property of determinants, $\det(AB) = \det A \cdot \det B = 0$, a contradiction. $\square$

Example (Inverse of a shear matrix). Consider the shear matrix $S_\lambda =$

$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. We have $\det S_\lambda = 1$, so S_λ is invertible. The cofactors are

$$\begin{array}{lll} \Delta_{11} = 1 & \Delta_{12} = 0 & \Delta_{13} = 0 \\ \Delta_{21} = -\lambda & \Delta_{22} = 1 & \Delta_{23} = 0 \\ \Delta_{31} = 0 & \Delta_{32} = 0 & \Delta_{33} = 1 \end{array}$$

So $S_\lambda^{-1} = \begin{pmatrix} 1 & -\lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Remark (Computational complexity of matrix inversion). How many arithmetic operations are involved in calculating the inverse of an $n \times n$ matrix? We count multiplication operations since they are the most time-consuming. Suppose that calculating $\det A$ takes f_n multiplications. This involves n determinants of $(n-1) \times (n-1)$ matrices, and n additional multiplications to combine them. Thus $f_n = nf_{n-1} + n$, which gives $f_n = O(n!)$ (in fact $f_n \approx (1+e)n!$).

To find the inverse, we must calculate n^2 cofactors. Each cofactor is an $(n-1) \times (n-1)$ determinant, requiring $O((n-1)!)$ operations. Hence the total time complexity is $O(n^2(n-1)!) = O(n \cdot n!)$.

This is prohibitively slow for large matrices. While it is theoretically possible to solve systems of linear equations by inverting a matrix, this approach is impractical in general. Instead, we develop more efficient methods such as Gaussian elimination, which has complexity $O(n^3)$.

4.3 Homogeneous and inhomogeneous equations

Definition (Homogeneous and inhomogeneous systems). Consider a system of linear equations $A\mathbf{x} = \mathbf{b}$, where A is an $n \times n$ matrix, and $\mathbf{x}, \mathbf{b} \in \mathbb{R}^n$ are column vectors. The system is called *homogeneous* if $\mathbf{b} = \mathbf{0}$, and *inhomogeneous* otherwise.

Proposition (Unique solution for invertible matrices). Let A be an $n \times n$ matrix with $\det A \neq 0$, and let $\mathbf{b} \in \mathbb{R}^n$. Then the system $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$. In particular, the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.

Proof. We first show existence. Since $\det A \neq 0$, the columns of A are linearly independent. The columns are the images of the standard basis vectors under A , i.e. $A\mathbf{e}_i$ for $i = 1, \dots, n$. Thus these images form a basis of $\mathbb{R}^n$, so the image of A is all of $\mathbb{R}^n$. In particular, $\mathbf{b}$ lies in the image of A , meaning a solution exists.

We now show uniqueness. Suppose $\mathbf{x}$ and $\mathbf{x}'$ are both solutions, so $A\mathbf{x} = A\mathbf{x}' = \mathbf{b}$. Then $A(\mathbf{x} - \mathbf{x}') = \mathbf{0}$, so $\mathbf{x} - \mathbf{x}'$ lies in the kernel of A . Since the rank of A is n , by the rank-nullity theorem the nullity is 0, so the kernel is trivial.

Hence $\mathbf{x} - \mathbf{x}' = \mathbf{0}$, i.e. $\mathbf{x} = \mathbf{x}'$. □

4.3.1 Gaussian elimination

Remark (The Gaussian elimination algorithm). Consider a general system of m linear equations in n unknowns:

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &= d_1 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &= d_2 \\ &\vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &= d_m. \end{aligned}$$

Assuming $A_{11} \neq 0$ (which can be achieved by reordering the equations if necessary), we use the first equation to eliminate x_1 from the remaining $(m-1)$ equations. We then use the second equation to eliminate x_2 from the remaining $(m-2)$ equations (reordering if needed), and repeat. After this process, we obtain a system in *row echelon form*:

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + A_{13}x_3 + \cdots + A_{1n}x_n &= d_1 \\ A_{22}^{(2)}x_2 + A_{23}^{(2)}x_3 + \cdots + A_{2n}^{(2)}x_n &= d_2^{(2)} \\ &\vdots \\ A_{rr}^{(r)}x_r + \cdots + A_{rn}^{(r)}x_n &= d_r^{(r)} \\ 0 &= d_{r+1}^{(r)} \\ &\vdots \\ 0 &= d_m^{(r)}, \end{aligned}$$

where $A_{ii}^{(i)} \neq 0$ for $i = 1, \dots, r$ (achievable by reordering), and the superscript (i) denotes the coefficient after i elimination steps.

Definition (Classification of linear systems). A system of linear equations is called:

- (i) *Overdetermined* if it has no solution (inconsistent);
- (ii) *Determined* if it has exactly one solution;
- (iii) *Underdetermined* if it has infinitely many solutions.

Proposition (Classification via row echelon form). Consider a system of m equations in n unknowns, reduced to row echelon form with r nonzero rows (where $r \leq \min(m, n)$). Then:

- (i) If $r < m$ and at least one of $d_{r+1}^{(r)}, \dots, d_m^{(r)}$ is nonzero, the system is overdetermined.

- (ii) If $r = n \leq m$ and $d_{r+i}^{(r)} = 0$ for all $i \geq 1$, the system is determined, with unique solution found by back substitution.
- (iii) If $r < n$ and $d_{r+i}^{(r)} = 0$ for all $i \geq 1$, the system is underdetermined, with $x_{r+1}, \dots, x_n$ as free parameters.

Example (Overdetermined system). Consider the system

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 3 \\ 6x_1 + 3x_2 + 3x_3 &= 0 \\ 6x_1 + 2x_2 + 4x_3 &= 6. \end{aligned}$$

Eliminating x_1 from the second and third equations gives

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 3 \\ -x_2 + x_3 &= -6 \\ -2x_2 + 2x_3 &= 0. \end{aligned}$$

Eliminating x_2 from the third equation gives

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 3 \\ -x_2 + x_3 &= -6 \\ 0 &= 12. \end{aligned}$$

Since $d_3^{(3)} = 12 \neq 0$, the system has no solution.

Example (Determined system). Consider the system

$$\begin{aligned} 2x_1 + 5x_2 &= 2 \\ 4x_1 + 3x_2 &= 11. \end{aligned}$$

Eliminating x_1 from the second equation gives

$$\begin{aligned} 2x_1 + 5x_2 &= 2 \\ -7x_2 &= 7. \end{aligned}$$

Back substitution yields $x_2 = -1$ and $x_1 = 7/2$.

Example (Underdetermined system). Consider the system

$$\begin{aligned} x_1 + x_2 &= 1 \\ 2x_1 + 2x_2 &= 2. \end{aligned}$$

Eliminating x_1 from the second equation gives

$$\begin{aligned}x_1 + x_2 &= 1 \\ 0 &= 0.\end{aligned}$$

Thus x_2 is a free parameter, and $x_1 = 1 - x_2$ for any choice of x_2 .

Remark (Computational complexity). For an $n \times n$ system, Gaussian elimination requires $O(n^3)$ arithmetic operations, which is far more efficient than computing the inverse via cofactors.

Remark (Connection to determinants). For a square system ($m = n$), Gaussian elimination is closely related to the determinant. Since row operations of the form “row i minus a multiple of row j ” do not change the determinant, and swapping rows introduces a factor of (-1) , we have

$$\det A = (-1)^k \begin{vmatrix} A_{11} & A_{12} & \cdots & \cdots & \cdots & A_{1n} \\ 0 & A_{22}^{(2)} & \cdots & \cdots & \cdots & A_{2n}^{(2)} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & A_{rr}^{(r)} & \cdots & A_{rn}^{(r)} \\ 0 & 0 & \cdots & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{vmatrix},$$

where k is the number of row swaps performed.

Lemma (Determinant of upper triangular matrix). Let T be an $n \times n$ upper triangular matrix (i.e., $T_{ij} = 0$ for all $i > j$). Then

$$\det T = T_{11}T_{22} \cdots T_{nn}.$$

Corollary. For a square $n \times n$ system reduced to row echelon form with r nonzero rows:

- (i) If $r < n$, then $\det A = 0$.
- (ii) If $r = n$, then $\det A = (-1)^k A_{11}A_{22}^{(2)} \cdots A_{nn}^{(n)} \neq 0$.

4.4 Matrix rank

Definition (Rank of a matrix). Let $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map with associated matrix A . The *rank* of A , denoted $r(A)$, is the dimension of the image of α .

Proposition (Rank equals number of independent columns). Let A be an $m \times n$ matrix. Then $r(A)$ equals the number of linearly independent columns of A .

Proof. Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ denote the standard basis of $\mathbb{R}^n$. The image of A is spanned by $A\mathbf{e}_1, \dots, A\mathbf{e}_n$. Since $A\mathbf{e}_j$ is the j th column of A , the dimension of the image equals the number of linearly independent columns. $\square$

Definition (Column rank and row rank). Let A be an $m \times n$ matrix. The *column rank* of A is the maximum number of linearly independent columns. The *row rank* of A is the maximum number of linearly independent rows.

Theorem (Column rank equals row rank). For any $m \times n$ matrix A , the column rank and row rank are equal.

Proof. Let r denote the row rank of A , and let $\mathbf{v}_1^T, \mathbf{v}_2^T, \dots, \mathbf{v}_r^T$ be a maximal set of linearly independent rows, where $\mathbf{v}_k^T = (v_{k1}, v_{k2}, \dots, v_{kn})$ for $k = 1, \dots, r$.

Denote the i th row of A by $\mathbf{r}_i^T = (A_{i1}, A_{i2}, \dots, A_{in})$. Since $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is a maximal linearly independent set of rows, every row of A can be written as a linear combination of these vectors:

$$\mathbf{r}_i^T = \sum_{k=1}^r C_{ik} \mathbf{v}_k^T$$

for some coefficients C_{ik} with $1 \leq i \leq m$ and $1 \leq k \leq r$.

The (i, j) -entry of A is therefore

$$A_{ij} = (\mathbf{r}_i)_j = \sum_{k=1}^r C_{ik} (\mathbf{v}_k)_j.$$

Writing this in terms of the j th column of A , we have

$$\begin{pmatrix} A_{1j} \\ A_{2j} \\ \vdots \\ A_{mj} \end{pmatrix} = \sum_{k=1}^r (\mathbf{v}_k)_j \begin{pmatrix} C_{1k} \\ C_{2k} \\ \vdots \\ C_{mk} \end{pmatrix}.$$

Thus every column of A is a linear combination of the r vectors $(C_{1k}, C_{2k}, \dots, C_{mk})^T$ for $k = 1, \dots, r$. Hence the column rank of A is at most r , the row rank.

Applying the same argument to A^T shows that the row rank is at most the column rank. Therefore the two ranks are equal. $\square$

4.5 Homogeneous problem $A\mathbf{x} = \mathbf{0}$

Remark. We now focus on the square case, where the number of unknowns equals the number of equations. Let A be an $n \times n$ matrix, and consider the homogeneous system $A\mathbf{x} = \mathbf{0}$.

Proposition (Nontrivial solutions require singular matrix). Let A be an $n \times n$ matrix. If $\det A \neq 0$, then $\mathbf{x} = \mathbf{0}$ is the unique solution to $A\mathbf{x} = \mathbf{0}$. Equivalently, if $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\mathbf{x} \neq \mathbf{0}$, then $\det A = 0$.

Proof. If $\det A \neq 0$, then A^{-1} exists and $\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}$. $\square$

4.5.1 Geometrical interpretation

Remark. Consider a 3×3 matrix A with rows $\mathbf{r}_1^T, \mathbf{r}_2^T, \mathbf{r}_3^T$:

$$A = \begin{pmatrix} \mathbf{r}_1^T \\ \mathbf{r}_2^T \\ \mathbf{r}_3^T \end{pmatrix}.$$

The equation $A\mathbf{x} = \mathbf{0}$ is equivalent to $\mathbf{r}_i \cdot \mathbf{x} = 0$ for $i = 1, 2, 3$. Each equation $\mathbf{r}_i \cdot \mathbf{x} = 0$ represents a plane through the origin with normal vector $\mathbf{r}_i$. Thus the solution set is the intersection of these three planes.

Proposition (Classification of solutions in $\mathbb{R}^3$). Let A be a 3×3 matrix with rows $\mathbf{r}_1^T, \mathbf{r}_2^T, \mathbf{r}_3^T$. The kernel of A is determined by the rank as follows:

- (i) If $r(A) = 3$ (equivalently, $\det A \neq 0$), then $\ker A = \{\mathbf{0}\}$ and $n(A) = 0$.
- (ii) If $r(A) = 2$, then $\ker A$ is a line through the origin, and $n(A) = 1$.
- (iii) If $r(A) = 1$, then $\ker A$ is a plane through the origin, and $n(A) = 2$.
- (iv) If $r(A) = 0$, then A is the zero matrix and $\ker A = \mathbb{R}^3$.

Proof. The dimension of the kernel follows from the rank-nullity theorem: $n(A) = 3 - r(A)$.

For case (i), if $\det A = [\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3] \neq 0$, then $\text{span}\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3\} = \mathbb{R}^3$, so $r(A) = 3$ and the kernel is trivial.

For case (ii), assume without loss of generality that $\mathbf{r}_1, \mathbf{r}_2$ are linearly independent. The solution $\mathbf{x}$ must lie on the intersection of the planes $\mathbf{r}_1 \cdot \mathbf{x} = 0$ and $\mathbf{r}_2 \cdot \mathbf{x} = 0$. This intersection is the line $\{\lambda(\mathbf{r}_1 \times \mathbf{r}_2) : \lambda \in \mathbb{R}\}$, since $\mathbf{r}_1 \times \mathbf{r}_2$ is perpendicular to both $\mathbf{r}_1$ and $\mathbf{r}_2$. Since $\mathbf{r}_3$ is a linear combination of $\mathbf{r}_1$ and $\mathbf{r}_2$, every point on this line also satisfies $\mathbf{r}_3 \cdot \mathbf{x} = 0$.

For case (iii), the rows $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ are all parallel. Thus $\mathbf{r}_1 \cdot \mathbf{x} = 0$ implies $\mathbf{r}_2 \cdot \mathbf{x} = \mathbf{r}_3 \cdot \mathbf{x} = 0$, so the kernel is the plane $\{\mathbf{x} \in \mathbb{R}^3 : \mathbf{r}_1 \cdot \mathbf{x} = 0\}$. $\square$

4.5.2 Linear mapping view of $A\mathbf{x} = \mathbf{0}$

Remark. In the general case, consider a linear map $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $\mathbf{x} \mapsto A\mathbf{x}$. The kernel $\ker(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$ has dimension $n(A)$, the nullity of A .

Proposition (Structure of the kernel). Let A be an $n \times n$ matrix with nullity $n(A)$.

- (i) If $n(A) = 0$, then $A\mathbf{e}_1, A\mathbf{e}_2, \dots, A\mathbf{e}_n$ are linearly independent, and $r(A) = n$.
- (ii) If $n(A) > 0$, then the image of A is a proper subspace of $\mathbb{R}^n$. If $\{\mathbf{u}_1, \dots, \mathbf{u}_{n(A)}\}$ is a basis of the kernel, extend it to a basis $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ of $\mathbb{R}^n$. Then $\{A\mathbf{u}_{n(A)+1}, \dots, A\mathbf{u}_n\}$ forms a basis of the image.

4.6 General solution of $A\mathbf{x} = \mathbf{d}$

Remark. We now consider the general inhomogeneous equation $A\mathbf{x} = \mathbf{d}$, where A is an $n \times n$ matrix and $\mathbf{x}, \mathbf{d} \in \mathbb{R}^n$ are column vectors.

Theorem (General solution of linear systems). Let A be an $n \times n$ matrix and $\mathbf{d} \in \mathbb{R}^n$. The solutions to $A\mathbf{x} = \mathbf{d}$ are characterised as follows:

- (i) If $\det A \neq 0$, then A^{-1} exists, $n(A) = 0$, and $r(A) = n$. For any $\mathbf{d} \in \mathbb{R}^n$, there is a unique solution $\mathbf{x} = A^{-1}\mathbf{d}$.
- (ii) If $\det A = 0$, then A^{-1} does not exist, $n(A) > 0$, and $r(A) < n$. The image of A is a proper subspace of $\mathbb{R}^n$, and:
 - (a) If $\mathbf{d} \notin \text{im } A$, there is no solution.
 - (b) If $\mathbf{d} \in \text{im } A$, the general solution is $\mathbf{x} = \mathbf{x}_0 + \mathbf{y}$, where $\mathbf{x}_0$ is any particular solution satisfying $A\mathbf{x}_0 = \mathbf{d}$, and $\mathbf{y} \in \ker A$.

Proof. Case (i) follows from the existence of A^{-1} when $\det A \neq 0$.

For case (ii), if $\mathbf{d} \notin \text{im } A$, then by definition no $\mathbf{x}$ satisfies $A\mathbf{x} = \mathbf{d}$. If $\mathbf{d} \in \text{im } A$, let $\mathbf{x}_0$ be a particular solution. For any $\mathbf{y} \in \ker A$, we have $A(\mathbf{x}_0 + \mathbf{y}) = A\mathbf{x}_0 + A\mathbf{y} = \mathbf{d} + \mathbf{0} = \mathbf{d}$. Conversely, if $\mathbf{x}$ is any solution, then $A(\mathbf{x} - \mathbf{x}_0) = \mathbf{d} - \mathbf{d} = \mathbf{0}$, so $\mathbf{x} - \mathbf{x}_0 \in \ker A$. $\square$

Corollary. In case (ii)(b) above, if $\{\mathbf{u}_1, \dots, \mathbf{u}_{n(A)}\}$ is a basis for $\ker A$, then the general solution can be written as

$$\mathbf{x} = \mathbf{x}_0 + \sum_{j=1}^{n(A)} \mu_j \mathbf{u}_j$$

for arbitrary scalars $\mu_j \in \mathbb{R}$. In particular, there are infinitely many solutions.

Example (A 2×2 system with parameter). Consider the system

$$\begin{pmatrix} 1 & 1 \\ a & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ b \end{pmatrix}$$

We have $\det A = 1 - a$. If $a \neq 1$, then A^{-1} exists and

$$A^{-1} = \frac{1}{1-a} \begin{pmatrix} 1 & -1 \\ -a & 1 \end{pmatrix}.$$

Then

$$\mathbf{x} = \frac{1}{1-a} \begin{pmatrix} 1-b \\ -a+b \end{pmatrix}.$$

If $a = 1$, then

$$A\mathbf{x} = \begin{pmatrix} x_1 + x_2 \\ x_1 + x_2 \end{pmatrix} = (x_1 + x_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

So $\operatorname{im} A = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ and $\ker A = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$. If $b \neq 1$, then $\begin{pmatrix} 1 \\ b \end{pmatrix} \notin \operatorname{im} A$ and there is no solution. If $b = 1$, then $\begin{pmatrix} 1 \\ b \end{pmatrix} \in \operatorname{im} A$.

A particular solution is $\mathbf{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, so the general solution is

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Example (A 3×3 system with parameters). Consider the system

$$\begin{pmatrix} a & a & b \\ b & a & a \\ a & b & a \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ c \\ 1 \end{pmatrix}$$

We have $\det A = (a-b)^2(2a+b)$. If $a \neq b$ and $b \neq -2a$, then the inverse exists and there is a unique solution for any c . Otherwise, the possible cases are

(i) $a = b, b \neq -2a$. So $a \neq 0$. The kernel is the plane $x + y + z = 0$

which is $\operatorname{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$. We extend this basis to $\mathbb{R}^3$ by

adding $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

So the image is the span of $\begin{pmatrix} a \\ a \\ a \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Hence if $c \neq 1$, then

$\begin{pmatrix} 1 \\ c \\ 1 \end{pmatrix}$ is not in the image and there is no solution. If $c = 1$, then a

particular solution is $\begin{pmatrix} \frac{1}{a} \\ 0 \\ 0 \end{pmatrix}$ and the general solution is

$$\mathbf{x} = \begin{pmatrix} \frac{1}{a} \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

(ii) If $a \neq b$ and $b = -2a$, then $a \neq 0$. The kernel satisfies

$$\begin{aligned} x + y - 2z &= 0 \\ -2x + y + z &= 0 \\ x - 2y + z &= 0 \end{aligned}$$

This can be solved to give $x = y = z$, and the kernel is $\text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$. We add $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ to form a basis of $\mathbb{R}^3$.

So the image is the span of $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$.

If $\begin{pmatrix} 1 \\ c \\ 1 \end{pmatrix}$ is in the image, then

$$\begin{pmatrix} 1 \\ c \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}.$$

Then the only solution is $\mu = 0, \lambda = 1, c = -2$. Thus there is no solution if $c \neq -2$, and when $c = -2$, pick a particular solution

$\begin{pmatrix} \frac{1}{a} \\ 0 \\ 0 \end{pmatrix}$ and the general solution is

$$\mathbf{x} = \begin{pmatrix} \frac{1}{a} \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

(iii) If $a = b$ and $b = -2a$, then $a = b = 0$ and $\ker A = \mathbb{R}^3$. So there is no solution for any c .

5 Eigenvalues and eigenvectors

Remark. The study of eigenvalues and eigenvectors is one of the most important topics in linear algebra, with numerous applications in mathematics, physics, and engineering. Intuitively, given a matrix A representing a linear map, an eigenvector is a nonzero vector $\mathbf{x} \neq \mathbf{0}$ that satisfies $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ called the associated eigenvalue. In other words, the linear map A transforms the eigenvector $\mathbf{x}$ into a multiple of itself: the direction of $\mathbf{x}$ is preserved, only its length is modified by the factor λ .

In this section, we will study the properties of eigenvectors and eigenvalues, and see their fundamental importance in the diagonalization of matrices.

5.1 Preliminaries and definitions

Theorem (Fundamental theorem of algebra). Let $p(z)$ be a polynomial of degree $m \geq 1$, i.e.

$$p(z) = \sum_{j=0}^m c_j z^j,$$

where $c_j \in \mathbb{C}$ and $c_m \neq 0$.

Then $p(z) = 0$ has precisely m (not necessarily distinct) roots in the complex plane, accounting for multiplicity.

Remark. The phrase “accounting for multiplicity” means that repeated roots are counted multiple times. For example, $x^2 - 2x + 1 = 0$ has only one distinct root, 1, but this root has multiplicity 2 and is thus counted twice.

Definition (Multiplicity of root). The root $z = \omega$ has *multiplicity* k if $(z - \omega)^k$ is a factor of $p(z)$ but $(z - \omega)^{k+1}$ is not.

Example (Roots with multiplicity). Let $p(z) = z^3 - z^2 - z + 1 = (z - 1)^2(z + 1)$. Then $p(z) = 0$ has roots 1, 1, -1, where $z = 1$ has multiplicity 2 and $z = -1$ has multiplicity 1.

Definition (Eigenvector and eigenvalue). Let $\alpha : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear map with associated matrix A . Then $\mathbf{x} \neq \mathbf{0}$ is an *eigenvector* of A if

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some λ . λ is the associated *eigenvalue*. This means that the direction of the eigenvector is preserved by the mapping, but is scaled up by λ .

Theorem (Eigenvalue determinant condition). Let A be an $n \times n$ matrix. Then λ is an eigenvalue of A if and only if

$$\det(A - \lambda I) = 0.$$

Proof. ($\Rightarrow$) Suppose λ is an eigenvalue with associated eigenvector $\mathbf{x} \neq \mathbf{0}$. From $A\mathbf{x} = \lambda\mathbf{x}$, we obtain $(A - \lambda I)\mathbf{x} = \mathbf{0}$, so $\mathbf{x} \in \ker(A - \lambda I)$. Since $\mathbf{x} \neq \mathbf{0}$, the kernel is non-trivial, hence $\det(A - \lambda I) = 0$.

($\Leftarrow$) Conversely, if $\det(A - \lambda I) = 0$, then $\ker(A - \lambda I)$ is non-trivial, so there exists $\mathbf{x} \neq \mathbf{0}$ with $(A - \lambda I)\mathbf{x} = \mathbf{0}$, i.e., $A\mathbf{x} = \lambda\mathbf{x}$. $\square$

Definition (Characteristic equation and polynomial). Let A be an $n \times n$ matrix. The *characteristic polynomial* of A is

$$p_A(\lambda) = \det(A - \lambda I),$$

and the *characteristic equation* of A is $p_A(\lambda) = 0$, i.e., $\det(A - \lambda I) = 0$.

Proposition (Properties of characteristic polynomial). Let A be an $n \times n$ matrix with characteristic polynomial $p_A(\lambda) = \det(A - \lambda I)$ and eigenvalues $\lambda_1, \dots, \lambda_n$ (accounting for multiplicity). Then:

- (i) $p_A(\lambda)$ has degree n and has n roots. So an $n \times n$ matrix has n eigenvalues (accounting for multiplicity).
- (ii) If A is real, then eigenvalues are either real or come in complex conjugate pairs.
- (iii) The trace and determinant satisfy

$$\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$$

and

$$\det A = \lambda_1 \lambda_2 \cdots \lambda_n.$$

Proof. From the definition of the determinant,

$$\begin{aligned} p_A(\lambda) &= \det(A - \lambda I) \\ &= \varepsilon_{j_1 j_2 \dots j_n} (A_{j_1 1} - \lambda \delta_{j_1 1}) \cdots (A_{j_n n} - \lambda \delta_{j_n n}) \\ &= c_0 + c_1 \lambda + \dots + c_n \lambda^n \end{aligned}$$

for some constants $c_0, \dots, c_n$.

(1) follows immediately from the fundamental theorem of algebra.

(2) If A is real, then all $c_i \in \mathbb{R}$, so complex roots must come in conjugate pairs.

(3) From the expansion, $c_n = (-1)^n$ and $c_{n-1} = (-1)^{n-1}(A_{11} + A_{22} + \dots + A_{nn}) = (-1)^{n-1} \operatorname{tr}(A)$. But c_{n-1} is also the sum of roots, i.e. $c_{n-1} = (-1)^{n-1}(\lambda_1 + \lambda_2 + \dots + \lambda_n)$. Therefore $\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$.

Finally, $c_0 = p_A(0) = \det(A)$. Also c_0 is the product of all roots, i.e. $c_0 = \lambda_1 \lambda_2 \cdots \lambda_n$. Therefore $\det A = \lambda_1 \lambda_2 \cdots \lambda_n$. $\square$

Proposition (Eigenspace is a subspace). For any eigenvalue λ of a matrix A , the kernel of $A - \lambda I$ is a vector subspace.

Proof. The kernel of the matrix $A - \lambda I$ is the set $\{\mathbf{x} : A\mathbf{x} = \lambda\mathbf{x}\}$. This is a vector subspace because the kernel of any linear map is always a subspace. $\square$

Definition (Eigenspace). The *eigenspace* denoted by E_λ is the kernel of the matrix $A - \lambda I$, i.e. the set of all vectors $\mathbf{v}$ such that $A\mathbf{v} = \lambda\mathbf{v}$ (consisting of the eigenvectors with eigenvalue λ together with the zero vector).

Definition (Algebraic multiplicity of eigenvalue). The *algebraic multiplicity* $M(\lambda)$ or M_λ of an eigenvalue λ is the multiplicity of λ in $p_A(\lambda) = 0$. By the fundamental theorem of algebra,

$$\sum_{\lambda} M(\lambda) = n.$$

If $M(\lambda) > 1$, then the eigenvalue is *degenerate*.

Definition (Geometric multiplicity of eigenvalue). The *geometric multiplicity* $m(\lambda)$ or m_λ of an eigenvalue λ is the dimension of the eigenspace, i.e. the maximum number of linearly independent eigenvectors with eigenvalue λ .

Definition (Defect of eigenvalue). The *defect* Δ_λ of eigenvalue λ is

$$\Delta_\lambda = M(\lambda) - m(\lambda).$$

It can be proven that $\Delta_\lambda \geq 0$, i.e. the geometric multiplicity is never greater than the algebraic multiplicity.

5.2 Linearly independent eigenvectors

Theorem (Distinct eigenvalues give independent eigenvectors). Suppose $n \times n$ matrix A has *distinct* eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Then the corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are linearly independent.

Proof. Proof by contradiction: Suppose $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are linearly dependent. Then we can find non-zero constants d_i for $i = 1, 2, \dots, r$, such that

$$d_1\mathbf{x}_1 + d_2\mathbf{x}_2 + \dots + d_r\mathbf{x}_r = \mathbf{0}.$$

Suppose that this is the shortest non-trivial linear combination that gives $\mathbf{0}$ (we may need to re-order $\mathbf{x}_i$).

Now apply $(A - \lambda_1 I)$ to the whole equation to obtain

$$d_1(\lambda_1 - \lambda_1)\mathbf{x}_1 + d_2(\lambda_2 - \lambda_1)\mathbf{x}_2 + \cdots + d_r(\lambda_r - \lambda_1)\mathbf{x}_r = \mathbf{0}.$$

We know that the first term is $\mathbf{0}$, while the others are not (since we assumed $\lambda_i \neq \lambda_j$ for $i \neq j$). So

$$d_2(\lambda_2 - \lambda_1)\mathbf{x}_2 + \cdots + d_r(\lambda_r - \lambda_1)\mathbf{x}_r = \mathbf{0},$$

and we have found a shorter linear combination that gives $\mathbf{0}$. Contradiction. $\square$

Example.

(i) $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then $p_A(\lambda) = \lambda^2 + 1 = 0$. So $\lambda_1 = i$ and $\lambda_2 = -i$.

To solve $(A - \lambda_1 I)\mathbf{x} = \mathbf{0}$, we obtain

$$\begin{pmatrix} -i & 1 \\ -1 & -i \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \mathbf{0}.$$

So we obtain

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

to be an eigenvector. Clearly any scalar multiple of $\begin{pmatrix} 1 \\ i \end{pmatrix}$ is also a solution, but still in the same eigenspace $E_i = \text{span} \begin{pmatrix} 1 \\ i \end{pmatrix}$

Solving $(A - \lambda_2 I)\mathbf{x} = \mathbf{0}$ gives

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

So $E_{-i} = \text{span} \begin{pmatrix} 1 \\ -i \end{pmatrix}$.

Note that $M(\pm i) = m(\pm i) = 1$, so $\Delta_{\pm i} = 0$. Also note that the two eigenvectors are linearly independent and form a basis of $\mathbb{C}^2$.

(ii) Consider

$$A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}$$

Then $\det(A - \lambda I) = 0$ gives $45 + 21\lambda - \lambda^2 - \lambda^3$. So $\lambda_1 = 5, \lambda_2 = \lambda_3 = -3$.

The eigenvector with eigenvalue 5 is

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

We can find that the eigenvectors with eigenvalue -3 are

$$\mathbf{x} = \begin{pmatrix} -2x_2 + 3x_3 \\ x_2 \\ x_3 \end{pmatrix}$$

for any x_2, x_3 . This gives two linearly independent eigenvectors,

$$\text{say } \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}.$$

So $M(5) = m(5) = 1$ and $M(-3) = m(-3) = 2$, and there is no defect for both of them. Note that these three eigenvectors form a basis of $\mathbb{C}^3$.

(iii) Let

$$A = \begin{pmatrix} -3 & -1 & 1 \\ -1 & -3 & 1 \\ -2 & -2 & 0 \end{pmatrix}$$

Then $0 = p_A(\lambda) = -(\lambda + 2)^3$. So $\lambda = -2, -2, -2$. To find the eigenvectors, we have

$$(A + 2I)\mathbf{x} = \begin{pmatrix} -1 & -1 & 1 \\ -1 & -1 & 1 \\ -2 & -2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \mathbf{0}$$

The general solution is thus $x_1 + x_2 - x_3 = 0$, and the gen-

eral solution is thus $x = \begin{pmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{pmatrix}$. The eigenspace $E_{-2} =$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Hence $M(-2) = 3$ and $m(-2) = 2$. Thus the defect $\Delta_{-2} = 1$. So the eigenvectors do not form a basis of $\mathbb{C}^3$.

(iv) Consider the reflection R in the plane with normal $\mathbf{n}$. Clearly $R\mathbf{n} = -\mathbf{n}$. The eigenvalue is -1 and the eigenvector is $\mathbf{n}$. Then $E_{-1} = \text{span}\{\mathbf{n}\}$. So $M(-1) = m(-1) = 1$.

If $\mathbf{p}$ is any vector in the plane, $R\mathbf{p} = \mathbf{p}$. So this has an eigenvalue of 1 and eigenvectors being any vector in the plane. So $M(1) = m(1) = 2$.

So the eigenvectors form a basis of $\mathbb{R}^3$.

(v) Consider a rotation R by θ about $\mathbf{n}$. Since $R\mathbf{n} = \mathbf{n}$, we have an eigenvalue of 1 and eigenspace $E_1 = \text{span}\{\mathbf{n}\}$.

We know that there are no other real eigenvalues since rotation changes the direction of any other vector. The other eigenvalues turn out to be $e^{\pm i\theta}$. If $\theta \neq 0$, there are 3 distinct eigenvalues and the eigenvectors form a basis of $\mathbb{C}^3$.

(vi) Consider a shear

$$A = \begin{pmatrix} 1 & \mu \\ 0 & 1 \end{pmatrix}$$

The characteristic equation is $(1 - \lambda)^2 = 0$ and $\lambda = 1$. The eigenvectors corresponding to $\lambda = 1$ is $\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. We have $M(1) = 2$ and $m(1) = 1$. So $\Delta_1 = 1$.

Proposition (Diagonal form in eigenvector basis). If an $n \times n$ matrix A has n distinct eigenvalues $\lambda_1, \dots, \lambda_n$, and hence has n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, then with respect to this eigenvector basis, A is diagonal:

$$A = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Proof. In this eigenvector basis, $\mathbf{v}_1 = (1, 0, \dots, 0)$ etc. We know that $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ (no summation). So the image of the i th basis vector is λ_i times the i th basis vector. Since the columns of A are simply the images of the basis vectors, the result follows. $\square$

The fact that A can be diagonalized by changing the basis is an important observation. We will now look at how we can change bases and see how we can make use of this.

5.3 Transformation matrices

In this section, we study how the components of vectors and matrices change when we change the basis. Throughout, let V denote either $\mathbb{R}^n$ or $\mathbb{C}^n$.

Definition (Transformation matrix). Let $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ and $\tilde{\mathcal{B}} = \{\tilde{\mathbf{e}}_1, \tilde{\mathbf{e}}_2, \dots, \tilde{\mathbf{e}}_n\}$ be two bases of V . The *transformation matrix* (or *change of basis matrix*) from $\mathcal{B}$ to $\tilde{\mathcal{B}}$ is the $n \times n$ matrix P defined by

$$\tilde{\mathbf{e}}_j = \sum_{i=1}^n P_{ij} \mathbf{e}_i$$

for $j = 1, 2, \dots, n$. That is, P_{ij} is the i th component of $\tilde{\mathbf{e}}_j$ with respect to the basis $\mathcal{B}$.

Remark. The columns of P are the vectors $\tilde{\mathbf{e}}_j$ expressed in terms of $\mathcal{B}$. Thus we may write $P = (\tilde{\mathbf{e}}_1 \ \tilde{\mathbf{e}}_2 \ \cdots \ \tilde{\mathbf{e}}_n)$, and P satisfies $P(\mathbf{e}_i) = \tilde{\mathbf{e}}_i$ for each i .

Remark. Note that the sum in the definition is $\sum_i P_{ij} \mathbf{e}_i$ (summing over the first index), not $\sum_j P_{ij} \mathbf{e}_j$. This convention differs from the standard matrix multiplication formula.

Proposition (Inverse of transformation matrix). Let P be the transformation matrix from basis $\mathcal{B}$ to basis $\tilde{\mathcal{B}}$, and let Q be the transformation matrix from $\tilde{\mathcal{B}}$ to $\mathcal{B}$. Then $Q = P^{-1}$.

Proof. By definition, we have

$$\mathbf{e}_i = \sum_{k=1}^n Q_{ki} \tilde{\mathbf{e}}_k$$

for each i . Substituting the expression $\tilde{\mathbf{e}}_j = \sum_{i=1}^n P_{ij} \mathbf{e}_i$ into the above, we obtain

$$\tilde{\mathbf{e}}_j = \sum_{i=1}^n P_{ij} \mathbf{e}_i = \sum_{i=1}^n P_{ij} \left(\sum_{k=1}^n Q_{ki} \tilde{\mathbf{e}}_k \right) = \sum_{k=1}^n \tilde{\mathbf{e}}_k \left(\sum_{i=1}^n Q_{ki} P_{ij} \right).$$

Since $\tilde{\mathbf{e}}_1, \tilde{\mathbf{e}}_2, \dots, \tilde{\mathbf{e}}_n$ are linearly independent, we must have

$$\sum_{i=1}^n Q_{ki} P_{ij} = \delta_{kj},$$

which is equivalent to $QP = I$. Hence $Q = P^{-1}$. $\square$

5.3.1 Transformation law for vectors

Definition (Component vectors). Let $\mathbf{u} \in V$ be a vector. With respect to basis $\mathcal{B} = \{\mathbf{e}_i\}$, we write $\mathbf{u} = \sum_{i=1}^n u_i \mathbf{e}_i$ and denote the column vector of components by $\mathbf{u} = (u_1, u_2, \dots, u_n)^\top$. With respect to basis $\tilde{\mathcal{B}} = \{\tilde{\mathbf{e}}_i\}$, we write $\mathbf{u} = \sum_{i=1}^n \tilde{u}_i \tilde{\mathbf{e}}_i$ and denote the column vector of components by $\tilde{\mathbf{u}} = (\tilde{u}_1, \tilde{u}_2, \dots, \tilde{u}_n)^\top$. Note that $\mathbf{u}$ and $\tilde{\mathbf{u}}$ represent the *same* vector but with components expressed in different bases.

Theorem (Basis transformation of vectors). Let P be the transformation matrix from basis $\mathcal{B}$ to basis $\tilde{\mathcal{B}}$. Let $\mathbf{u}$ and $\tilde{\mathbf{u}}$ denote the component vectors of a vector with respect to $\mathcal{B}$ and $\tilde{\mathcal{B}}$, respectively. Then

$$\mathbf{u} = P\tilde{\mathbf{u}} \quad \text{and} \quad \tilde{\mathbf{u}} = P^{-1}\mathbf{u}.$$

Proof. Using the definition of the transformation matrix, we have

$$\mathbf{u} = \sum_{j=1}^n \tilde{u}_j \tilde{\mathbf{e}}_j = \sum_{j=1}^n \tilde{u}_j \sum_{i=1}^n P_{ij} \mathbf{e}_i = \sum_{i=1}^n \left(\sum_{j=1}^n P_{ij} \tilde{u}_j \right) \mathbf{e}_i.$$

Comparing with $\mathbf{u} = \sum_{i=1}^n u_i \mathbf{e}_i$, we find $u_i = \sum_{j=1}^n P_{ij} \tilde{u}_j$, which in matrix form is $\mathbf{u} = P\tilde{\mathbf{u}}$. The second equation follows by multiplying both sides by P^{-1} . $\square$

Example. Let $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2\}$ be the standard basis of $\mathbb{R}^2$, where $\mathbf{e}_1 = (1, 0)^\top$ and $\mathbf{e}_2 = (0, 1)^\top$. Let $\tilde{\mathcal{B}} = \{\tilde{\mathbf{e}}_1, \tilde{\mathbf{e}}_2\}$ with $\tilde{\mathbf{e}}_1 = (1, 1)^\top$ and $\tilde{\mathbf{e}}_2 = (-1, 1)^\top$.

Since $\tilde{\mathbf{e}}_1 = \mathbf{e}_1 + \mathbf{e}_2$ and $\tilde{\mathbf{e}}_2 = -\mathbf{e}_1 + \mathbf{e}_2$, the transformation matrix is

$$P = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

For an arbitrary vector $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$, we can compute the components in $\tilde{\mathcal{B}}$ directly:

$$\begin{aligned} \mathbf{u} &= u_1\mathbf{e}_1 + u_2\mathbf{e}_2 = u_1 \cdot \frac{1}{2}(\tilde{\mathbf{e}}_1 - \tilde{\mathbf{e}}_2) + u_2 \cdot \frac{1}{2}(\tilde{\mathbf{e}}_1 + \tilde{\mathbf{e}}_2) \\ &= \frac{1}{2}(u_1 + u_2)\tilde{\mathbf{e}}_1 + \frac{1}{2}(-u_1 + u_2)\tilde{\mathbf{e}}_2. \end{aligned}$$

Alternatively, using Theorem 5.3.1:

$$\tilde{\mathbf{u}} = P^{-1}\mathbf{u} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(u_1 + u_2) \\ \frac{1}{2}(-u_1 + u_2) \end{pmatrix},$$

which agrees with the direct computation.

5.3.2 Transformation law for matrices

Theorem (Basis transformation of matrices). Let $\alpha : V \rightarrow V$ be a linear map with associated matrix A with respect to basis $\mathcal{B}$, and let $\tilde{A}$ be the matrix of α with respect to basis $\tilde{\mathcal{B}}$. If P is the transformation matrix from $\mathcal{B}$ to $\tilde{\mathcal{B}}$, then

$$\tilde{A} = P^{-1}AP.$$

Proof. Let $\mathbf{u}' = \alpha(\mathbf{u}) = A\mathbf{u}$, where $\mathbf{u}$ and $\mathbf{u}'$ denote component vectors with respect to $\mathcal{B}$. Let $\tilde{\mathbf{u}}$ and $\tilde{\mathbf{u}}'$ denote the component vectors with respect to $\tilde{\mathcal{B}}$. By Theorem 5.3.1, we have $\mathbf{u} = P\tilde{\mathbf{u}}$ and $\mathbf{u}' = P\tilde{\mathbf{u}}'$. Substituting:

$$P\tilde{\mathbf{u}}' = AP\tilde{\mathbf{u}} \implies \tilde{\mathbf{u}}' = P^{-1}AP\tilde{\mathbf{u}} = \tilde{A}\tilde{\mathbf{u}}.$$

Hence $\tilde{A} = P^{-1}AP$. □

Example. Consider the shear transformation on $\mathbb{R}^3$ given by the matrix

$$S_\lambda = \begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with respect to the standard basis $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. Define a new basis $\tilde{\mathcal{B}}$ by rotating by angle θ about the $\mathbf{e}_3$ -axis:

$$\tilde{\mathbf{e}}_1 = \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2, \quad \tilde{\mathbf{e}}_2 = -\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2, \quad \tilde{\mathbf{e}}_3 = \mathbf{e}_3.$$

The transformation matrix and its inverse are

$$P = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

By Theorem 5.3.2, the shear matrix in the new basis is

$$\tilde{S}_\lambda = P^{-1} S_\lambda P = \begin{pmatrix} 1 + \lambda \sin \theta \cos \theta & \lambda \cos^2 \theta & 0 \\ -\lambda \sin^2 \theta & 1 - \lambda \sin \theta \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This is considerably more complicated than the original matrix S_λ , illustrating that the choice of basis significantly affects the simplicity of a matrix representation.

Remark. The transformation law generalizes to linear maps between different spaces. Let $\alpha : \mathbb{C}^m \rightarrow \mathbb{C}^n$ be a linear map with associated $n \times m$ matrix A . Suppose $\mathbb{C}^m$ has bases $\mathcal{B} = \{\mathbf{e}_i\}$ and $\tilde{\mathcal{B}} = \{\tilde{\mathbf{e}}_i\}$ with transformation matrix P (an $m \times m$ matrix), and $\mathbb{C}^n$ has bases $\mathcal{F} = \{\mathbf{f}_i\}$ and $\tilde{\mathcal{F}} = \{\tilde{\mathbf{f}}_i\}$ with transformation matrix R (an $n \times n$ matrix). Then the matrix $\tilde{A}$ of α with respect to bases $\tilde{\mathcal{B}}$ and $\tilde{\mathcal{F}}$ is given by

$$\tilde{A} = R^{-1}AP.$$

Derivation. Let $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{x}' = A\mathbf{x} \in \mathbb{C}^n$ (with components in bases $\mathcal{B}$ and $\mathcal{F}$). By Theorem 5.3.1, we have $\mathbf{x} = P\tilde{\mathbf{x}}$ and $\mathbf{x}' = R\tilde{\mathbf{x}}'$. Substituting:

$$R\tilde{\mathbf{x}}' = AP\tilde{\mathbf{x}} \implies \tilde{\mathbf{x}}' = R^{-1}AP\tilde{\mathbf{x}}.$$

Hence $\tilde{A} = R^{-1}AP$. □

Example. Consider the linear map $\alpha : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with matrix

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 6 & 3 \end{pmatrix}$$

with respect to the standard bases $\mathcal{B}$ in $\mathbb{R}^3$ and $\mathcal{F}$ in $\mathbb{R}^2$. Keep the standard basis in $\mathbb{R}^3$ (so $P = I$) and change to the basis $\tilde{\mathcal{F}} = \{\tilde{\mathbf{f}}_1, \tilde{\mathbf{f}}_2\}$ in $\mathbb{R}^2$, where

$$\tilde{\mathbf{f}}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \tilde{\mathbf{f}}_2 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

The transformation matrix for $\mathbb{R}^2$ and its inverse are

$$R = \begin{pmatrix} 2 & 1 \\ 1 & 5 \end{pmatrix}, \quad R^{-1} = \frac{1}{9} \begin{pmatrix} 5 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore,

$$\tilde{A} = R^{-1}AP = \frac{1}{9} \begin{pmatrix} 5 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 3 & 4 \\ 1 & 6 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 17/9 \\ 0 & 1 & 2/9 \end{pmatrix}.$$

Alternatively, we can verify this by computing where the basis vectors map. Since $\tilde{\mathbf{e}}_i = \mathbf{e}_i$:

$$\begin{aligned}\tilde{\mathbf{e}}_1 = \mathbf{e}_1 &\mapsto 2\mathbf{f}_1 + \mathbf{f}_2 = \tilde{\mathbf{f}}_1, \\ \tilde{\mathbf{e}}_2 = \mathbf{e}_2 &\mapsto 3\mathbf{f}_1 + 6\mathbf{f}_2 = \tilde{\mathbf{f}}_1 + \tilde{\mathbf{f}}_2, \\ \tilde{\mathbf{e}}_3 = \mathbf{e}_3 &\mapsto 4\mathbf{f}_1 + 3\mathbf{f}_2 = \frac{17}{9}\tilde{\mathbf{f}}_1 + \frac{2}{9}\tilde{\mathbf{f}}_2,\end{aligned}$$

from which we can construct $\tilde{A}$ column by column.

5.4 Similar matrices

The transformation law for matrices (Theorem 5.3.2) motivates the following important equivalence relation on matrices.

Definition (Similar matrices). Two $n \times n$ matrices A and B are *similar* if there exists an invertible matrix P such that

$$B = P^{-1}AP.$$

Equivalently, A and B are similar if and only if they represent the same linear map with respect to different bases.

Remark. Similarity is an equivalence relation on the set of $n \times n$ matrices. In the language of group theory, similar matrices lie in the same conjugacy class under the action of the general linear group.

Proposition (Invariants of similar matrices). Let A and B be similar $n \times n$ matrices. Then:

- (i) $\det A = \det B$;
- (ii) $\operatorname{tr} A = \operatorname{tr} B$;
- (iii) $p_A(\lambda) = p_B(\lambda)$, where p_A and p_B denote the characteristic polynomials.

Remark. Since the determinant and trace appear as coefficients of the characteristic polynomial, property (iii) implies properties (i) and (ii). Nevertheless, we provide direct proofs of all three.

Proof. Suppose $B = P^{-1}AP$ for some invertible matrix P .

- (i) Using the multiplicativity of the determinant:

$$\det B = \det(P^{-1}AP) = (\det P)^{-1}(\det A)(\det P) = \det A.$$

(ii) Using index notation with the Einstein summation convention:

$$\begin{aligned}\operatorname{tr} B = B_{ii} &= (P^{-1})_{ij} A_{jk} P_{ki} = A_{jk} P_{ki} (P^{-1})_{ij} \\ &= A_{jk} (PP^{-1})_{kj} = A_{jk} \delta_{kj} = A_{jj} = \operatorname{tr} A.\end{aligned}$$

(iii) For the characteristic polynomial:

$$\begin{aligned}p_B(\lambda) &= \det(B - \lambda I) = \det(P^{-1}AP - \lambda I) \\ &= \det(P^{-1}AP - \lambda P^{-1}IP) = \det(P^{-1}(A - \lambda I)P) \\ &= \det(A - \lambda I) = p_A(\lambda).\end{aligned}\quad \square$$

5.5 Diagonalizable matrices

Definition (Diagonalizable matrix). An $n \times n$ matrix A is *diagonalizable* if it is similar to a diagonal matrix, i.e., if there exists an invertible matrix P such that $P^{-1}AP$ is diagonal.

Proposition (Eigenvector characterization of diagonalizability). An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors, i.e., if and only if the eigenvectors of A form a basis of $\mathbb{C}^n$.

Proof. If A has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$, let $P = (\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n)$. Then $P^{-1}AP = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Conversely, if $P^{-1}AP = D$ is diagonal, then the columns of P are eigenvectors of A . $\square$

Remark. Having n distinct eigenvalues is a *sufficient* condition for diagonalizability (since eigenvectors corresponding to distinct eigenvalues are linearly independent), but it is *not necessary*. A matrix with repeated eigenvalues may still be diagonalizable if the eigenspaces are sufficiently large.

Example. Consider the matrix

$$A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}.$$

This matrix has eigenvalues $\lambda_1 = 5$ and $\lambda_2 = \lambda_3 = -3$ (a repeated eigenvalue). The corresponding eigenvectors are

$$\tilde{\mathbf{e}}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \tilde{\mathbf{e}}_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad \tilde{\mathbf{e}}_3 = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}.$$

Forming P with these eigenvectors as columns:

$$P = \begin{pmatrix} 1 & -2 & 3 \\ 2 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \frac{1}{8} \begin{pmatrix} 1 & 2 & -3 \\ -2 & 4 & 6 \\ 1 & 2 & 5 \end{pmatrix}.$$

Then

$$P^{-1}AP = \begin{pmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix},$$

confirming that A is diagonalizable despite having a repeated eigenvalue.

Theorem (Linear independence of eigenspace bases). Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_r$ (where $r \leq n$). For each i , let $\mathcal{B}_i$ be a basis for the eigenspace E_{λ_i} . Then the union $\mathcal{B} = \bigcup_{i=1}^r \mathcal{B}_i$ is a linearly independent set.

Remark. This theorem generalizes the result that eigenvectors corresponding to distinct eigenvalues are linearly independent. The key technique in the proof is to apply a carefully chosen polynomial in A to isolate the contribution from a single eigenspace.

Proof. For each i , write $\mathcal{B}_i = \{\mathbf{x}_1^{(i)}, \mathbf{x}_2^{(i)}, \dots, \mathbf{x}_{m_i}^{(i)}\}$, where $m_i = \dim E_{\lambda_i}$. Consider a linear combination

$$\sum_{i=1}^r \sum_{j=1}^{m_i} \alpha_{ij} \mathbf{x}_j^{(i)} = \mathbf{0}.$$

For any fixed $K \in \{1, \dots, r\}$, apply the matrix $\prod_{k \neq K} (A - \lambda_k I)$ to both sides. Since $(A - \lambda_k I)\mathbf{x} = (\lambda_i - \lambda_k)\mathbf{x}$ for any $\mathbf{x} \in E_{\lambda_i}$, this annihilates all terms with $i \neq K$ and yields

$$\sum_{j=1}^{m_K} \alpha_{Kj} \left[\prod_{k \neq K} (\lambda_K - \lambda_k) \right] \mathbf{x}_j^{(K)} = \mathbf{0}.$$

The scalar $\prod_{k \neq K} (\lambda_K - \lambda_k)$ is nonzero since the eigenvalues are distinct. Since $\mathcal{B}_K$ is a basis (hence linearly independent), we conclude $\alpha_{Kj} = 0$ for all j . As K was arbitrary, all coefficients α_{ij} vanish, proving that $\mathcal{B}$ is linearly independent. $\square$

Theorem (Diagonalizability criterion). An $n \times n$ matrix A is diagonalizable if and only if every eigenvalue has zero defect, i.e., the geometric multiplicity equals the algebraic multiplicity for each eigenvalue.

Proof. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues with algebraic multiplicities a_i and geometric multiplicities $g_i = \dim E_{\lambda_i}$.

By Theorem 5.5, the union of bases for all eigenspaces is linearly independent, giving at most $\sum_{i=1}^r g_i$ linearly independent eigenvectors. Since $g_i \leq a_i$ for each i and $\sum_{i=1}^r a_i = n$, we have $\sum_{i=1}^r g_i \leq n$.

By Proposition 5.5, A is diagonalizable if and only if there are n linearly independent eigenvectors, which occurs if and only if $\sum_{i=1}^r g_i = n$. This happens if and only if $g_i = a_i$ for all i , i.e., every eigenvalue has zero defect. $\square$

5.6 Canonical (Jordan normal) form

By Theorem 5.5, a matrix is diagonalizable if and only if all its eigenvalues have zero defect. When some eigenvalue has nonzero defect, the matrix cannot be diagonalized, but it can still be reduced to a nearly diagonal form called the *Jordan normal form*.

Theorem (Jordan normal form for 2×2 matrices). Every 2×2 complex matrix A is similar to exactly one of the following canonical forms:

- (i) $\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ with $\lambda_1 \neq \lambda_2$ (two distinct eigenvalues);
- (ii) $\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \lambda I$ (repeated eigenvalue, zero defect);
- (iii) $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ (repeated eigenvalue, defect one).

Proof. We consider each case based on the eigenvalue structure.

Case 1: Suppose A has two distinct eigenvalues $\lambda_1 \neq \lambda_2$. Then the corresponding eigenvectors are linearly independent. Taking P to have these eigenvectors as columns, we obtain $P^{-1}AP = \text{diag}(\lambda_1, \lambda_2)$.

Case 2: Suppose A has a repeated eigenvalue $\lambda_1 = \lambda_2 = \lambda$ with $\dim E_\lambda = 2$. Then $E_\lambda = \mathbb{C}^2$, so we can choose any basis $\{\mathbf{u}, \mathbf{v}\}$ of eigenvectors. With $P = (\mathbf{u} \ \mathbf{v})$, we have $P^{-1}AP = \lambda I$.

Note that $P^{-1}AP = \lambda I$ implies $A = P(\lambda I)P^{-1} = \lambda I$. Thus A itself equals λI and is unchanged by any similarity transformation.

Case 3: Suppose $\lambda_1 = \lambda_2 = \lambda$ with $\dim E_\lambda = 1$, so $E_\lambda = \text{span}\{\mathbf{v}\}$. Choose any $\mathbf{w} \in \mathbb{C}^2 \setminus E_\lambda$ to form a basis $\{\mathbf{v}, \mathbf{w}\}$ of $\mathbb{C}^2$.

Since $A\mathbf{w} \in \mathbb{C}^2$, we can write $A\mathbf{w} = \alpha\mathbf{v} + \beta\mathbf{w}$ for some scalars α, β . In the basis $\{\mathbf{v}, \mathbf{w}\}$, the matrix becomes

$$\tilde{A} = P^{-1}AP = \begin{pmatrix} \lambda & \alpha \\ 0 & \beta \end{pmatrix}.$$

Since similar matrices have the same characteristic polynomial and A has λ as a repeated eigenvalue, we must have $\beta = \lambda$.

To normalize α to 1, define $\mathbf{u} = (\tilde{A} - \lambda I)\mathbf{w} = \alpha\mathbf{v}$. Since $\mathbf{w} \notin E_\lambda$, we have $\mathbf{u} \neq \mathbf{0}$. Then $\mathbf{u}$ is an eigenvector:

$$(\tilde{A} - \lambda I)\mathbf{u} = (\tilde{A} - \lambda I)^2\mathbf{w} = \begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix}^2 \mathbf{w} = \mathbf{0}.$$

From $\mathbf{u} = \tilde{A}\mathbf{w} - \lambda\mathbf{w}$, we get $\tilde{A}\mathbf{w} = \mathbf{u} + \lambda\mathbf{w}$. In the basis $\{\mathbf{u}, \mathbf{w}\}$, the matrix is

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

The full similarity transformation is achieved by the matrix PQ , where P changes to the basis $\{\mathbf{v}, \mathbf{w}\}$ and Q changes to $\{\mathbf{u}, \mathbf{w}\}$. $\square$

Definition (Jordan block). A *Jordan block* of size k with eigenvalue λ is the $k \times k$ matrix

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & 0 & 0 & \lambda \end{pmatrix}.$$

That is, $J_k(\lambda)$ has λ on the diagonal, 1 on the superdiagonal, and 0 elsewhere.

Theorem (Jordan normal form). Every $n \times n$ complex matrix A is similar to a block diagonal matrix

$$J = \text{diag}(J_{k_1}(\lambda_1), J_{k_2}(\lambda_2), \dots, J_{k_r}(\lambda_r))$$

where each $J_{k_i}(\lambda_i)$ is a Jordan block. The matrix J is called the *Jordan normal form* of A and is unique up to permutation of the blocks.

Remark. The proof of the general Jordan normal form theorem requires more sophisticated techniques and is deferred to the IB Linear Algebra course. However, the 2×2 case (Theorem 5.6) illustrates the key ideas: we construct a basis using eigenvectors and, when needed, *generalized eigenvectors* (vectors $\mathbf{w}$ satisfying $(A - \lambda I)^k \mathbf{w} = \mathbf{0}$ for some $k > 1$).

Example. Consider the matrix

$$A = \begin{pmatrix} -3 & -1 & 1 \\ -1 & -3 & 1 \\ -2 & -2 & 0 \end{pmatrix}.$$

The characteristic polynomial is $p_A(\lambda) = -(\lambda + 2)^3$, so $\lambda = -2$ is the only eigenvalue with algebraic multiplicity 3. Solving $(A + 2I)\mathbf{x} = \mathbf{0}$, we find that $\dim E_{-2} = 2$, with basis

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Since the geometric multiplicity (2) is less than the algebraic multiplicity (3), the matrix is not diagonalizable.

To find the Jordan form, we need a generalized eigenvector. Choose

$\mathbf{w} = (1, 0, 0)^\top \notin E_{-2}$ and compute

$$\mathbf{u} = (A + 2I)\mathbf{w} = \begin{pmatrix} -1 & -1 & 1 \\ -1 & -1 & 1 \\ -2 & -2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ -2 \end{pmatrix}.$$

One can verify that $A\mathbf{u} = -2\mathbf{u}$ (so $\mathbf{u}$ is an eigenvector) and $A\mathbf{w} = \mathbf{u} - 2\mathbf{w}$. Using the basis $\{\mathbf{u}, \mathbf{w}, \mathbf{v}_2\}$ with

$$P = \begin{pmatrix} -1 & 1 & 1 \\ -1 & 0 & 0 \\ -2 & 0 & 1 \end{pmatrix},$$

the Jordan normal form is

$$P^{-1}AP = \begin{pmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} = \text{diag}(J_2(-2), J_1(-2)).$$

This consists of one 2×2 Jordan block and one 1×1 Jordan block (i.e., a single eigenvalue on the diagonal).

5.7 Cayley-Hamilton Theorem

Theorem (Cayley-Hamilton theorem). Every $n \times n$ complex matrix A satisfies its own characteristic equation. That is, if $p_A(\lambda) = \det(A - \lambda I)$ is the characteristic polynomial of A , then $p_A(A) = 0$.

Proof for diagonalizable matrices. We prove the theorem for diagonalizable matrices; the general case is deferred to IB Linear Algebra.

Suppose A is diagonalizable, so there exists an invertible matrix P such that $D = P^{-1}AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . Note that for any positive integer i ,

$$D^i = (P^{-1}AP)^i = P^{-1}A^iP.$$

Therefore, for any polynomial p , we have $p(D) = P^{-1}p(A)P$.

Since similar matrices have the same characteristic polynomial (Proposition 5.4), we have $p_A = p_D$. Thus

$$p_A(D) = P^{-1}p_A(A)P.$$

Now, since D is diagonal, $D^i = \text{diag}(\lambda_1^i, \lambda_2^i, \dots, \lambda_n^i)$, and hence

$$p_A(D) = \text{diag}(p_A(\lambda_1), p_A(\lambda_2), \dots, p_A(\lambda_n)) = 0,$$

because each eigenvalue λ_i is a root of p_A . Therefore $P^{-1}p_A(A)P = 0$, which implies $p_A(A) = 0$. $\square$

Corollary (Computing the inverse via Cayley-Hamilton). Let A be an invertible $n \times n$ matrix with characteristic polynomial $p_A(\lambda) = c_0 + c_1\lambda + c_2\lambda^2 + \cdots + c_n\lambda^n$. Then

$$A^{-1} = -\frac{1}{c_0}(c_1I + c_2A + c_3A^2 + \cdots + c_nA^{n-1}).$$

Proof. By the Cayley-Hamilton theorem, $p_A(A) = c_0I + c_1A + c_2A^2 + \cdots + c_nA^n = 0$. Multiplying both sides by A^{-1} :

$$c_0A^{-1} + c_1I + c_2A + \cdots + c_nA^{n-1} = 0.$$

Since A is invertible, $c_0 = (-1)^n \det A \neq 0$. Solving for A^{-1} gives the result. $\square$

Definition (Matrix exponential). For an $n \times n$ matrix A , the *matrix exponential* is defined by the power series

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots.$$

This series converges for all matrices A .

Proposition (Matrix exponential of diagonalizable matrices). Let A be a diagonalizable matrix with $P^{-1}AP = D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$. Then

$$e^A = P \text{diag}(e^{\lambda_1}, e^{\lambda_2}, \dots, e^{\lambda_n}) P^{-1}.$$

Proof. Using $A^k = PD^kP^{-1}$ for all $k \geq 0$:

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = \sum_{k=0}^{\infty} \frac{PD^kP^{-1}}{k!} = P \left(\sum_{k=0}^{\infty} \frac{D^k}{k!} \right) P^{-1} = Pe^DP^{-1}.$$

Since $D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$, we have $e^D = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})$. $\square$

Example (Cayley-Hamilton for 2×2 Jordan blocks). Consider a 2×2 matrix similar to the Jordan block $B = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. The characteristic polynomial is $p_B(z) = (\lambda - z)^2$. We verify directly:

$$p_B(B) = (\lambda I - B)^2 = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Combined with the proof for diagonalizable matrices and Theorem 5.6, this verifies the Cayley-Hamilton theorem for all 2×2 matrices.

Remark. The proof of the Cayley-Hamilton theorem for general (non-diagonalizable) matrices requires different techniques and will be covered in IB Linear Algebra. One approach uses the Jordan normal form; another uses the adjugate matrix.

5.8 Eigenvalues and eigenvectors of a Hermitian matrix

5.8.1 Eigenvalues and eigenvectors

Theorem (Hermitian matrices have real eigenvalues). Let H be an $n \times n$ Hermitian matrix. Then all eigenvalues of H are real.

Proof. Suppose H has eigenvalue λ with eigenvector $\mathbf{v} \neq \mathbf{0}$, so $H\mathbf{v} = \lambda\mathbf{v}$. Pre-multiplying by $\mathbf{v}^\dagger$ gives

$$\mathbf{v}^\dagger H\mathbf{v} = \lambda \mathbf{v}^\dagger \mathbf{v}. \quad (*)$$

Taking the Hermitian conjugate of (*): the left-hand side becomes

$$(\mathbf{v}^\dagger H\mathbf{v})^\dagger = \mathbf{v}^\dagger H^\dagger \mathbf{v} = \mathbf{v}^\dagger H\mathbf{v}$$

since $H^\dagger = H$, while the right-hand side becomes $(\lambda \mathbf{v}^\dagger \mathbf{v})^\dagger = \lambda^* \mathbf{v}^\dagger \mathbf{v}$. Thus

$$\mathbf{v}^\dagger H\mathbf{v} = \lambda^* \mathbf{v}^\dagger \mathbf{v}.$$

Comparing with (*), we obtain $\lambda \mathbf{v}^\dagger \mathbf{v} = \lambda^* \mathbf{v}^\dagger \mathbf{v}$. Since $\mathbf{v} \neq \mathbf{0}$, we have $\mathbf{v}^\dagger \mathbf{v} = \|\mathbf{v}\|^2 > 0$, so $\lambda = \lambda^*$, i.e., λ is real. $\square$

Theorem (Orthogonality of Hermitian eigenvectors). Let H be a Hermitian matrix. Eigenvectors of H corresponding to distinct eigenvalues are orthogonal.

Proof. Let $\mathbf{v}_i$ and $\mathbf{v}_j$ be eigenvectors with distinct eigenvalues $\lambda_i \neq \lambda_j$:

$$H\mathbf{v}_i = \lambda_i \mathbf{v}_i, \quad H\mathbf{v}_j = \lambda_j \mathbf{v}_j.$$

Pre-multiplying the first equation by $\mathbf{v}_j^\dagger$ gives

$$\mathbf{v}_j^\dagger H\mathbf{v}_i = \lambda_i \mathbf{v}_j^\dagger \mathbf{v}_i. \quad (i)$$

Pre-multiplying the second equation by $\mathbf{v}_i^\dagger$ and taking the Hermitian conjugate:

$$\mathbf{v}_j^\dagger H^\dagger \mathbf{v}_i = \lambda_j^* \mathbf{v}_j^\dagger \mathbf{v}_i.$$

Since H is Hermitian and its eigenvalues are real (Theorem 5.8.1), this becomes

$$\mathbf{v}_j^\dagger H\mathbf{v}_i = \lambda_j \mathbf{v}_j^\dagger \mathbf{v}_i. \quad (ii)$$

Subtracting (ii) from (i): $(\lambda_i - \lambda_j) \mathbf{v}_j^\dagger \mathbf{v}_i = 0$. Since $\lambda_i \neq \lambda_j$, we conclude $\mathbf{v}_j^\dagger \mathbf{v}_i = 0$, i.e., $\mathbf{v}_i$ and $\mathbf{v}_j$ are orthogonal. $\square$

Remark. If a Hermitian matrix has n distinct eigenvalues, then by Theorem 5.8.1 its eigenvectors are mutually orthogonal and hence form an orthogonal basis of $\mathbb{C}^n$ (which can be normalized to an orthonormal basis). When eigenvalues are repeated (degenerate), the eigenvectors within each eigenspace need not be orthogonal, but can be orthogonalized using the Gram-Schmidt process.

5.8.2 Gram-Schmidt orthogonalization (non-examinable)

Definition (Orthogonal projection). The *orthogonal projection* of a vector $\mathbf{w}$ onto a nonzero vector $\mathbf{v}$ is

$$\text{proj}_{\mathbf{v}}(\mathbf{w}) = \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v} = \frac{\mathbf{v}^\dagger \mathbf{w}}{\mathbf{v}^\dagger \mathbf{v}} \mathbf{v}.$$

Proposition (Gram-Schmidt orthogonalization). Let $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_r\}$ be a set of linearly independent vectors. Define vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ iteratively by

$$\begin{aligned} \mathbf{v}_1 &= \mathbf{w}_1, \\ \mathbf{v}_k &= \mathbf{w}_k - \sum_{j=1}^{k-1} \text{proj}_{\mathbf{v}_j}(\mathbf{w}_k) \quad \text{for } k = 2, \dots, r. \end{aligned}$$

Then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ is an orthogonal set spanning the same subspace as $\{\mathbf{w}_1, \dots, \mathbf{w}_r\}$.

Sketch of proof. At each step, we subtract from $\mathbf{w}_k$ its components in the directions of $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$. For orthogonality, note that for $i < k$:

$$\langle \mathbf{v}_i, \mathbf{v}_k \rangle = \langle \mathbf{v}_i, \mathbf{w}_k \rangle - \frac{\langle \mathbf{v}_i, \mathbf{w}_k \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \langle \mathbf{v}_i, \mathbf{v}_i \rangle = 0.$$

Normalizing each $\mathbf{v}_k$ to $\hat{\mathbf{v}}_k = \mathbf{v}_k / \|\mathbf{v}_k\|$ yields an orthonormal set. $\square$

5.8.3 Unitary transformations

Proposition (Orthonormal columns imply unitary). Let U be an $n \times n$ matrix whose columns $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ form an orthonormal set, i.e., $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$. Then U is unitary: $U^\dagger U = I$.

Proof. Writing $U = (\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_n)$, we compute

$$(U^\dagger U)_{ij} = (U^\dagger)_{ik} U_{kj} = U_{ki}^* U_{kj} = (\mathbf{u}_i)_k^* (\mathbf{u}_j)_k = \mathbf{u}_i^\dagger \mathbf{u}_j = \langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}.$$

Thus $U^\dagger U = I$, so U is unitary. $\square$

Remark. Consequently, any change of basis from one orthonormal basis to another is represented by a unitary matrix. For real vector spaces, this specializes to orthogonal matrices.

5.8.4 Diagonalization of $n \times n$ Hermitian matrices

Theorem (Spectral theorem for Hermitian matrices). Every $n \times n$ Hermitian matrix H has n orthonormal eigenvectors. Consequently, H is unitarily diagonalizable: there exists a unitary matrix U such that

$$U^\dagger H U = D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n),$$

where $\lambda_1, \dots, \lambda_n$ are the (real) eigenvalues of H .

Proof (non-examinable). Let $\lambda_1, \lambda_2, \dots, \lambda_r$ be the distinct eigenvalues of H (where $r \leq n$), with corresponding orthonormal eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$. Extend this to a basis of $\mathbb{C}^n$:

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{n-r}\}.$$

Apply Gram-Schmidt (Proposition 5.8.2) to obtain an orthonormal basis

$$\tilde{\mathcal{B}} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{n-r}\}.$$

Form the matrix $P = (\mathbf{v}_1 \cdots \mathbf{v}_r \mathbf{u}_1 \cdots \mathbf{u}_{n-r})$. By Proposition 5.8.3, P is unitary, so $P^{-1} = P^\dagger$. The matrix $P^\dagger H P$ has the block form

$$P^\dagger H P = \begin{pmatrix} \Lambda & 0 \\ 0 & C \end{pmatrix},$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$ and C is an $(n-r) \times (n-r)$ Hermitian matrix. The eigenvalues of C are eigenvalues of H (with remaining multiplicities), since

$$\det(H - \lambda I) = \det(P^\dagger H P - \lambda I) = (\lambda_1 - \lambda) \cdots (\lambda_r - \lambda) \det(C - \lambda I).$$

Repeating this process on C , we eventually obtain n orthonormal eigenvectors. If $\mathbf{w}_{r+1}, \dots, \mathbf{w}_n$ are orthonormal eigenvectors of C , define

$$Q = \begin{pmatrix} I_r & 0 \\ 0 & W \end{pmatrix},$$

where $W = (\mathbf{w}_{r+1} \cdots \mathbf{w}_n)$. Then Q is unitary and $Q^\dagger (P^\dagger H P) Q = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Setting $U = PQ$, which is unitary as a product of unitary matrices, we obtain $U^\dagger H U = D$. The columns of U are the n orthonormal eigenvectors. $\square$

Corollary (Unitary diagonalization of Hermitian matrices). For any Hermitian matrix H , we have $H = U D U^\dagger$, where U is unitary and D is the diagonal matrix of eigenvalues.

Remark. For a real symmetric matrix S (a special case of Hermitian matrices with real entries), the unitary matrix U can be chosen to be a real orthogonal matrix Q . Thus

$$S = QDQ^\top \quad \text{and} \quad D = Q^\top SQ.$$

Example. Let $S = \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix}$ with $\beta \neq 0$ be a real symmetric matrix.

We find an orthogonal matrix that diagonalizes S .

The characteristic equation $\det(S - \lambda I) = (1 - \lambda)^2 - \beta^2 = 0$ gives eigenvalues $\lambda = 1 \pm \beta$.

The corresponding normalized eigenvectors are

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (\lambda = 1 + \beta), \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (\lambda = 1 - \beta).$$

The orthogonal transformation matrix is

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

which represents a rotation by $\pi/4$. Then $S = QDQ^\top$ with $D = \text{diag}(1 + \beta, 1 - \beta)$.

5.8.5 Normal matrices

Definition (Normal matrix). An $n \times n$ matrix N is *normal* if it commutes with its Hermitian conjugate:

$$NN^\dagger = N^\dagger N.$$

Remark. The class of normal matrices includes many important special cases:

- Hermitian matrices ($N^\dagger = N$);
- Real symmetric matrices ($N^\top = N$ with real entries);
- Skew-Hermitian matrices ($N^\dagger = -N$);
- Real skew-symmetric matrices ($N^\top = -N$ with real entries);
- Unitary matrices ($N^\dagger = N^{-1}$);
- Orthogonal matrices ($N^\top = N^{-1}$ with real entries).

Theorem (Spectral theorem for normal matrices). Let N be a normal matrix. Then:

- (i) If λ is an eigenvalue of N , then λ^* is an eigenvalue of $N^\dagger$.
- (ii) Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- (iii) N is unitarily diagonalizable: there exists a unitary matrix U and a diagonal matrix D such that $N = UDU^\dagger$.

Remark. The proof of Theorem 5.8.5 is beyond the scope of this course. The key observation is that normality is precisely the condition needed to generalize the spectral theorem from Hermitian to complex matrices.

6 Quadratic forms and conics

In this section, we study homogeneous polynomials of degree two in several variables, such as $x_1^2 + x_2^2$ or $3x_1^2 + 2x_1x_2 + 4x_2^2$. These arise naturally in the study of conic sections and their higher-dimensional generalizations. Such expressions can be written in matrix form as $\mathbf{x}^\dagger A \mathbf{x}$ for an appropriate matrix A .

Definition (Sesquilinear, Hermitian, and quadratic forms). Let A be an $n \times n$ complex matrix and let $\mathbf{x} \in \mathbb{C}^n$. The expression

$$F(\mathbf{x}) = \mathbf{x}^\dagger A \mathbf{x} = \sum_{i,j=1}^n x_i^* A_{ij} x_j$$

is called a *sesquilinear form*. If A is Hermitian, then F is called a *Hermitian form*. If A is real symmetric and $\mathbf{x} \in \mathbb{R}^n$, then F is called a *quadratic form*.

Proposition (Hermitian forms are real-valued). If H is a Hermitian matrix, then $\mathbf{x}^\dagger H \mathbf{x} \in \mathbb{R}$ for all $\mathbf{x} \in \mathbb{C}^n$.

Proof. We show that $\mathbf{x}^\dagger H \mathbf{x}$ equals its own complex conjugate:

$$(\mathbf{x}^\dagger H \mathbf{x})^* = (\mathbf{x}^\dagger H \mathbf{x})^\dagger = \mathbf{x}^\dagger H^\dagger \mathbf{x} = \mathbf{x}^\dagger H \mathbf{x},$$

where we used $(AB)^\dagger = B^\dagger A^\dagger$ and $H^\dagger = H$. Hence $\mathbf{x}^\dagger H \mathbf{x}$ is real. $\square$

Proposition (Diagonalization of Hermitian forms). Let $F(\mathbf{x}) = \mathbf{x}^\dagger H \mathbf{x}$ be a Hermitian form, where H has eigenvalues $\lambda_1, \dots, \lambda_n$. Then there exists a unitary matrix U such that, writing $\mathbf{x}' = U^\dagger \mathbf{x}$,

$$F(\mathbf{x}) = \sum_{i=1}^n \lambda_i |x'_i|^2.$$

Proof. By the spectral theorem (Theorem 5.8.4), $H = U D U^\dagger$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and U is unitary. Then

$$F(\mathbf{x}) = \mathbf{x}^\dagger H \mathbf{x} = \mathbf{x}^\dagger U D U^\dagger \mathbf{x} = (U^\dagger \mathbf{x})^\dagger D (U^\dagger \mathbf{x}) = (\mathbf{x}')^\dagger D \mathbf{x}' = \sum_{i=1}^n \lambda_i |x'_i|^2. \quad \square$$

Definition (Principal axes). The eigenvectors of the matrix H (equivalently, the columns of U) are called the *principal axes* of the Hermitian form F .

Remark. For a real quadratic form $\mathbf{x}^\top S \mathbf{x}$, we can always assume S is symmetric. If A is any matrix, then $\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top S \mathbf{x}$ where $S = \frac{1}{2}(A + A^\top)$, since $\mathbf{x}^\top A_{\text{skew}} \mathbf{x} = 0$ for any skew-symmetric matrix A_{skew} .

Example. Consider the quadratic form $F = 2x^2 - 4xy + 5y^2$. Writing $\mathbf{x} = (x, y)^\top$, we have $F = \mathbf{x}^\top S \mathbf{x}$ where

$$S = \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}.$$

The eigenvalues of S are $\lambda_1 = 1$ and $\lambda_2 = 6$, with corresponding normalized eigenvectors

$$\mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

The orthogonal matrix $Q = (\mathbf{v}_1 \ \mathbf{v}_2) = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$ diagonalizes S . In the new coordinates $\mathbf{x}' = Q^\top \mathbf{x}$, we have

$$F = (x')^2 + 6(y')^2.$$

Thus the level set $F = c$ (for $c > 0$) is an ellipse with principal axes along the eigenvectors of S .

6.1 Quadrics and conics

6.1.1 Quadrics

Definition (Quadric). A *quadric* (or *quadric hypersurface*) in $\mathbb{R}^n$ is the set of points $\mathbf{x}$ satisfying a general quadratic equation

$$\mathbf{x}^\top A \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c = 0,$$

where A is a real $n \times n$ matrix, $\mathbf{b} \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

Remark. By the observation that $\mathbf{x}^\top A_{\text{skew}} \mathbf{x} = 0$ for any skew-symmetric matrix, we may assume without loss of generality that A is symmetric: replace A by $S = \frac{1}{2}(A + A^\top)$.

Proposition (Standard form of a quadric). Any quadric with invertible quadratic part can be reduced to the standard form

$$\sum_{i=1}^n \lambda_i x_i^2 = k$$

by a rotation (orthogonal transformation) followed by a translation.

Proof. Since S is real symmetric, we can diagonalize it: $S = QDQ^\top$ with Q orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Setting $\mathbf{x}' = Q^\top \mathbf{x}$ and $\mathbf{b}' = Q^\top \mathbf{b}$, the

equation becomes

$$(\mathbf{x}')^\top D \mathbf{x}' + (\mathbf{b}')^\top \mathbf{x}' + c = 0.$$

If S (equivalently D) is invertible, we can eliminate the linear term by the translation $\mathbf{x}'' = \mathbf{x}' + \frac{1}{2}D^{-1}\mathbf{b}'$. This yields

$$(\mathbf{x}'')^\top D \mathbf{x}'' = k$$

for some constant k (absorbing the constant terms). $\square$

6.1.2 Conic sections ($n = 2$)

In two dimensions, the standard form of a quadric is

$$\lambda_1 x_1^2 + \lambda_2 x_2^2 = k.$$

The type of curve depends on the signs of the eigenvalues λ_1, λ_2 .

Proposition (Classification of conics). The non-degenerate conic sections are classified as follows:

- (i) **Ellipse** ($\lambda_1 \lambda_2 > 0$, k has the same sign as λ_i): The axes of the ellipse coincide with the eigenvectors of S .
- (ii) **Hyperbola** ($\lambda_1 \lambda_2 < 0$): Writing $\lambda_1 = k/a^2 > 0$ and $\lambda_2 = -k/b^2 < 0$, we obtain

$$\frac{x_1^2}{a^2} - \frac{x_2^2}{b^2} = 1.$$

- (iii) **Parabola** ($\lambda_1 \lambda_2 = 0$, one eigenvalue zero): The matrix S is singular, so we cannot complete the reduction as above. Suppose $\lambda_2 = 0$ and $\lambda_1 \neq 0$. The equation in rotated coordinates is

$$\lambda_1 (x'_1)^2 + b'_1 x'_1 + b'_2 x'_2 + c = 0.$$

If $b'_2 \neq 0$, completing the square in x'_1 and translating gives

$$\lambda_1 x_1^2 + b_2 x_2 = 0,$$

which is a parabola.

Remark (Degenerate cases). If $\lambda_2 = 0$ and $b'_2 = 0$, the equation $\lambda_1 (x'_1)^2 + b'_1 x'_1 + c = 0$ is a quadratic in x'_1 alone (with x'_2 arbitrary). This yields 0, 1, or 2 parallel lines, depending on the discriminant.

Remark. The conic sections arise as intersections of a plane with a double cone, hence the name. They also describe the trajectories of objects under an inverse-square gravitational force, as studied in IA Dynamics and Relativity.

6.2 Focus-directrix property

Conic sections admit an alternative geometric definition in terms of a focus and directrix.

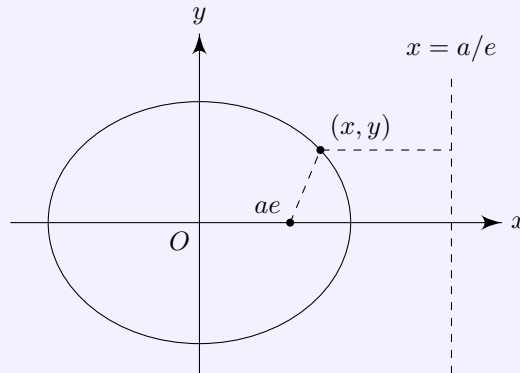
Definition (Focus, directrix, and eccentricity). Let $e \geq 0$ be a non-negative real number called the *eccentricity*, and let $a > 0$ be a positive real number called the *scale parameter*. A *conic section* with eccentricity e and scale a is the locus of points P such that

$$\frac{\text{distance from } P \text{ to focus}}{\text{distance from } P \text{ to directrix}} = e.$$

The *foci* are located at $(\pm ae, 0)$ and the *directrices* are the lines $x = \pm a/e$ (for $e \neq 0$).

Proposition (Focus-directrix characterization of conics). The conic sections are classified by eccentricity as follows:

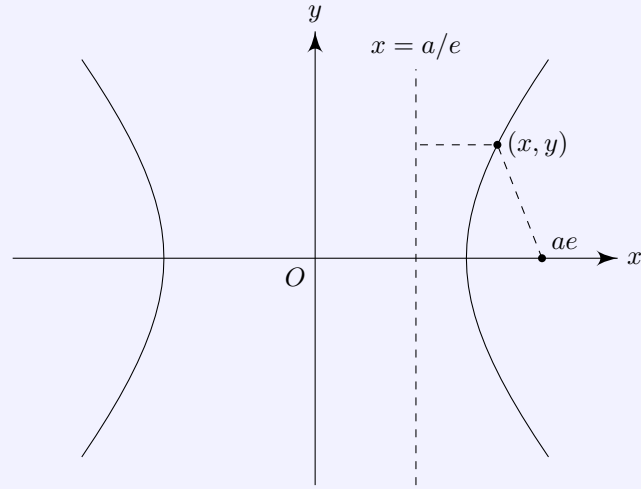
(i) **Ellipse** ($0 \leq e < 1$): The focus-directrix condition gives



$$\sqrt{(x - ae)^2 + y^2} = e \left(\frac{a}{e} - x \right) \implies \frac{x^2}{a^2} + \frac{y^2}{a^2(1 - e^2)} = 1.$$

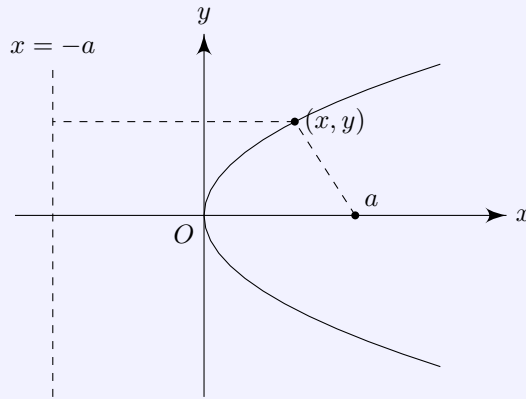
This is an ellipse with semi-major axis a and semi-minor axis $b = a\sqrt{1 - e^2}$. When $e = 0$, we obtain a circle.

(ii) **Hyperbola** ($e > 1$): The focus-directrix condition gives



$$\sqrt{(x - ae)^2 + y^2} = e \left(x - \frac{a}{e} \right) \Rightarrow \frac{x^2}{a^2} - \frac{y^2}{a^2(e^2 - 1)} = 1.$$

(iii) **Parabola** ($e = 1$): The focus-directrix condition gives



$$\sqrt{(x - a)^2 + y^2} = x + a \Rightarrow y^2 = 4ax.$$

Definition (Semi-latus rectum). The *semi-latus rectum* l is defined by $l = a|1 - e^2|$. Geometrically, l/e equals the distance from the focus to the directrix.

Proposition (Polar form of conics). In polar coordinates (r, θ) centered at a focus, the equation of a conic section is

$$r = \frac{l}{1 + e \cos \theta}.$$

Proof. The focus-directrix property states that $r = e \cdot d$, where d is the distance to the directrix. With the focus at the origin and the directrix at distance l/e , we have $d = l/e - r \cos \theta$. Thus

$$r = e \left(\frac{l}{e} - r \cos \theta \right) = l - er \cos \theta,$$

which rearranges to $r(1 + e \cos \theta) = l$. □

Remark. From the polar form, we see that $r \rightarrow \infty$ when $\cos \theta \rightarrow -1/e$. This is only possible when $e \geq 1$ (hyperbola or parabola). For ellipses ($e < 1$), we have $|-1/e| > 1$, so r remains bounded for all θ , as expected for a closed curve.

7 Transformation groups

In this section, we study matrices as transformations of space that preserve certain geometric properties. We have seen that orthogonal matrices arise as change-of-basis matrices between orthonormal bases. Here we take an alternative viewpoint: orthogonal matrices are precisely the linear transformations that preserve distances (lengths and angles).

This geometric characterization depends on our notion of distance, or *metric*. In special relativity, distances are measured using the Minkowski metric rather than the Euclidean metric. This leads to the *Lorentz transformations*, which we briefly introduce at the end of this section.

7.1 The orthogonal group

Proposition (Orthogonal matrices form a group). The set of all $n \times n$ orthogonal matrices forms a group under matrix multiplication.

Proof. We verify the group axioms:

- (i) **Closure:** If P, Q are orthogonal, then $(PQ)(PQ)^\top = PQQ^\top P^\top = PIP^\top = I$, so PQ is orthogonal.
- (ii) **Identity:** The identity matrix I satisfies $II^\top = I$, so I is orthogonal.
- (iii) **Inverses:** If P is orthogonal, then $P^{-1} = P^\top$, which satisfies $(P^\top)(P^\top)^\top = P^\top P = I$, so P^{-1} is orthogonal.
- (iv) **Associativity:** Matrix multiplication is associative. □

Definition (Orthogonal and special orthogonal groups). The *orthogonal group* $O(n)$ is the group of all $n \times n$ orthogonal matrices. The *special orthogonal group* $SO(n)$ is the subgroup of $O(n)$ consisting of orthogonal matrices with determinant $+1$.

Proposition (Structure of $O(2)$). Every matrix in $O(2)$ has one of the following forms:

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad \text{or} \quad S_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

The matrix R_θ is a rotation by angle θ (with $\det R_\theta = 1$), while S_θ is a reflection (with $\det S_\theta = -1$). Thus $SO(2)$ consists of all rotations.

7.2 Length-preserving transformations

Theorem (Characterizations of orthogonal matrices). For an $n \times n$ matrix P , the following are equivalent:

- (i) P is orthogonal (i.e., $P^\top P = I$).

- (ii) P preserves lengths: $|P\mathbf{x}| = |\mathbf{x}|$ for all $\mathbf{x} \in \mathbb{R}^n$.
- (iii) P preserves the inner product: $(P\mathbf{x}) \cdot (P\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
- (iv) P preserves orthonormality: if $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is orthonormal, so is $\{P\mathbf{v}_1, \dots, P\mathbf{v}_n\}$.
- (v) The columns of P are orthonormal.

Proof. We prove the cyclic implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (v) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii): $|P\mathbf{x}|^2 = (P\mathbf{x})^\top (P\mathbf{x}) = \mathbf{x}^\top P^\top P \mathbf{x} = \mathbf{x}^\top \mathbf{x} = |\mathbf{x}|^2$.

(ii) $\Rightarrow$ (iii): Using the polarization identity, $|P(\mathbf{x} + \mathbf{y})|^2 = |\mathbf{x} + \mathbf{y}|^2$ expands to

$$|P\mathbf{x}|^2 + |P\mathbf{y}|^2 + 2(P\mathbf{x}) \cdot (P\mathbf{y}) = |\mathbf{x}|^2 + |\mathbf{y}|^2 + 2\mathbf{x} \cdot \mathbf{y}.$$

Since $|P\mathbf{x}| = |\mathbf{x}|$ and $|P\mathbf{y}| = |\mathbf{y}|$, we conclude $(P\mathbf{x}) \cdot (P\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$.

(iii) $\Rightarrow$ (iv): If $\{\mathbf{v}_i\}$ is orthonormal, then $(P\mathbf{v}_i) \cdot (P\mathbf{v}_j) = \mathbf{v}_i \cdot \mathbf{v}_j = \delta_{ij}$, so $\{P\mathbf{v}_i\}$ is orthonormal.

(iv) $\Rightarrow$ (v): Apply (iv) to the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$. Then $\{P\mathbf{e}_1, \dots, P\mathbf{e}_n\}$ is orthonormal, and these are precisely the columns of P .

(v) $\Rightarrow$ (i): Let $\mathbf{p}_i$ denote the i th column of P . Then $(P^\top P)_{ij} = \mathbf{p}_i \cdot \mathbf{p}_j = \delta_{ij}$, so $P^\top P = I$. $\square$

Corollary. The orthogonal group $O(n)$ is precisely the group of linear transformations of $\mathbb{R}^n$ that preserve the Euclidean length (or equivalently, the Euclidean inner product).

7.3 Lorentz transformations

In special relativity, spacetime is equipped with a different notion of distance than Euclidean space. We briefly introduce the matrices that preserve this relativistic distance in $(1+1)$ -dimensional spacetime (one space dimension and one time dimension).

Definition (Minkowski inner product). The *Minkowski inner product* on $\mathbb{R}^2$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle_J = \mathbf{x}^\top J \mathbf{y}, \quad \text{where} \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Explicitly, $\langle \mathbf{x}, \mathbf{y} \rangle_J = x_1 y_1 - x_2 y_2$.

Remark. Compare this to the Euclidean inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top I \mathbf{y} = x_1 y_1 + x_2 y_2$. The Minkowski inner product has a crucial sign difference in the second term.

Proposition (Characterization of Minkowski-preserving matrices). A matrix M preserves the Minkowski inner product (i.e., $\langle M\mathbf{x}, M\mathbf{y} \rangle_J = \langle \mathbf{x}, \mathbf{y} \rangle_J$ for all

$\mathbf{x}, \mathbf{y}$) if and only if

$$M^\top JM = J.$$

Proof. We have $\langle M\mathbf{x}, M\mathbf{y} \rangle_J = (M\mathbf{x})^\top J(M\mathbf{y}) = \mathbf{x}^\top M^\top JM\mathbf{y}$. This equals $\mathbf{x}^\top J\mathbf{y}$ for all $\mathbf{x}, \mathbf{y}$ if and only if $M^\top JM = J$. $\square$

Proposition (Structure of Minkowski-preserving matrices). The matrices satisfying $M^\top JM = J$ with $M_{11} > 0$ take one of two forms:

$$H_\alpha = \begin{pmatrix} \cosh \alpha & \sinh \alpha \\ \sinh \alpha & \cosh \alpha \end{pmatrix} \quad \text{or} \quad K_\alpha = \begin{pmatrix} \cosh \alpha & -\sinh \alpha \\ \sinh \alpha & -\cosh \alpha \end{pmatrix},$$

where H_α is a *hyperbolic rotation* (or *boost*) and K_α is a *hyperbolic reflection*.

Remark. The condition $M_{11} > 0$ excludes transformations that reverse the direction of time. In physics, such time-reversing transformations are typically excluded, giving the *proper orthochronous Lorentz group*.

Definition (Lorentz boost). A *Lorentz boost* with velocity v (where $|v| < 1$ in units where the speed of light $c = 1$) is the matrix

$$B_v = \frac{1}{\sqrt{1-v^2}} \begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix} = \begin{pmatrix} \gamma & \gamma v \\ \gamma v & \gamma \end{pmatrix},$$

where $\gamma = 1/\sqrt{1-v^2}$ is the *Lorentz factor*. We have $B_v = H_\alpha$ with $\alpha = \tanh^{-1}(v)$.

Definition (Lorentz group). The *Lorentz group* (in $1+1$ dimensions) is the group of all Lorentz boosts under matrix multiplication.

Proposition (Velocity addition formula). Lorentz boosts satisfy $B_{v_1}B_{v_2} = B_{v_3}$, where

$$v_3 = \frac{v_1 + v_2}{1 + v_1 v_2}.$$

This is the relativistic velocity addition formula.

Proof. Since $B_v = H_{\tanh^{-1}(v)}$ and hyperbolic rotations satisfy $H_{\alpha_1}H_{\alpha_2} = H_{\alpha_1+\alpha_2}$, we have

$$v_3 = \tanh(\tanh^{-1}(v_1) + \tanh^{-1}(v_2)) = \frac{v_1 + v_2}{1 + v_1 v_2},$$

using the addition formula for hyperbolic tangent. $\square$